

8 Reset and Clock Control (RCC)

The RCC block manages the clock and reset generation for the whole microcontroller.

The RCC block is located in the D3 domain (refer to [Section 6: Power control \(PWR\)](#) for a detailed description).

The operating modes this section refers to are defined in [Section 6.6.1: Operating modes](#) of the PWR block.

8.1 RCC main features

Reset block

- Generation of local and system reset
- Bidirectional pin reset allowing to reset the microcontroller or external devices
- WWDG reset supported

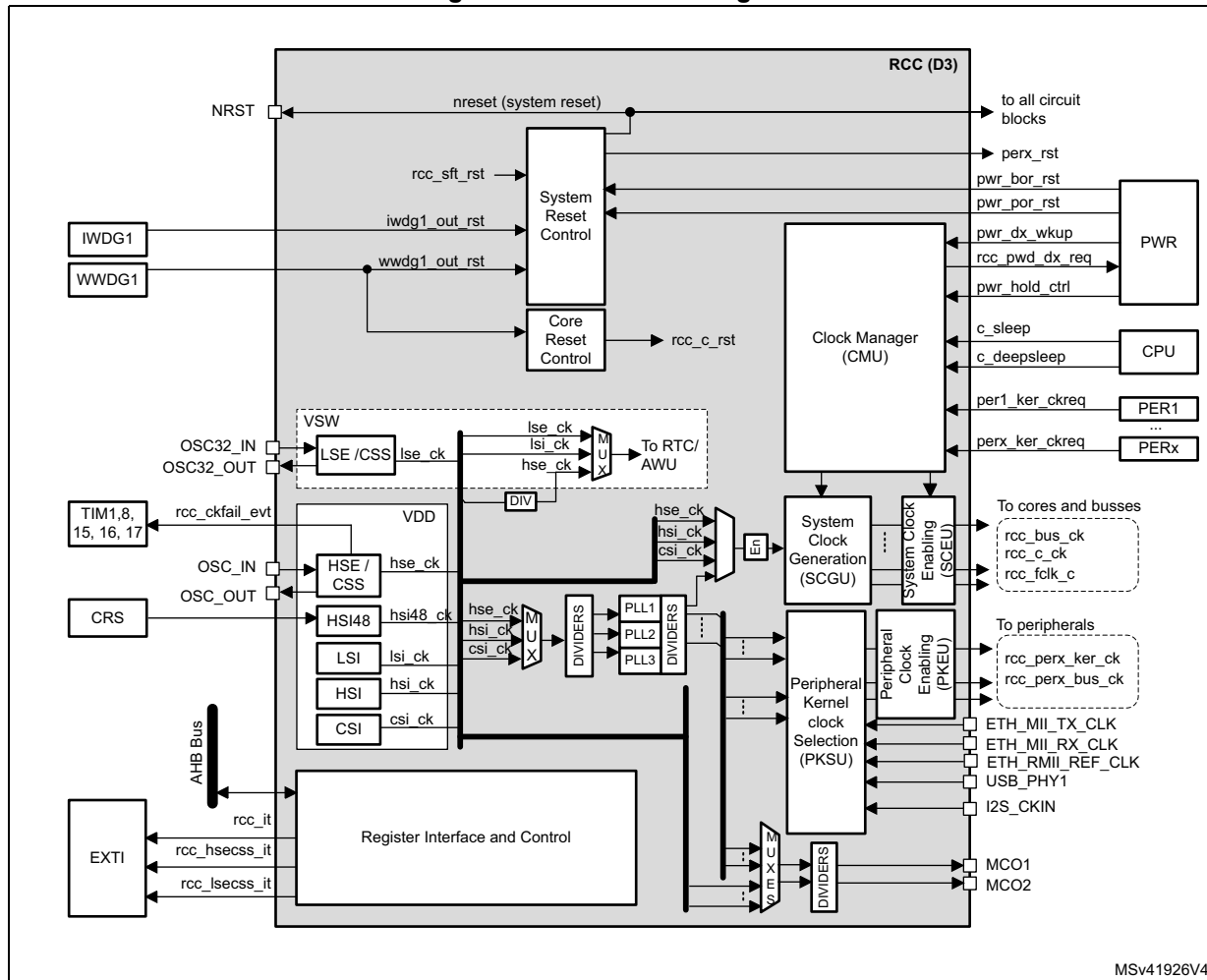
Clock generation block

- Generation and dispatching of clocks for the complete device
- 3 separate PLLs using integer or fractional ratios
- Possibility to change the PLL fractional ratios on-the-fly
- Smart clock gating to reduce power dissipation
- 2 external oscillators:
 - High-speed external oscillator (HSE) supporting a wide range of crystals from 4 to 48 MHz frequency
 - Low-speed external oscillator (LSE) for the 32 kHz crystals
- 4 internal oscillators
 - High-speed internal oscillator (HSI)
 - 48 MHz RC oscillator (HSI48)
 - Low-power Internal oscillator (CSI)
 - Low-speed internal oscillator (LSI)
- Buffered clock outputs for external devices
- Generation of two types of interrupts lines:
 - Dedicated interrupt lines for clock security management
 - One general interrupt line for other events
- Clock generation handling in Stop and Standby mode
- D3 domain Autonomous mode

8.2 RCC block diagram

Figure 42 shows the RCC block diagram.

Figure 42. RCC Block diagram



8.3 RCC pins and internal signals

Table 52 lists the RCC inputs and output signals connected to package pins or balls.

Table 52. RCC input/output signals connected to package pins or balls

Signal name	Signal type	Description
NRST	I/O	System reset, can be used to provide reset to external devices
OSC32_IN	I	32 kHz oscillator input
OSC32_OUT	O	32 kHz oscillator output
OSC_IN	I	System oscillator input

Table 52. RCC input/output signals connected to package pins or balls (continued)

Signal name	Signal type	Description
OSC_OUT	O	System oscillator output
MCO1	O	Output clock 1 for external devices
MCO2	O	Output clock 2 for external devices
I2S_CKIN	I	External kernel clock input for digital audio interfaces: SPI/I2S, SAI, and DFSDM
ETH_MII_TX_CLK	I	External TX clock provided by the Ethernet MII interface
ETH_MII_RX_CLK	I	External RX clock provided by the Ethernet MII interface
ETH_RMII_REF_CLK	I	External reference clock provided by the Ethernet RMII interface
USB_PHY1	I	USB clock input provided by the external USB PHY (OTG_HS_ULPI_CK)

The RCC exchanges a lot of internal signals with all components of the product, for that reason, the [Table 52](#) only shows the most significant internal signals.

Table 53. RCC internal input/output signals

New Signal name	Signal type	Description
rcc_it	O	General interrupt request line
rcc_hsecss_it	O	HSE clock security failure interrupt
rcc_lsecss_it	O	LSE clock security failure interrupt
rcc_ckfail_evt	O	Event indicating that a HSE clock security failure is detected. This signal is connected to TIMERS
nreset	I/O	System reset
iwdg1_out_rst	I	Reset line driven by the IWDG1, indicating that a timeout occurred.
wwdg1_out_rst	I	Reset line driven by the WWDG1, indicating that a timeout occurred.
pwr_bor_rst	I	Brownout reset generated by the PWR block
pwr_por_rst	I	Power-on reset generated by the PWR block
pwr_vsw_rst	I	Power-on reset of the VSW domain generated by the PWR block
rcc_perx_rst	O	Reset generated by the RCC for the peripherals.
pwr_d[3:1]_wkup	I	Wake-up domain request generated by the PWR. Generally used to restore the clocks a domain when this domain exits from DStop
rcc_pwd_d[3:1]_req	O	Low-Power request generated by the RCC. Generally used to ask to the PWR to set a domain into low-power mode, when a domain is in DStop.
pwr_hold_ctrl	I	Signals generated by the PWR, in order to set the processor into CStop when exiting from system Stop mode.
c_sleep	I	Signal generated by the CPU, indicating if the CPU is in CRun, CSleep or CStop.
c_deepsleep	I	

Table 53. RCC internal input/output signals (continued)

New Signal name	Signal type	Description
perx_ker_ckreq	I	Signal generated by some peripherals in order to request the activation of their kernel clock.
rcc_perx_ker_ck	O	Kernel clock signals generated by the RCC, for some peripherals.
rcc_perx_bus_ck	O	Bus interface clock signals generated by the RCC for peripherals.
rcc_bus_ck	O	Clocks for APB (rcc_apb_ck), AHB (rcc_ahb_ck) and AXI (rcc_axi_ck) bridges generated by the RCC.
rcc_c_ck	O	Clock for the CPU, generated by the RCC.
rcc_fclk_c	O	

8.4 RCC reset block functional description

Several sources can generate a reset:

- An external device via NRST pin
- A failure on the supply voltage applied to V_{DD}
- A watchdog timeout
- A software command

The reset scope depends on the source that generates the reset. Three reset categories exist:

- Power-on/off reset
- System reset
- Local resets

8.4.1 Power-on/off reset

The power-on/off reset (**pwr_por_rst**) is generated by the power controller block (PWR). It is activated when the input voltage (V_{DD}) is below a threshold level. This is the most complete reset since it resets the whole circuit, except the backup domain. The power-on/off reset function can be disabled through PDR_ON pin (see [Section 6.5: Power supply supervision](#)).

Refer to [Table 54: Reset distribution summary](#) for details.

8.4.2 System reset

A system reset (**nreset**) resets all registers to their reset values unless otherwise specified in the register description.

A system reset can be generated from one of the following sources:

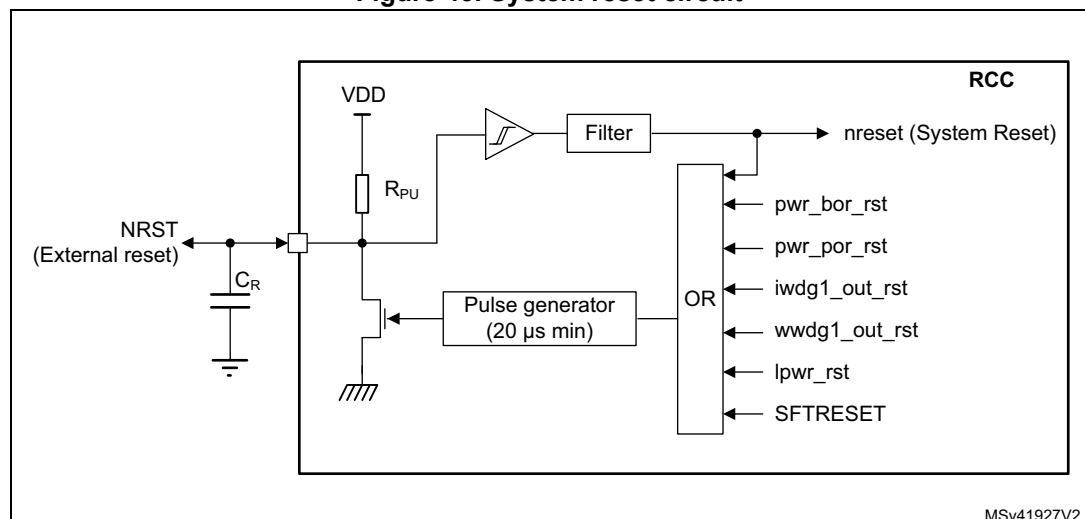
- A reset from NRST pin (external reset)
- A reset from the power-on/off reset block (**pwr_por_rst**)
- A reset from the brownout reset block (**pwr_bor_rst**)
Refer to [Section 6.5.2: Brownout reset \(BOR\)](#) for a detailed description of the BOR function.
- A reset from the independent watchdogs (**iwdg1_out_rst**)
- A software reset from the Cortex®-M7 core
It is generated via the SYSRESETREQ signal issued by the Cortex®-M7 core. This signal is also named SFTRESET in this document.
- A reset from the window watchdogs depending on WWDG configuration (**wwdg1_out_rst**)
- A reset from the low-power mode security reset, depending on option byte configuration (**lpwr[2:1]_rst**)

Note: The SYSRESETREQ bit in Cortex®-M7 Application Interrupt and Reset Control Register must be set to force a software reset on the device. Refer to the Cortex®-M7 with FPU technical reference manual for more details (see <http://infocenter.arm.com>).

As shown in [Figure 43](#), some internal sources (such as **pwr_por_rst**, **pwr_bor_rst**, **iwdg1_out_rst**) perform a system reset of the circuit, which is also propagated to the NRST pin to reset the connected external devices. The pulse generator guarantees a minimum reset pulse duration of 20 µs for each internal reset source. In case of an external reset, the reset pulse is generated while the NRST pin is asserted Low.

Note: It is not recommended to let the NRST pin unconnected. When it is not used, connect this pin to ground via a 10 to 100 nF capacitor (C_R in [Figure 43](#)).

Figure 43. System reset circuit



8.4.3 Local resets

CPU reset

The CPU can reset itself by means of the CPURST bit in [RCC AHB3 Reset Register \(RCC_AHB3RSTR\)](#).

Domain reset

Some resets also dependent on the domain status. For example, when D1 domain exits from DStandby, it is reset (**d1_rst**). The same mechanism applies to D2.

When the system exits from Standby mode, a **stby_rst** reset is applied. The **stby_rst** signal generates a reset of the complete V_{CORE} domain as long the V_{CORE} voltage provided by the internal regulator is not valid.

[Table 54](#) gives a detailed overview of reset sources and scopes.

Table 54. Reset distribution summary

Reset source	Reset name	D1 CPU	D1 Interconnect	D1 Peripherals	D1 Debug	WWDG1	D2 Interconnect	D2 Peripherals	D3 Peripherals	IWDG1	FLASH	RTC domain	Backup RAM	System Supply	NRST pin	Comments
Pin	NRST	x	x	x	-	x	x	x	x	x	-	-	-	-	x	– Resets D1 and D2 domains, and all their peripherals – Resets D3 domain peripherals – Resets V_{DD} domain: IWDG1, LDO... – Debug features, Flash memory, RTC and backup RAM are not reset
PWR	pwr_bor_rst	x	x	x	-	x	x	x	x	x	-	-	-	-	x	– Same as pin reset. The pin is asserted as well.
	pwr_por_rst	x	x	x	x	x	x	x	x	x	x	-	-	x	x	– Same as pwr_bor_rst reset, plus: Reset of the Flash memory digital block (including the option byte loading). – Reset of the debug block
	lpwr_rst	x	x	x	-	x	x	x	x	x	-	-	-	-	x	– The low-power mode security reset has the same scope than pwr_por_rst. Refer to Section 8.4.5: Low-power mode security reset (lpwr_rst) for additional information.

Table 54. Reset distribution summary (continued)

Reset source	Reset name	D1 CPU	D1 Interconnect	D1 Peripherals	D1 Debug	WWDG1	D2 Interconnect	D2 Peripherals	D3 Peripherals	IWDG1	FLASH	RTC domain	Backup RAM	System Supply	NRST pin	Comments
RCC	BDRST	-	-	-	-	-	-	-	-	-	-	x	-	-	-	– The backup domain reset can be triggered by software. Refer to Section 8.4.6: Backup domain reset for additional information
	d1_rst	x	x	x	x	x	-	-	-	-	-	-	-	-	-	– Resets D1 domain, and all its peripherals, when the domain exits DStandby mode.
	d2_rst	-	-	-	-	-	x	x	-	-	-	-	-	-	-	– Resets D2 domain, and all its peripherals, when the domain exits DStandby mode.
	stby_rst	x	x	x	x	x	x	x	x	-	-	-	-	-	-	– When the device exits Standby mode, a reset of the complete V _{CORE} domain is performed as long the V _{CORE} voltage is not valid. The V _{CORE} is supplied by the internal regulator. NRST signal is not asserted.
	CPURST	x	-	-	-	x	-	-	-	-	-	-	-	-	-	– This reset is generated by software through the bit located into RCC AHB3 Reset Register (RCC_AHB3RSTR) . – Resets the CPU, and the WWDG1 block
CPU	SFTRESET	x	x	x	-	x	x	x	x	x	-	-	-	-	x	– This reset is generated by software when writing SYSRESETREQ bit located into AIRCR register of the Cortex [®] -M7 core. – Same scope as pwr_bor_rst reset.
Backup domain	pwr_vsw_rst	-	-	-	-	-	-	-	-	-	-	x	-	-	-	– This reset is generated by the backup domain when the V _{SW} supply voltage is outside the operating range.
IWDG1	iwdg1_out_rst	x	x	x	-	x	x	x	x	x	-	-	-	-	x	– Same as pwr_bor_rst reset.
WWDG1	wwdg1_out_rst	x	x	x	-	x	x	x	x	x	-	-	-	-	x	– Same as pwr_bor_rst reset.

8.4.4 Reset source identification

The CPU can identify the reset source by checking the reset flags in the RCC_RSR (or RCC_C1_RSR) register.

The CPU can reset the flags by setting RMVF bit.

[Table 55](#) shows how the status bits of RCC_RSR (or RCC_C1_RSR) register behaves, according to the situation that generated the reset. For example when an IWDG1 timeout occurs (line #10), if the CPU is reading the RCC_RSR (or RCC_C1_RSR) register during the boot phase, both PINRSTF and IWDG1RSTF bits are set, indicating that the IWDG1 also generated a pin reset.

Table 55. Reset source identification (RCC_RSR)⁽¹⁾

#	Situations Generating a Reset	LPWRRSTF	WWDG1RSTF	IWDG1RSTF	SFTRSTF	PORRSTF	PINRSTF	BORRSTF	D2RSTF	D1RSTF	CPURSTF
1	Power-on reset (pwr_por_rst)	0	0	0	0	1	1	1	1	1	1
2	Pin reset (NRST)	0	0	0	0	0	1	0	0	0	1
3	Brownout reset (pwr_bor_rst)	0	0	0	0	0	1	1	0	0	1
4	System reset generated by CPU (SFTRESET)	0	0	0	1	0	1	0	0	0	1
5	CPU reset (CPURST)	0	0	0	0	0	0	0	0	0	1
6	WWDG1 reset (wwdg1_out_rst)	0	1	0	0	0	1	0	0	0	1
8	IWDG1 reset (iwdg1_out_rst)	0	0	1	0	0	1	0	0	0	1
10	D1 exits DStandby mode	0	0	0	0	0	0	0	0	1	0
11	D2 exits DStandby mode	0	0	0	0	0	0	0	1	0	0
12	D1 erroneously enters DStandby mode or CPU erroneously enters CStop mode	1	0	0	0	0	1	0	0	0	1

1. Grayed cells highlight the register bits that are set.

8.4.5 Low-power mode security reset (lpwr_rst)

To prevent critical applications from mistakenly enter a low-power mode, two low-power mode security resets are available. When enabled through nRST_STOP_D1 option bytes, a system reset is generated if the following conditions are met:

- CPU accidentally enters CStop mode
This type of reset is enabled by resetting nRST_STOP_D1 user option byte. In this case, whenever the CPU CStop mode entry sequence is successfully executed, a system reset is generated.
- D1 domain accidentally enters DStandby mode
This type of reset is enabled by resetting nRST_STDBY_D1 user option byte. In this case, whenever a D1 domain DStandby mode entry sequence is successfully executed, a system reset is generated.

LPWRRSTF bits in [RCC Reset Status Register \(RCC_RSR\)](#) indicates that a low-power mode security reset occurred (see line #12 in [Table 55](#)).

lpwr_rst is activated when a low-power mode security reset due to D1 or CPU occurred.

Refer to [Section 4.4: FLASH option bytes](#) for additional information.

8.4.6 Backup domain reset

A backup domain reset is generated when one of the following events occurs:

- A software reset, triggered by setting BDRST bit in the [RCC Backup Domain Control Register \(RCC_BDCR\)](#). All RTC registers and the RCC_BDCR register are reset to their default values. The backup RAM is not affected.
- V_{SW} voltage is outside the operating range. All RTC registers and the RCC_BDCR register are reset to their default values. In this case the content of the backup RAM is no longer valid.

There are two ways to reset the backup RAM:

- through the Flash memory interface by requesting a protection level change from 1 to 0
- when a tamper event occurs.

Refer to [Section 6.4.4: Backup domain](#) section of PWR block for additional information.

8.4.7 Power-on and wakeup sequences

For detailed diagrams refer to [Section 6.4.1: System supply startup](#) in the PWR section.

The time interval between the event which exits the product from a low-power and the moment where the CPU is able to execute code, depends on the system state and on its configuration. [Figure 44](#) shows the most usual examples.

Power-on wakeup sequence

The power-on wakeup sequence shown in [Figure 44](#) gives the most significant phases of the power-on sequence. It is the longest sequence since the circuit was not powered. Note that this sequence remains unchanged, whatever V_{BAT} was present or not.

Boot from pin reset

When a pin reset occurs, V_{DD} is still present. As a result:

- The regulator settling time is faster since the reference voltage is already stable.
- The HSI restart delay may be needed if the HSI was not enabled when the NRST occurred, otherwise this restart delay phase is skipped.
- The Flash memory power recovery delay can also be skipped if the Flash memory was enabled when the NRST occurred.

Note: The boot sequence is similar for `pwr_bor_rst`, `lpwr_rst`, `STFxRESET`, `iwdg1_out_rst` and `wwdg1_out_rst` (if `WW1RSC` = '1').

Boot from system Standby

When waking up from system Standby, the reference voltage is stable since V_{DD} has not been removed. As a result, the regulator settling time is fast. Since V_{CORE} was not present, the restart delay for the HSI, the Flash memory power recovery and the option byte reloading cannot be skipped.

Restart from system Stop

When restarting from system Stop, V_{DD} is still present. As a result, the sequence is mainly composed of two steps:

1. Regulator settling time to reach VOS3 (default voltage)
2. HSI/CSI restart delay. This step can be skipped if `HSIKERON` or `CSIKERON` bit, in [RCC Source Control Register \(RCC_CR\)](#) is set to '1'.

Boot from domain DStandby

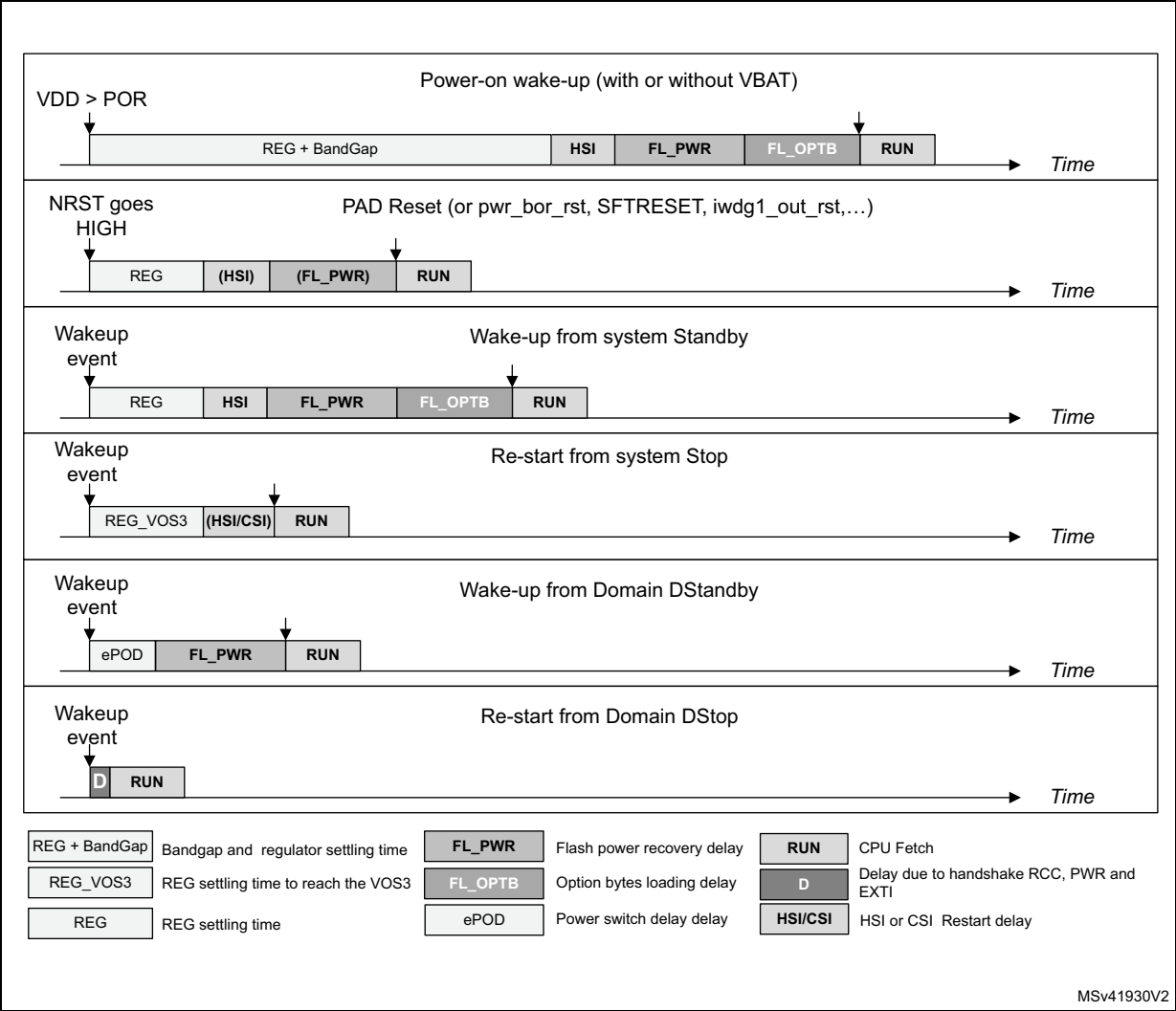
The boot sequence of a domain from domain DStandby is mainly composed of two steps:

1. The power switch settling time (the regulator is already activated).
2. The Flash memory power recovery.

Restart from domain DStop

The restart sequence of a domain from domain DStop is mainly composed of the handshake between the RCC, EXTI and PWR blocks.

Figure 44. Boot sequences versus system states



8.5 RCC clock block functional description

The RCC provides a wide choice of clock generators:

- HSI (High-speed internal oscillator) clock: ~ 8, 16, 32 or 64 MHz
- HSE (High-speed external oscillator) clock: 4 to 48 MHz
- LSE (Low-speed external oscillator) clock: 32 kHz
- LSI (Low-speed internal oscillator) clock: ~ 32 kHz
- CSI (Low-power internal oscillator) clock: ~4 MHz
- HSI48 (High-speed 48 MHz internal oscillator) clock: ~48 MHz

It offers a high flexibility for the application to select the appropriate clock for CPU and peripherals, in particular for peripherals that require a specific clock such as Ethernet, USB OTG-FS and HS, SPI/I2S, SAI and SDMMC.

To optimize the power consumption, each clock source can be switched ON or OFF independently.

The RCC provides up to 3 PLLs; each of them can be configured with integer or fractional ratios.

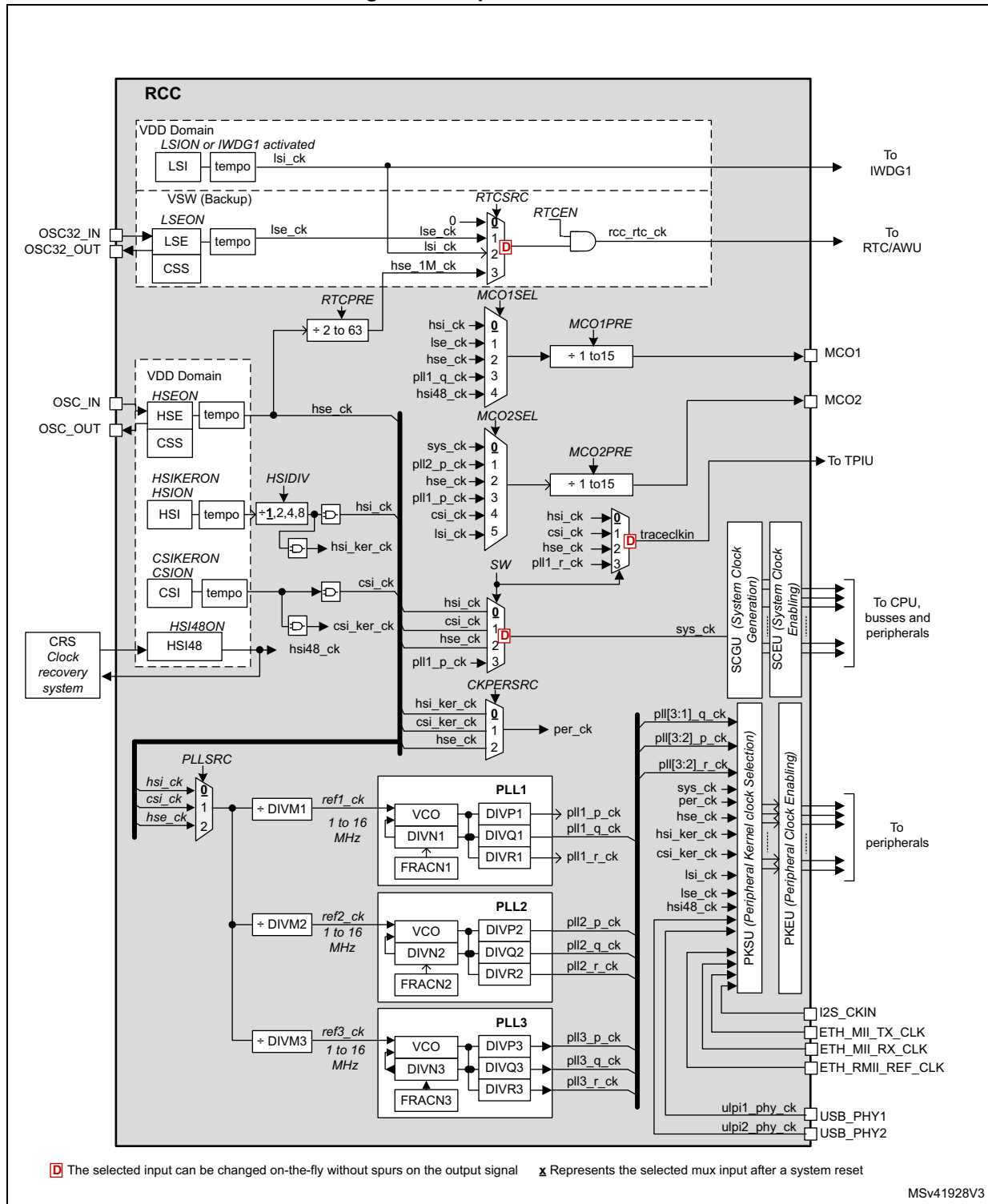
As shown in the [Figure 45](#), the RCC offers 2 clock outputs (MCO1 and MCO2), with a great flexibility on the clock selection and frequency adjustment.

The SCGU block (System Clock Generation Unit) contains several prescalers used to configure the CPU and bus matrix clock frequencies.

The PKSU block (Peripheral Kernel clock Selection Unit) provides several dynamic switches allowing a large choice of kernel clock distribution to peripherals.

The PKEU (Peripheral Kernel clock Enable Unit) and SCEU (System Clock Enable Unit) blocks perform the peripheral kernel clock gating, and the bus interface/cores/bus matrix clock gating, respectively.

Figure 45. Top-level clock tree



8.5.1 Clock naming convention

The RCC provides clocks to the complete circuit. To avoid misunderstanding, the following terms are used in this document:

- **Peripheral clocks**
The peripheral clocks are the clocks provided by the RCC to the peripherals. Two kinds of clock are available:
 - The bus interface clocks
 - The kernel clocks

A peripheral receives from the RCC a bus interface clock in order to access its registers, and thus control the peripheral operation. This clock is generally the AHB, APB or AXI clock depending on which bus the peripheral is connected to. Some peripherals only need a bus interface clock (e.g. RNG, TIMx).

Some peripherals also require a dedicated clock to handle the interface function. This clock is named “kernel clock”. As an example, peripherals such as SAI have to generate specific and accurate master clock frequencies, which require dedicated kernel clock frequencies. Another advantage of decoupling the bus interface clock from the specific interface needs, is that the bus clock can be changed without reprogramming the peripheral.
- **CPU clocks**
The CPU clock is the clock provided to the CPU. It is derived from the system clock (**sys_ck**).
- **Bus matrix clocks**
The bus matrix clocks are the clocks provided to the different bridges (APB, AHB or AXI). These clocks are derived from the system clock (**sys_ck**).

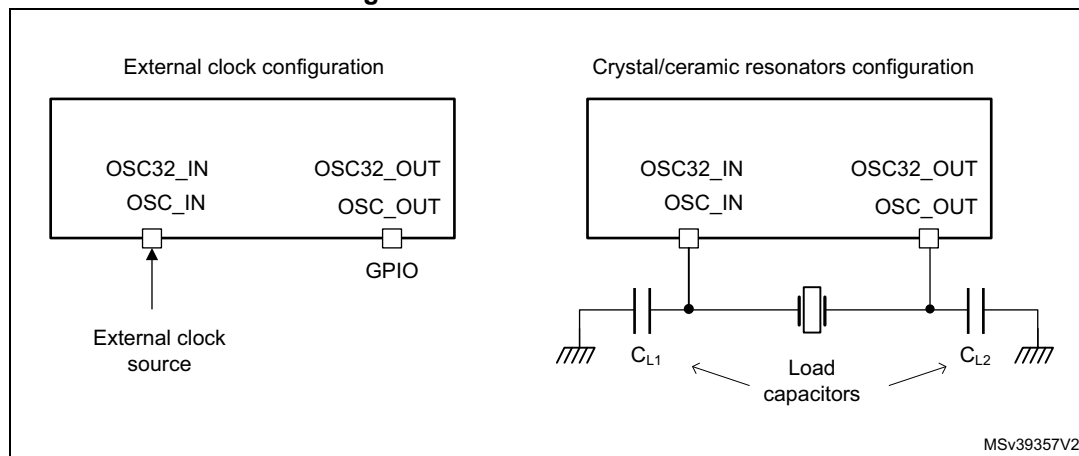
8.5.2 Oscillators description

HSE oscillator

The HSE block can generate a clock from two possible sources:

- External crystal/ceramic resonator
- External clock source

Figure 46. HSE/LSE clock source



External clock source (HSE bypass)

In this mode, an external clock source must be provided to OSC_IN pin. This mode is selected by setting the HSEBYP and HSEON bits of the [RCC Source Control Register \(RCC_CR\)](#) to '1'. The external clock source (square, sinus or triangle) with ~50% duty cycle has to drive the OSC_IN pin.

External crystal/ceramic resonator

The oscillator is enabled by setting the HSEBYP bit to '0' and HSEON bit to '1'.

The HSE can be used when the product requires a very accurate high-speed clock.

The associated hardware configuration is shown in [Figure 46](#): the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. The loading capacitance values must be adjusted according to the selected crystal or ceramic resonator. Refer to the electrical characteristics section of the datasheet for more details.

The HSERDY flag of the [RCC Source Control Register \(RCC_CR\)](#), indicates whether the HSE oscillator is stable or not. At startup, the **hse_ck** clock is not released until this bit is set by hardware. An interrupt can be generated if enabled in the [RCC Clock Source Interrupt Enable Register \(RCC_CIER\)](#).

The HSE can be switched ON and OFF through the HSEON bit. Note that the HSE cannot be switched OFF if one of the two conditions is met:

- The HSE is used directly (via software mux) as system clock
- The HSE is selected as reference clock for PLL1, with PLL1 enabled and selected to provide the system clock (via software mux).

In that case the hardware does not allow programming the HSEON bit to '0'.

The HSE is automatically disabled by hardware, when the system enters Stop or Standby mode (refer to [Section 8.5.7: Handling clock generators in Stop and Standby mode](#) for additional information).

In addition, the HSE clock can be driven to the MCO1 and MCO2 outputs and used as clock source for other application components.

LSE oscillator

The LSE block can generate a clock from two possible sources:

- External crystal/ceramic resonator
- External user clock

External clock source (LSE bypass)

In this mode, an external clock source must be provided to OSC32_IN pin. The input clock can have a frequency up to 1 MHz. This mode is selected by setting the LSEBYP and LSEON bits of [RCC Backup Domain Control Register \(RCC_BDCR\)](#) to '1'. The external clock signal (square, sinus or triangle) with ~50% duty cycle has to drive the OSC32_IN pin.

External crystal/ceramic resonator (LSE crystal)

The LSE clock is generated from a 32.768 kHz crystal or ceramic resonator. It has the advantage to provide a low-power highly accurate clock source to the real-time clock (RTC) for clock/calendar or other timing functions.

The LSERDY flag of the [RCC Backup Domain Control Register \(RCC_BDCR\)](#) indicates whether the LSE crystal is stable or not. At startup, the LSE crystal output clock signal is not released until this bit is set by hardware. An interrupt can be generated if enabled in the [RCC Clock Source Interrupt Enable Register \(RCC_CIER\)](#).

The LSE oscillator is switched ON and OFF using the LSEON bit. The LSE remains enabled when the system enters Stop or Standby mode.

In addition, the LSE clock can be driven to the MCO1 output and used as clock source for other application components.

The LSE also offers a programmable driving capability (LSEDRV[1:0]) that can be used to modulate the amplifier driving capability. The driving capability can be changed dynamically from high drive to medium high drive, and then to medium low drive.

HSI oscillator

The HSI block provides the default clock to the product.

The HSI is a high-speed internal RC oscillator which can be used directly as system clock, peripheral clock, or as PLL input. A predivider allows the application to select an HSI output frequency of 8, 16, 32 or 64 MHz. This predivider is controlled by the HSIDIV.

The HSI advantages are the following:

- Low-cost clock source since no external crystal is required
- Faster startup time than HSE (a few microseconds)

The HSI frequency, even with frequency calibration, is less accurate than an external crystal oscillator or ceramic resonator.

The HSI can be switched ON and OFF using the HSION bit. Note that the HSI cannot be switched OFF if one of the two conditions is met:

- The HSI is used directly (via software mux) as system clock
- The HSI is selected as reference clock for PLL1, with PLL1 enabled and selected to provide the system clock (via software mux).

In that case the hardware does not allow programming the HSION bit to '0'.

Note that the HSIDIV cannot be changed if the HSI is selected as reference clock for at least one enabled PLL (PLLxON bit set to '1'). In that case the hardware does not update the HSIDIV with the new value. However it is possible to change the HSIDIV if the HSI is used directly as system clock.

The HSIRDY flag indicates if the HSI is stable or not. At startup, the HSI output clock is not released until this bit is set by hardware.

The HSI clock can also be used as a backup source (auxiliary clock) if the HSE fails (refer to [Section : CSS on HSE](#)). The HSI can be disabled or not when the system enters Stop mode, please refer to [Section 8.5.7: Handling clock generators in Stop and Standby mode](#) for additional information.

In addition, the HSI clock can be driven to the MCO1 output and used as clock source for other application components.

Care must be taken when the HSI is used as kernel clock for communication peripherals, the application must take into account the following parameters:

- the time interval between the moment where the peripheral generates a kernel clock request and the moment where the clock is really available,
- the frequency accuracy.

Note: The HSI can remain enabled when the system is in Stop mode (see [Section 8.5.7](#) for additional information).

HSION, HSIRDY and HSIDIV bits are located in the [RCC Source Control Register \(RCC_CR\)](#).

HSI calibration

RC oscillator frequencies can vary from one chip to another due to manufacturing process variations. That is why each device is factory calibrated by STMicroelectronics to improve accuracy (refer to the product datasheet for more information).

After a power-on reset, the factory calibration value is loaded in the HSICAL[11:0] bits.

If the application is subject to voltage or temperature variations, this may affect the RC oscillator frequency. The user application can trim the HSI frequency using the HSITRIM bits.

Note: HSICAL and HSITRIM bits are located in [RCC Internal Clock Source Calibration Register \(RCC_ICSCR\)](#) for revision Y devices and in [RCC HSI configuration register \(RCC_HSI CFGR\)](#) for revision V devices.

CSI oscillator

The CSI is a low-power RC oscillator which can be used directly as system clock, peripheral clock, or PLL input.

The CSI advantages are the following:

- Low-cost clock source since no external crystal is required
- Faster startup time than HSE (a few microseconds)
- Very low-power consumption,

The CSI provides a clock frequency of about 4 MHz, while the HSI is able to provide a clock up to 64 MHz.

CSI frequency, even with frequency calibration, is less accurate than an external crystal oscillator or ceramic resonator.

The CSI can be switched ON and OFF through the CSION bit. The CSIRDY flag indicates whether the CSI is stable or not. At startup, the CSI output clock is not released until this bit is set by hardware.

The CSI cannot be switched OFF if one of the two conditions is met:

- The CSI is used directly (via software mux) as system clock
- The CSI is selected as reference clock for PLL1, with PLL1 enabled and selected to provide the system clock (via software mux).

In that case the hardware does not allow programming the CSION bit to '0'.

The CSI can be disabled or not when the system enters Stop mode (refer to [Section 8.5.7: Handling clock generators in Stop and Standby mode](#) for additional information).

In addition, the CSI clock can be driven to the MCO2 output and used as clock source for other application components.

Even if the CSI settling time is faster than the HSI, care must be taken when the CSI is used as kernel clock for communication peripherals: the application has to take into account the following parameters:

- the time interval between the moment where the peripheral generates a kernel clock request and the moment where the clock is really available,
- the frequency precision.

Note: *CSICAL* and *CSITRIM* bits are located into the [RCC Source Control Register \(RCC_CR\)](#) or revision Y devices and [RCC CSI configuration register \(RCC_CSICFGR\)](#) for revision V devices.

CSI calibration

RC oscillator frequencies can vary from one chip to another due to manufacturing process variations, this is why each device is factory calibrated by STMicroelectronics to achieve improve (refer to the product datasheet for more information).

After reset, the factory calibration value is loaded in the CSICAL[7:0] bits.

If the application is subject to voltage or temperature variations, this may affect the RC oscillator frequency. The user application can trim the CSI frequency using the CSITRIM bits.

Note: *CSICAL* and *CSITRIM* bits are located in [RCC Internal Clock Source Calibration Register \(RCC_ICSCR\)](#) for revision Y devices and in [RCC CSI configuration register \(RCC_CSICFGR\)](#) for revision V devices.

HSI48 oscillator

The HSI48 is an RC oscillator that delivers a 48 MHz clock that can be used directly as kernel clock for some peripherals.

The HSI48 oscillator mainly aims at providing a high precision clock to the USB peripheral by means of a special Clock Recovery System (CRS) circuitry, which could use the USB SOF signal, the LSE, or an external signal, to automatically adjust the oscillator frequency on-the-fly, in very small granularity.

The HSI48 oscillator is disabled as soon as the system enters Stop or Standby mode. When the CRS is not used, this oscillator is free running and thus subject to manufacturing process variations. That is why each device is factory calibrated by STMicroelectronics to achieve an accuracy of ACC_{HSI48} (refer to the product datasheet of the for more information).

For more details on how to configure and use the CRS, please refer to [Section 9: Clock recovery system \(CRS\)](#).

The HSI48RDY flag indicates whether the HSI48 oscillator is stable or not. At startup, the HSI48 output clock is not released until this bit is set by hardware.

The HSI48 can be switched ON and OFF using the HSI48ON bit.

The HSI48 clock can also be driven to the MCO1 multiplexer and used as clock source for other application components.

Note: *HSI48ON* and *HSI48RDY* bits are located in the [RCC Source Control Register \(RCC_CR\)](#).

LSI oscillator

The LSI acts as a low-power clock source that can be kept running when the system is in Stop or Standby mode for the independent watchdog (IWDG) and Auto-Wakeup Unit (AWU). The clock frequency is around 32 kHz. For more details, refer to the electrical characteristics section of the datasheet.

The LSI can be switched ON and OFF using the LSION bit. The LSIRDY flag indicates whether the LSI oscillator is stable or not. If an independent watchdog is started either by hardware or software, the LSI is forced ON and cannot be disabled.

The LSI remains enabled when the system enters Stop or Standby mode (refer to [Section 8.5.7: Handling clock generators in Stop and Standby mode](#) for additional information).

At LSI startup, the clock is not provided until the hardware sets the LSIRDY bit. An interrupt can be generated if enabled in the [RCC Clock Source Interrupt Enable Register \(RCC_CIER\)](#).

In addition, the LSI clock can be driven to the MCO2 output and used as a clock source for other application components.

Note: Bits LSION and LSIRDY are located into the [RCC Clock Control and Status Register \(RCC_CSR\)](#).

8.5.3 Clock Security System (CSS)

CSS on HSE

The clock security system can be enabled by software via the HSECSSON bit. The HSECSSON bit can be enabled even when the HSEON is set to '0'.

The CSS on HSE is enabled by the hardware when the HSE is enabled and ready, and HSECSSON set to '1'.

The CSS on HSE is disabled when the HSE is disabled. As a result, this function does not work when the system is in Stop mode.

It is not possible to clear directly the HSECSSON bit by software.

The HSECSSON bit is cleared by hardware when a system reset occurs or when the system enters Standby mode (see [Section 8.4.2: System reset](#)).

If a failure is detected on the HSE clock, the system automatically switches to the HSI in order to provide a safe clock. The HSE is then automatically disabled, a clock failure event is sent to the break inputs of advanced-control timers (TIM1, TIM8, TIM15, TIM16, and TIM17), and an interrupt is generated to inform the software about the failure (CSS interrupt: **rcc_hsecss_it**), thus allowing the MCU to perform rescue operations. If the HSE output was used as clock source for PLLs when the failure occurred, the PLLs are also disabled.

If an HSE clock failure occurs when the CSS is enabled, the CSS generates an interrupt which causes the automatic generation of an NMI. The HSECSSF flag in [RCC Clock Source Interrupt Flag Register \(RCC_CIFR\)](#) is set to '1' to allow the application to identify the failure source. The NMI routine is executed indefinitely until the HSECSSF bit is cleared. As a consequence, the application has to clear the HSECSSF flag in the NMI ISR by setting the HSECSSC bit in the [RCC Clock Source Interrupt Clear Register \(RCC_CICR\)](#).

CSS on LSE

A clock security system on the LSE oscillator can be enabled by software by programming the LSECSSON bit in the [RCC Backup Domain Control Register \(RCC_BDCR\)](#).

This bit can be disabled only by hardware when the following conditions are met:

- after a pwr_vsw_rst (V_{SW} software reset)
- or after a failure detection on LSE.

LSECSSON bit must be written after the LSE is enabled (LSEON bit set by software) and ready (LSERDY set by hardware), and after the RTC clock has been selected through the RTCSEL bit.

The CSS on LSE works in all modes (Run, Stop and Standby) except VBAT.

If an LSE failure is detected, the LSE clock is no more delivered to the RTC but the value of RTCSEL, LSECSSON and LSEON bits are not changed by the hardware.

A wakeup is generated in Standby mode. In other modes an interrupt (**rcc_lsecss_it**) can be sent to wake up the software. The software must then disable the LSECSSON bit, stop the defective LSE (clear LSEON bit), and can change the RTC clock source (no clock or LSI or HSE) through RTCSEL bits, or take any required action to secure the application.

8.5.4 Clock output generation (MCO1/MCO2)

Two micro-controller clock output (MCO) pins, MCO1 and MCO2, are available. A clock source can be selected for each output. The selected clock can be divided thanks to configurable prescaler (refer to [Figure 45](#) for additional information on signal selection).

MCO1 and MCO2 outputs are controlled via MCO1PRE[3:0], MCO1[2:0], MCO2PRE[3:0] and MCO2[2:0] located in the [RCC Clock Configuration Register \(RCC_CFGR\)](#).

The GPIO port corresponding to each MCO pin, has to be programmed in alternate function mode.

The clock provided to the MCOs outputs must not exceed the maximum pin speed (refer to the product datasheet for information on the supported pin speed).

8.5.5 PLL description

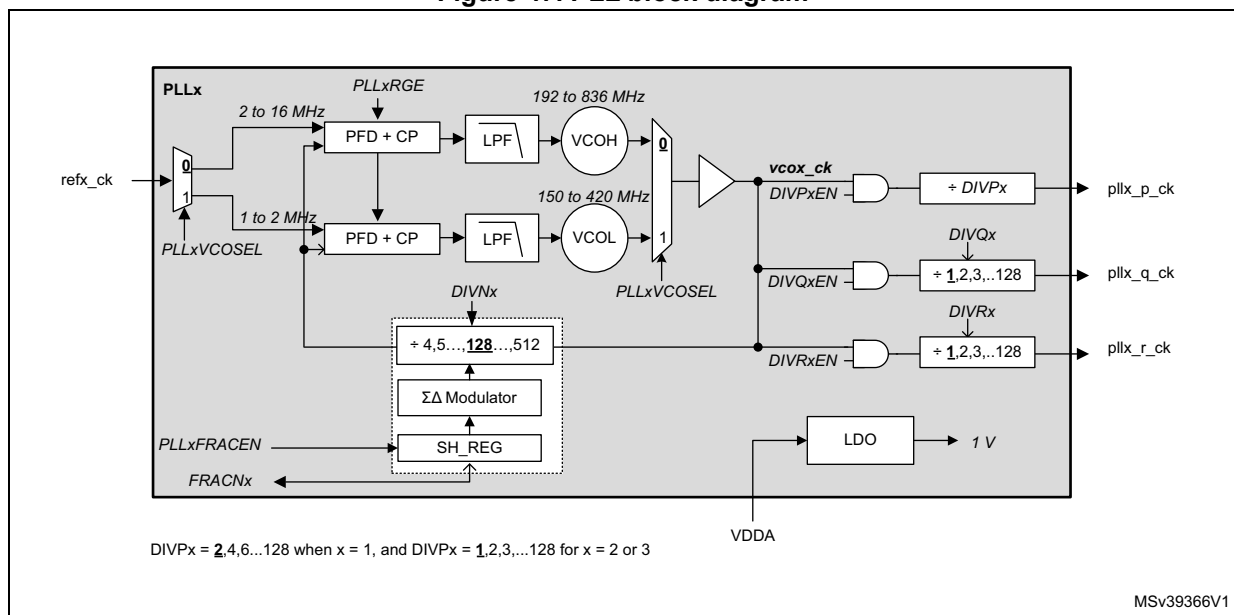
The RCC features three PLLs:

- A main PLL, PLL1, which is generally used to provide clocks to the CPU and to some peripherals.
- Two dedicated PLLs, PLL2 and PLL3, which are used to generate the kernel clock for peripherals.

The PLLs integrated into the RCC are completely independent. They offer the following features:

- Two embedded VCOs:
 - A wide-range VCO (VCOH)
 - A low-frequency VCO (VCOL)
- Input frequency range:
 - 1 to 2 MHz when VCOL is used
 - 2 to 16 MHz when VCOH is used
- Capability to work either in integer or Fractional mode
- 13-bit Sigma-Delta modulator, allowing to fine-tune the VCO frequency by steps of 11 to 0.3 ppm.
- The Sigma-Delta modulator can be updated on-the-fly, without generating frequency overshoots on PLLs outputs.
- Each PLL offer 3 outputs with post-dividers

Figure 47. PLL block diagram



The PLLs are controlled via RCC_PLLxDIVR, RCC_PLLxFRACR, RCC_PLLCFGR and RCC_CR registers.

The frequency of the reference clock provided to the PLLs (**refx_ck**) must range from 1 to 16 MHz. The user application has to program properly the DIVMx dividers of the [RCC PLLs Clock Source Selection Register \(RCC_PLLCKSELR\)](#) in order to match this condition. In addition, the PLLxRGE of the [RCC PLLs Configuration Register \(RCC_PLLCFGR\)](#) field must be set according to the reference input frequency to guarantee an optimal performance of the PLL.

The user application can then configure the proper VCO: if the frequency of the reference clock is lower or equal to 2 MHz, then VCOL must be selected, otherwise VCOH must be chosen. To reduce the power consumption, it is recommended to configure the VCO output to the lowest frequency.

DIVNx loop divider has to be programmed to achieve the expected frequency at VCO output. In addition, the VCO output range must be respected.

The PLLs operate in integer mode when the value of SH_REG (FRACNx shadow register) is set to '0'. The SH_REG is updated with the FRACNx value when PLLxFRACEN bit goes from '0' to '1'. The Sigma-Delta modulator is designed in order to minimize the jitter impact while allowing very small frequency steps.

The PLLs can be enabled by setting PLLxON to '1'. The bits PLLxRDY indicate that the PLL is ready (i.e. locked).

Note: *Before enabling the PLLs, make sure that the reference frequency (**refx_ck**) provided to the PLL is stable, so the hardware does not allow changing DIVMx when the PLLx is ON and it is also not possible to change PLLSRC when one of the PLL is ON.*

The hardware prevents writing PLL1ON to '0' if the PLL1 is currently used to deliver the system clock. There are other hardware protections on the clock generators (refer to sections [HSE oscillator](#), [HSI oscillator](#) and [CSI oscillator](#)).

The following PLL parameters cannot be changed once the PLL is enabled: DIVNx, PLLxRGE, PLLxVCOSEL, DIVPx, DIVQx, DIVRx, DIVPxEN, DIVQxEN and DIVRxEN.

To insure an optimal behavior of the PLL when one of the post-divider (DIVP, DIVQ or DIVR) is not used, the application shall set the enable bit (DIVyEN) as well as the corresponding post-divider bits (DIVP, DIVQ or DIVR) to '0'.

If the above rules are not respected, the PLL output frequency is not guaranteed.

Output frequency computation

When the PLL is configured in integer mode (SH_REG = '0'), the VCO frequency (F_{VCO}) is given by the following expression:

$$F_{VCO} = F_{REF_CK} \times DIVN$$

$$F_{PLL_y_CK} = (F_{VCO} / (DIVy + 1)) \text{ with } y = P, Q \text{ or } R$$

When the PLL is configured in fractional mode (SH_REG different from '0'), the DIVN divider must be initialized before enabling the PLLs. However, it is possible to change the value of FRACNx on-the-fly without disturbing the PLL output.

This feature can be used either to generate a specific frequency from any crystal value with a good accuracy, or to fine-tune the frequency on-the-fly.

For each PLL, the VCO frequency is given by the following formula:

$$F_{VCO} = F_{ref_ck} \times \left(DIVN + \frac{FRACN}{2^{(13)}} \right)$$

Note: For PLL1, DIVP bitfield can only take odd values.

The PLLs are disabled by hardware when:

- The system enters Stop or Standby mode.
- An HSE failure occurs when HSE or PLL (clocked by HSE) are used as system clock.

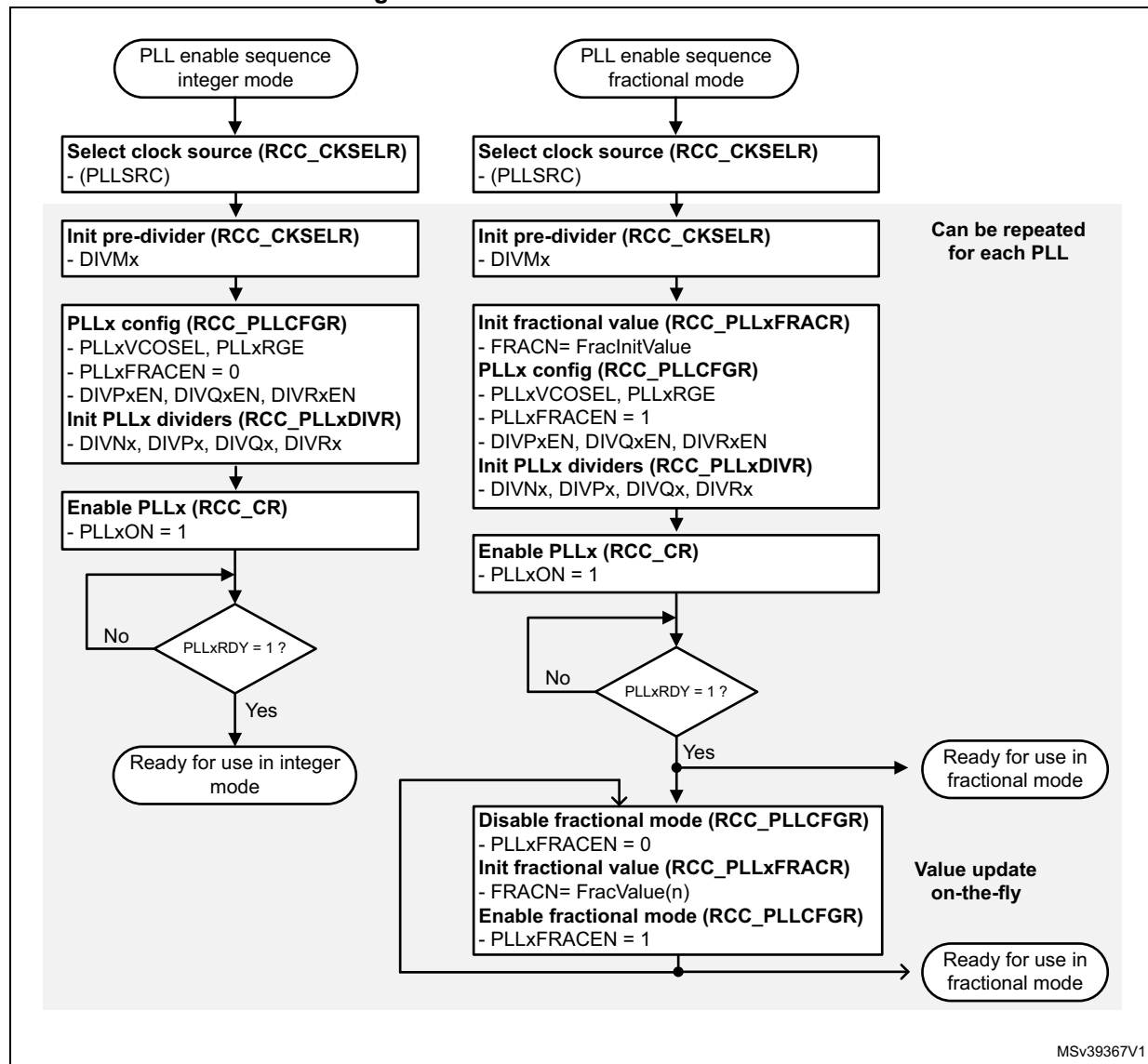
PLL initialization phase

Figure 48 shows the recommended PLL initialization sequence in integer and fractional mode. The PLLx are supposed to be disabled at the start of the initialization sequence:

1. Initialize the PLLs registers according to the required frequency.
 - Set PLLxFRACEN of *RCC PLLs Configuration Register (RCC_PLLCFGR)* to '0' for integer mode.
 - For fractional mode, set FRACN to the required initial value (FracInitValue) and then set PLLxFRACEN to '1'.
2. Once the PLLxON bit is set to '1', the user application has to wait until PLLxRDY bit is set to '1'. If the PLLx is in fractional mode, the PLLxFRACEN bit must not be set back to '0' as long as PLLxRDY = '0'.
3. Once the PLLxRDY bit is set to '1', the PLLx is ready to be used.
4. If the application intends to tune the PLLx frequency on-the-fly (possible only in fractional mode), then:
 - a) PLLxFRACEN must be set to '0',
When PLLxFRACEN = '0', the Sigma-Delta modulator is still operating with the value latched into SH_REG.
 - b) A new value must be uploaded into PLLxFRACR (FracValue(n)).
 - c) PLLxFRACEN must be set to '1', in order to latch the content of PLLxFRACR into its shadow register.

Note: When the PLLxRDY goes to '1', it means that the difference between the PLLx output frequency and the target value is lower than ±2%.

Figure 48. PLLs Initialization Flowchart



8.5.6 System clock (sys_ck)

System clock selection

After a system reset, the HSI is selected as system clock and all PLLs are switched OFF. When a clock source is used for the system clock, it is not possible for the software to disable the selected source via the xxxON bits.

Of course, the system clock can be stopped by the hardware when the System enters Stop or Standby mode.

When the system is running, the user application can select the system clock (**sys_ck**) among the 4 following sources:

- HSE
- HSI
- CSI
- or pll1_p_ck

This function is controlled by programming the [RCC Clock Configuration Register \(RCC_CFGR\)](#). A switch from one clock source to another occurs only if the target clock source is ready (clock stable after startup delay or PLL locked). If a clock source that is not yet ready is selected, the switch occurs when the clock source is ready.

The SWS status bits in the [RCC Clock Configuration Register \(RCC_CFGR\)](#) indicate which clock is currently used as system clock. The other status bits in the RCC_CR register indicate which clock(s) is (are) ready.

System clock generation

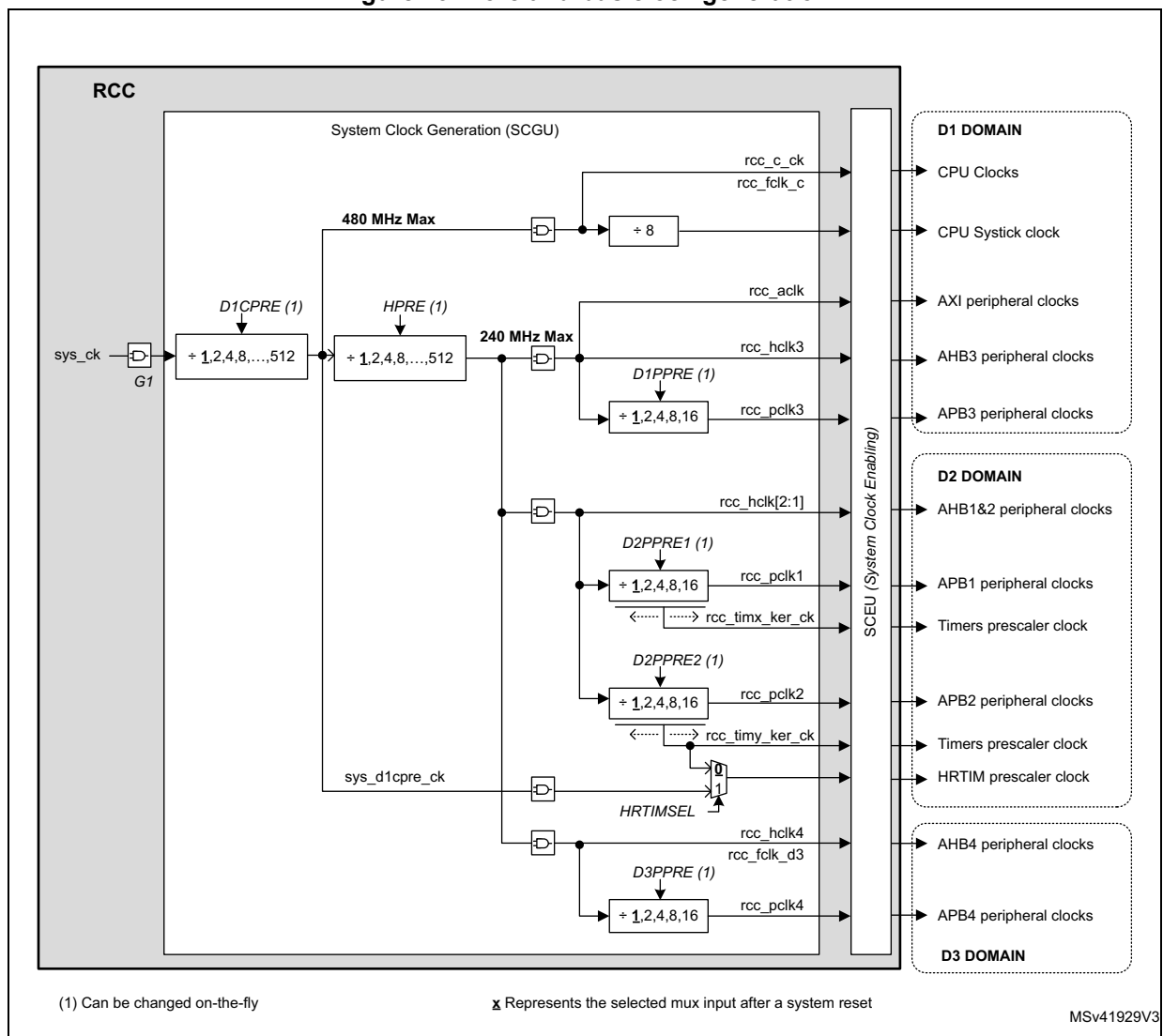
[Figure 49](#) shows a simplified view of the clock distribution for the CPU and busses. All the dividers shown in the block diagram can be changed on-the-fly without generating timing violations. This feature is a very simply solution to adapt the busses frequencies to the application needs, thus optimizing the power consumption.

The D1CPRE divider can be used to adjust the CPU clock. However this also impacts the clock frequency of all bus matrix and HRTIM.

In the same way, HPRE divider can be used to adjust the clock for D1 domain bus matrix, but this also impacts the clock frequency of bus matrix of D2 and D3 domains.

Most of the prescalers are controlled via RCC_D1CFGR, RCC_D2CFGR and RCC_D3CFGR registers.

Figure 49. Core and bus clock generation



This block also provides the clock for the timers (**rcc_timx_ker_ck** and **rcc_tiny_ker_ck**). The frequency of the timers clock depends on the APB prescaler corresponding to the bus to which the timer is connected, and on TIMPRE bit. [Table 56](#) shows how to select the timer clock frequency.

Table 56. Ratio between clock timer and pclk

D2PPRE1 (1) D2PPRE2	TIMPRE (2)		$F_{rcc_timx_ker_ck}$ $F_{rcc_tiny_ker_ck}$	F_{rcc_pclk1} F_{rcc_pclk2}	Comments
0xx	0	→	F_{rcc_hclk1}	F_{rcc_hclk1}	The timer clock is equal to the bus clock.
100	0	→	F_{rcc_hclk1}	$F_{rcc_hclk1} / 2$	The timer clock is twice as fast as the bus clock.
101	0	→	$F_{rcc_hclk1} / 2$	$F_{rcc_hclk1} / 4$	
110	0	→	$F_{rcc_hclk1} / 4$	$F_{rcc_hclk1} / 8$	
111	0	→	$F_{rcc_hclk1} / 8$	$F_{rcc_hclk1} / 16$	

Table 56. Ratio between clock timer and pclk (continued)

D2PPRE1 ⁽¹⁾ D2PPRE2	TIMPRE ⁽²⁾		$F_{\text{rcc_timx_ker_ck}}$ $F_{\text{rcc_tmy_ker_ck}}$	$F_{\text{rcc_pclk1}}$ $F_{\text{rcc_pclk2}}$	Comments
0xx	1	→	$F_{\text{rcc_hclk1}}$	$F_{\text{rcc_hclk1}}$	The timer clock is equal to the bus clock.
100	1	→	$F_{\text{rcc_hclk1}}$	$F_{\text{rcc_hclk1}} / 2$	The timer clock is twice as fast as the bus clock.
101	1	→	$F_{\text{rcc_hclk1}}$	$F_{\text{rcc_hclk1}} / 4$	The timer clock is 4 times faster than the bus clock.
110	1	→	$F_{\text{rcc_hclk1}} / 2$	$F_{\text{rcc_hclk1}} / 8$	
111	1	→	$F_{\text{rcc_hclk1}} / 4$	$F_{\text{rcc_hclk1}} / 16$	

1. D2PPRE1 and D2PPRE2 belong to [RCC Domain 2 Clock Configuration Register \(RCC_D2CFGR\)](#).

2. TIMPRE belongs to [RCC Clock Configuration Register \(RCC_CFGR\)](#).

8.5.7 Handling clock generators in Stop and Standby mode

When the whole system enters Stop mode, all the clocks (system and kernel clocks) are stopped as well as the following clock sources:

- CSI, HSI (depending on HSIKERON, and CSIKERON bits)
- HSE
- PLL1, PLL2 and PLL3
- HSI48

The content of the RCC registers is not altered except for PLL1ON, PLL2ON, PLL3ON, HSEON and HSI48ON which are set to '0'.

Exiting Stop mode

When the microcontroller exits system Stop mode via a wake-up event, the application can select which oscillator (HSI and/or CSI) will be used to restart. The STOPWUCK bit selects the oscillator used as system clock. The STOPKERWUCK bit selects the oscillator used as kernel clock for peripherals. The STOPKERWUCK bit is useful if after a system Stop a peripheral needs a kernel clock generated by an oscillator different from the one used for the system clock.

All these bits belong to the [RCC Clock Configuration Register \(RCC_CFGR\)](#). [Table 57](#) gives a detailed description of their behavior.

Table 57. STOPWUCK and STOPKERWUCK description

STOPWUCK	STOPKERWUCK		Activated oscillator when the system exits Stop mode	Distributed clocks when System exits Stop mode	
				System Clock	Kernel Clock
0	0	→	HSI	HSI	HSI
	1	→	HSI and CSI		HSI and/or CSI
1	0	→		CSI	CSI
	1	→			

During Stop mode

There are two specific cases where the HSI or CSI can be enabled during system Stop mode:

- When a dedicated peripheral requests the kernel clock:
In this case the peripheral will receive the HSI or CSI according to the kernel clock source selected for this peripheral (via PERxSRC).
- When the bits HSIKERON or CSIKERON are set:
In this case the HSI and CSI are kept running during Stop mode but the outputs are gated. In that way, the clock will be available immediately when the system exits Stop mode or when a peripheral requests the kernel clock (see [Table 58](#) for details).

HSIKERON and CSIKERON bits belong to [RCC Source Control Register \(RCC_CR\)](#). [Table 58](#) gives a detailed description of their behavior.

Table 58. HSIKERON and CSIKERON behavior

HSIKERON (CSIKERON)		HSI (CSI) state during Stop mode	HSI (CSI) Setting time
0	→	OFF	$t_{su(HSI)} (t_{su(CSI)})^{(1)}$
1	→	Running and Gated	Immediate

1. $t_{su(HSI)}$ and $t_{su(CSI)}$ are the startup times of the HSI and CSI oscillators (see refer to the product datasheet for the values of these parameters).

When the microcontroller exists system Standby mode, the HSI is selected as system and kernel clock, the RCC registers are reset to their initial values except for the RCC_RSR (or RCC_C1_RSR) and RCC_BDCR registers.

Note as well that the HSI and CSI outputs provide two clock paths (see [Figure 45](#)):

- one path for the system clock (**hsi_ck** or **csi_ck**)
- one path for the peripheral kernel clock (**hsi_ker_ck** or **csi_ker_ck**).

When a peripheral requests the kernel clock in system Stop mode, only the path providing the **hsi_ker_ck** or **csi_ker_ck** is activated.

Caution: It is not guaranteed that the CPU will get automatically the same clock frequencies when leaving CStop mode: this mainly depends on the System state. For example If the CPU goes to CStop, while the D3 domain is kept in CRun, when the CPU exits from CStop, the clock settings remain unchanged. If the D3 domain goes to CStop while the CPU is also in CStop, then when the CPU exits from CStop, the CPU will operate with HSI or CSI when it left the CStop mode.

8.5.8 Kernel clock selection

Some peripherals are designed to work with two different clock domains that operate asynchronously:

- a clock domain synchronous with the register and bus interface (**ckg_bus_perx** clock)
- and a clock domain generally synchronous with the peripheral (kernel clock).

The benefit of having peripherals supporting these two clock domains is that the user application has more freedom to choose optimized clock frequency for the CPU, bus matrix and for the kernel part of the peripheral.

As a consequence, the user application can change the bus frequency without reprogramming the peripherals. As an example an on-going transfer with UART will not be disturbed if its APB clock is changed on-the-fly.

[Table 59](#) shows the kernel clock that the RCC can deliver to the peripherals. Each row of [Table 59](#) represents a MUX and the peripherals connected to its output. The columns starting from number 5 represents the clock sources. Column 3 gives the maximum allowed frequency at each MUX output. It is up to the user to respect these requirements.



Table 59. Kernel clock distribution overview

Peripherals	Clock mux control bits	Maximum allowed frequency [MHz]				Domain	Clock Sources																			
		VOS0	VOS1	VOS2	VOS2		pll1_q_ck	pll2_p_ck	pll2_q_ck	pll2_r_ck	pll3_p_ck	pll3_q_ck	pll3_r_ck	sys_ck	bus clocks ⁽¹⁾	hse_ck	hsi_ker_ck	csi_ker_ck	hsi48_ck	lse_ck	lsi_ck	per_ck ⁽²⁾	I2S_CKIN	dsi_phy_ck	USB_PHY1	Disabled
LTDC	-	150	150	75	50	D1						x														
FMC	FMCSEL	300	250	200	133		1			2				0								3				
QUADSPI	QSPISEL	250	200	150	100		1			2				0								3				
SDMMC1	SDMMCSEL	250 ⁽⁴⁾	200 ⁽⁴⁾	150 ⁽⁴⁾	100 ⁽⁴⁾		0			1																
SDMMC2			250	200	150	100	D2	0	1		2										4	3				
DFSDM1 Aclk	SAI1SEL	250	200	150	100								1	0												
DFSDM1 clk	DFSDM1SEL	250	200	150	100																					
FDCAN	FDCANSEL	125	100	75	50	1			2						0											
HDMI-CEC	CECSEL	66	66	66	33												2 ⁽³⁾		0	1						
I2C1,2,3	I2C123SEL	125	100	75	50								1	0		2	3									
LPTIM1	LPTIM1SEL	125	100	75	50			1					2	0					3	4	5					
TIM[8:1][17:12]	-	250	200	150	100									x												
HRTIM	-	480	400	300	200									x												
RNG	RNGSEL	250	200	150	100	1												0	2	3						
SAI1	SAI1SEL	150	150	113	75	0		1			2										4	3				
SAI2	SAI23SEL	150	150	113	75	0		1			2										4	3				
SAI3			150	150	113	75																				
SPDIFRX	SPDIFSEL	250	200	150	100	0				1			2			3										
SPI(I2S)1,2,3	SPI123SEL	200	200	150	100	0		1			2										4	3				

Table 59. Kernel clock distribution overview (continued)

Peripherals	Clock mux control bits	Maximum allowed frequency [MHz]				Domain	Clock Sources																			
		VOS0	VOS1	VOS2	VOS2		pll1_q_ck	pll2_p_ck	pll2_q_ck	pll2_r_ck	pll3_p_ck	pll3_q_ck	pll3_r_ck	sys_ck	bus clocks ⁽¹⁾	hse_ck	hsi_ker_ck	csi_ker_ck	hsi48_ck	lse_ck	lsi_ck	per_ck ⁽²⁾	I2S_CKIN	dsi_phy_ck	USB_PHY1	Disabled
SPI4,5	SPI45SEL	125	100	75	50	D2			1			2			0	5	3	4								
SWPMI	SWPSEL	125	100	75	50										0		1									
USART1,6	USART16SEL	125	100	75	50				1			2			0		3	4		5						
USART2,3 UART4,5,7,8	USART234578SEL	125	100	75	50				1			2			0		3	4		5						
USB1OTG USB2OTG	USBSEL	66	66	66	63		1					2							3							0
USB1ULPI	-	60	60	60	60																			x		
ADC1,2, 3	ADCSEL	80 ⁽⁴⁾	80 ⁽⁴⁾	40	40 ⁽⁴⁾				0				1									2				
I2C4	I2C4SEL	125	100	75	50	D3						1		0		2	3									
LPTIM2	LPTIM2SEL	125	100	75	50			1					2		0					3	4	5				
LPTIM3,4,5	LPTIM345SEL	125	100	75	50			1					2		0					3	4	5				
LPUART1	LPUART1SEL	125	100	75	50				1			2			0		3	4		5						
SAI4_A	SAI4ASEL	150	150	75	75		0	1			2											4	3			
SAI4_B	SAI4BSEL	150	150	75	75		0	1			2											4	3			
SPI6	SPI6SEL	125	100	75	50				1			2			0	5	3	4								
RTC/AWU ⁽⁵⁾	RTCSEL	1				VSW									3 ⁽⁶⁾				1	2					0	

1. The bus clocks are the bus interface clocks to which the peripherals are connected, it can be APB, AHB or AXI clocks.

2. The per_ck clock could be hse_ck, hsi_ker_ck or csi_ker_ck according to CKPERSEL selection.

3. Clock CSI divided by 122.

4. With a duty cycle close to 50%, meaning that DIV[P/Q/R]x values shall be even. For SDMMCx, the duty cycle shall be 50% when supporting DDR.

5. The RTC is not functional in V_{BAT} mode when the clock source is lsi_ck or hse_ck.

6. Clock HSE divided by RTCPRE.

[Figure 50](#) to [Figure 59](#) provide a more detailed description of kernel clock distribution. To simplify the drawings, the bus interface clocks (pclk, hclk) are not represented, even if they are gated with enable signals. Refer to [Section 8.5.11: Peripheral clock gating control](#) for more details.

To reduce the amount of switches, some peripherals share the same kernel clock source. Nevertheless, all peripherals have their dedicated enable signal.

Peripherals dedicated to audio applications

The audio peripherals generally need specific accurate frequencies, except for SPDIFRX. As shown in [Figure 50](#), the kernel clock of the SAI or SPI(I2S)s can be generated by:

- The PLL1 when the amount of active PLLs has to be reduced
- The PLL2 or 3 for optimal flexibility in frequency generation
- HSE, HSI or CSI for use-cases where the current consumption is critical
- I2S_CKIN when an external clock reference need to be used.

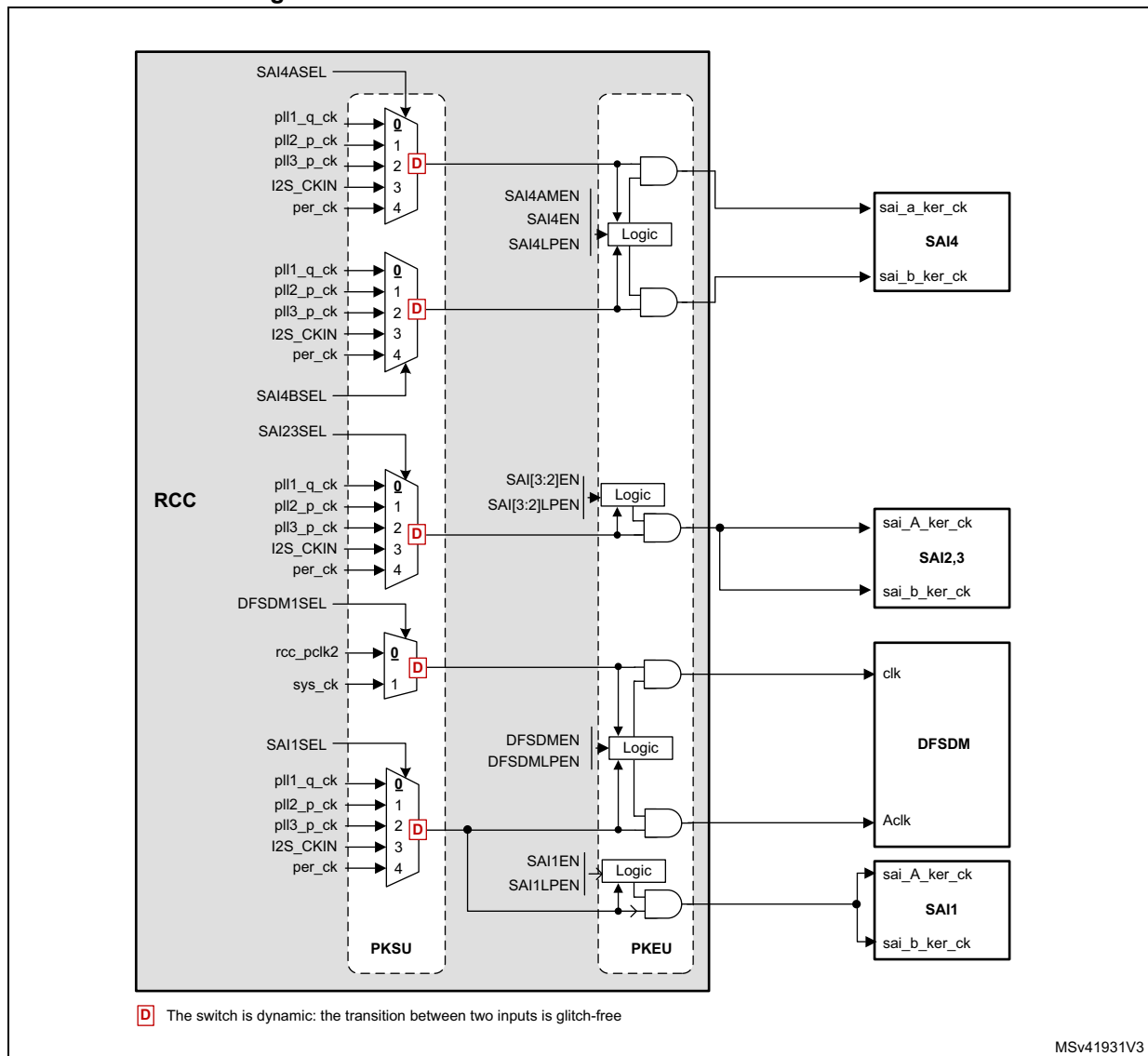
Note: The SPDIFRX does not require a specific frequency, but only a kernel clock frequency high enough to make the peripheral work properly. Refer to the SPDIFRX description for more details.

DFSDM1 can use the same clock as SAI1A. This is useful when DFSDM1 is used for audio applications.

To improve the flexibility, SAI4 can use different clock for each sub-block.

The SPI/I2S1, 2, and 3 share the same kernel clock source (see [Figure 51](#)).

Figure 50. Kernel clock distribution for SAI4 and DFSDM



1. **X** represents the selected MUX input after a system reset.
2. This figure does not show the connection of the bus interface clock to the peripherals. For details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

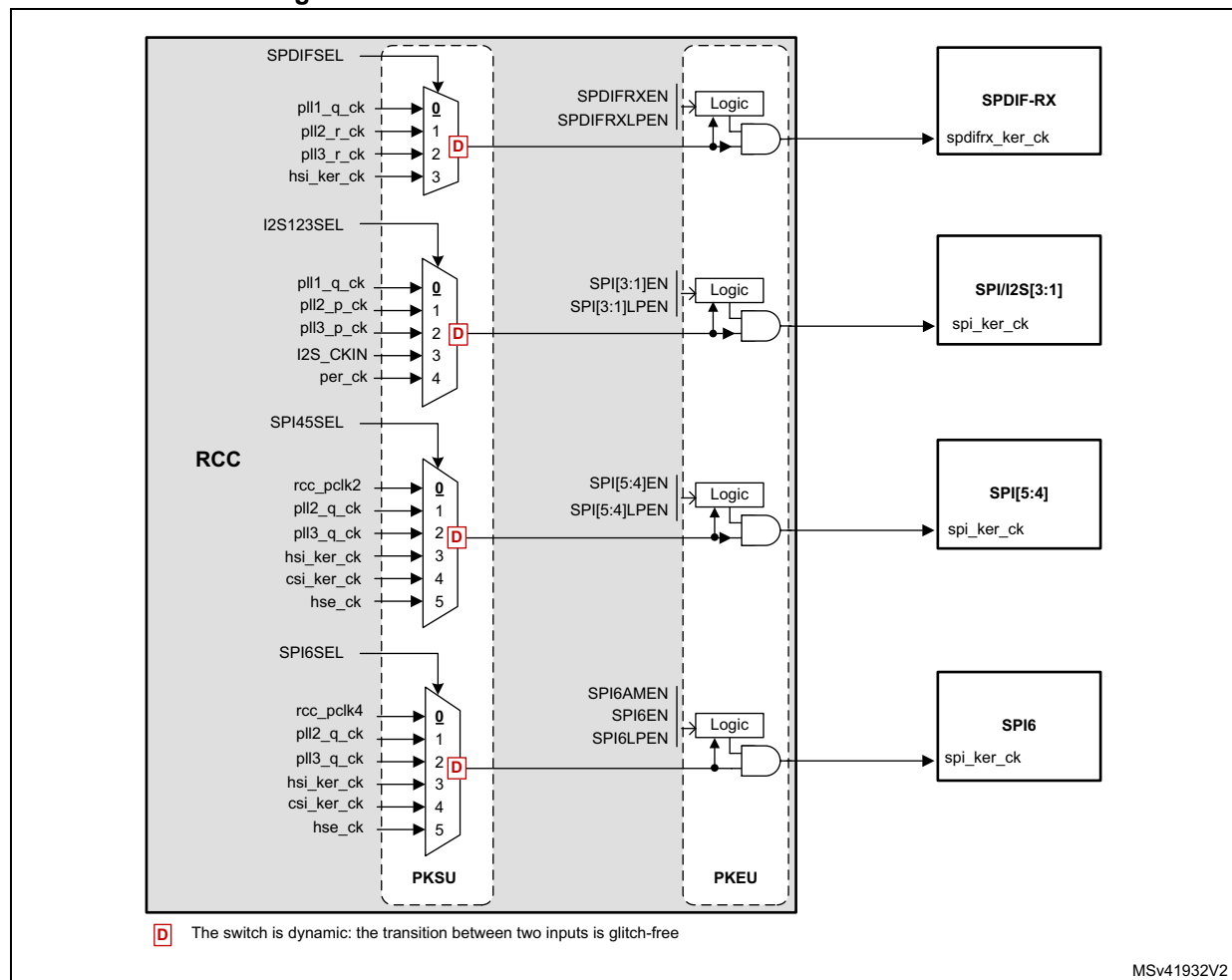
Peripherals dedicated to control and data transfer

Peripherals such as SPIs, I2Cs, UARTs do not need a specific kernel clock frequency but a clock fast enough to generate the correct baud rate, or the required bit clock on the serial interface. For that purpose the source can be selected among:

- PLL1 when the amount of active PLLs has to be reduced
- PLL2 or PLL3 if better flexibility is required. As an example, this solution allows changing the frequency bus via PLL1 without affecting the speed of some serial interfaces.
- HSI or CSI for low-power use-cases or when the peripheral has to quickly wake up from Stop mode (i.e. UART, I2C...).

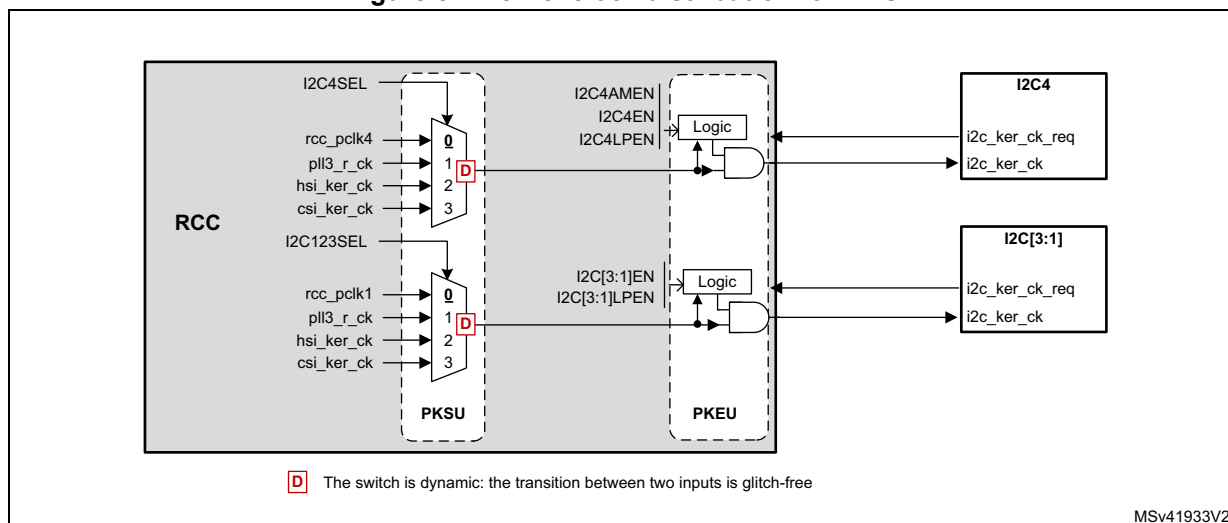
Note: UARTs also need the LSE clock when high baud rates are not required.

Figure 51. Kernel clock distribution for SPIs and SPI/I2S



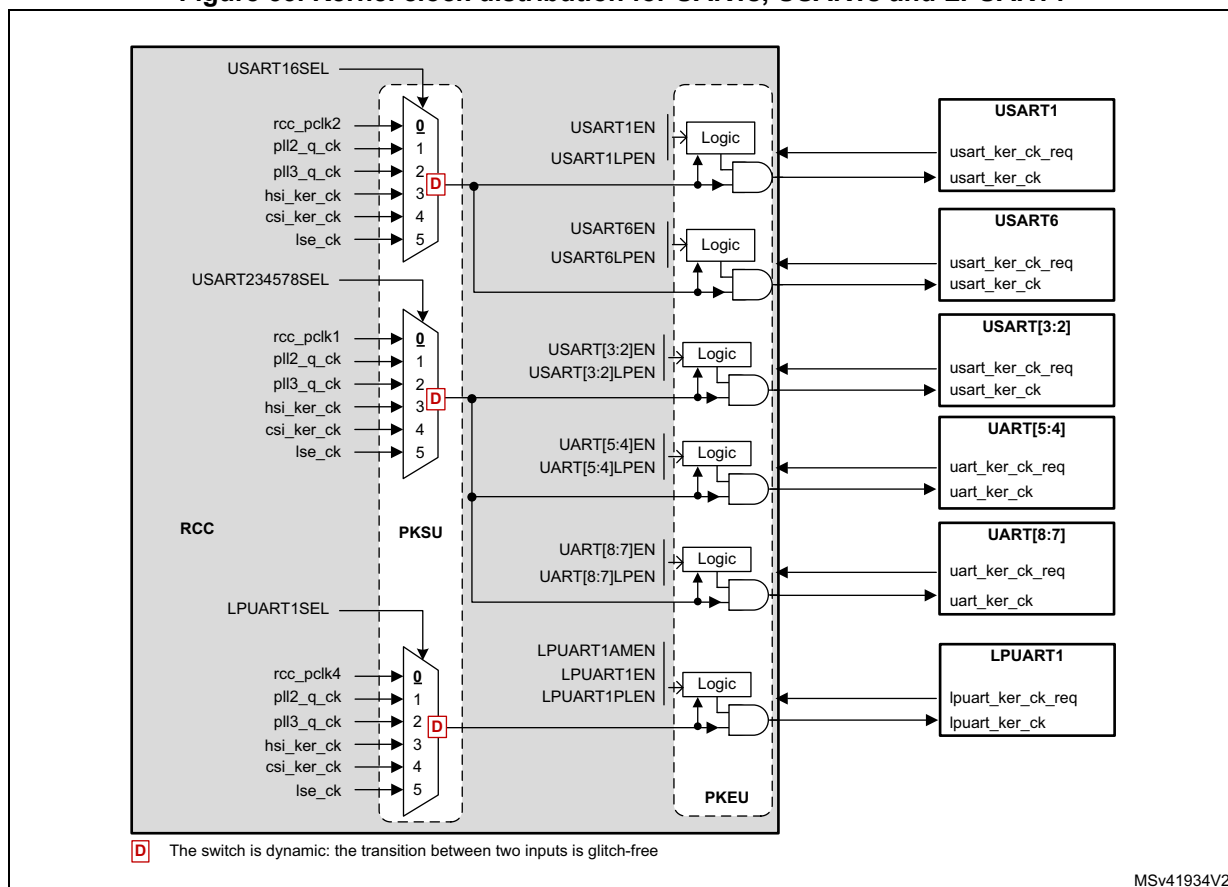
1. **X** represents the selected MUX input after a system reset.
2. This figure does not show the connection of the bus interface clock to the peripheral. For details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

Figure 52. Kernel clock distribution for I2Cs



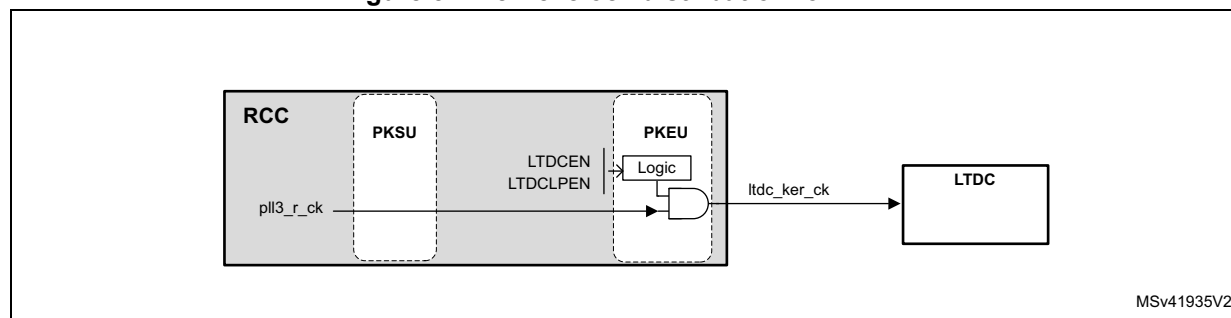
1. **X** represents the selected MUX input after a system reset
2. This figure does not show the connection of the bus interface clock to the peripheral, for details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

Figure 53. Kernel clock distribution for UARTs, USARTs and LPUART1



1. **X** represents the selected MUX input after a system reset.
2. This figure does not show the connection of the bus interface clock to the peripheral, for details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

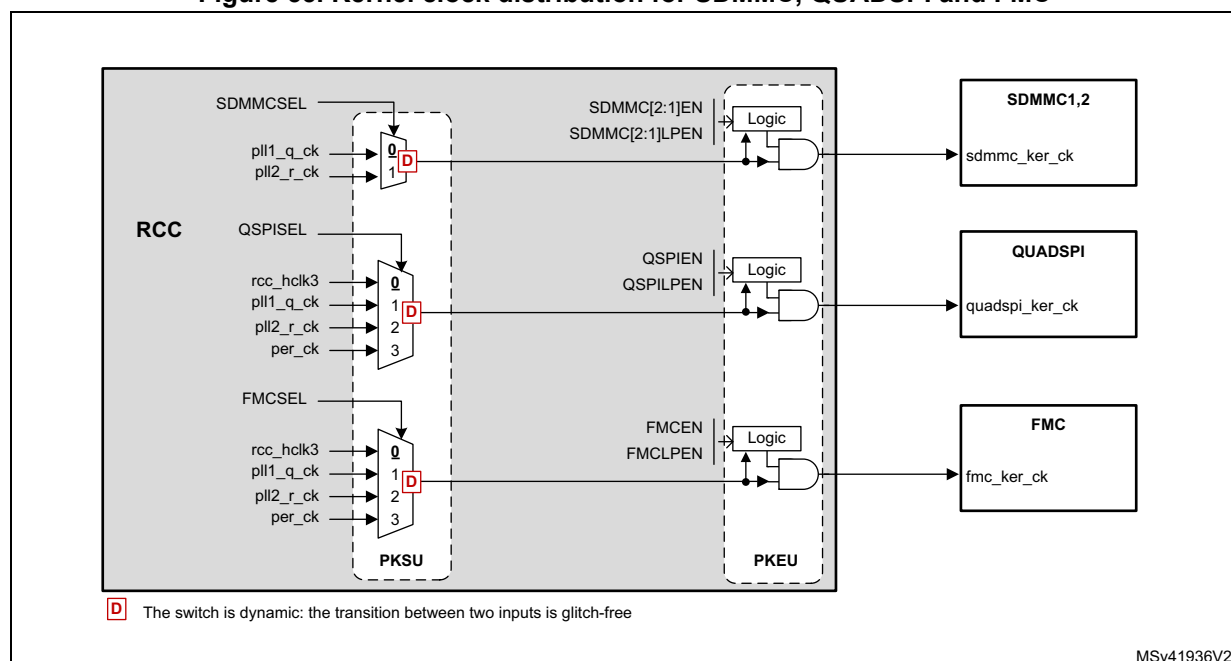
Figure 54. Kernel clock distribution for LTDC



1. **X** represents the selected MUX input after a system reset.
2. This figure does not show the connection of the bus interface clock to the peripheral. For details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

The FMC, QUADSPI and SDMMC1/2 can also use a clock different from the bus interface clock for more flexibility.

Figure 55. Kernel clock distribution for SDMMC, QUADSPI and FMC

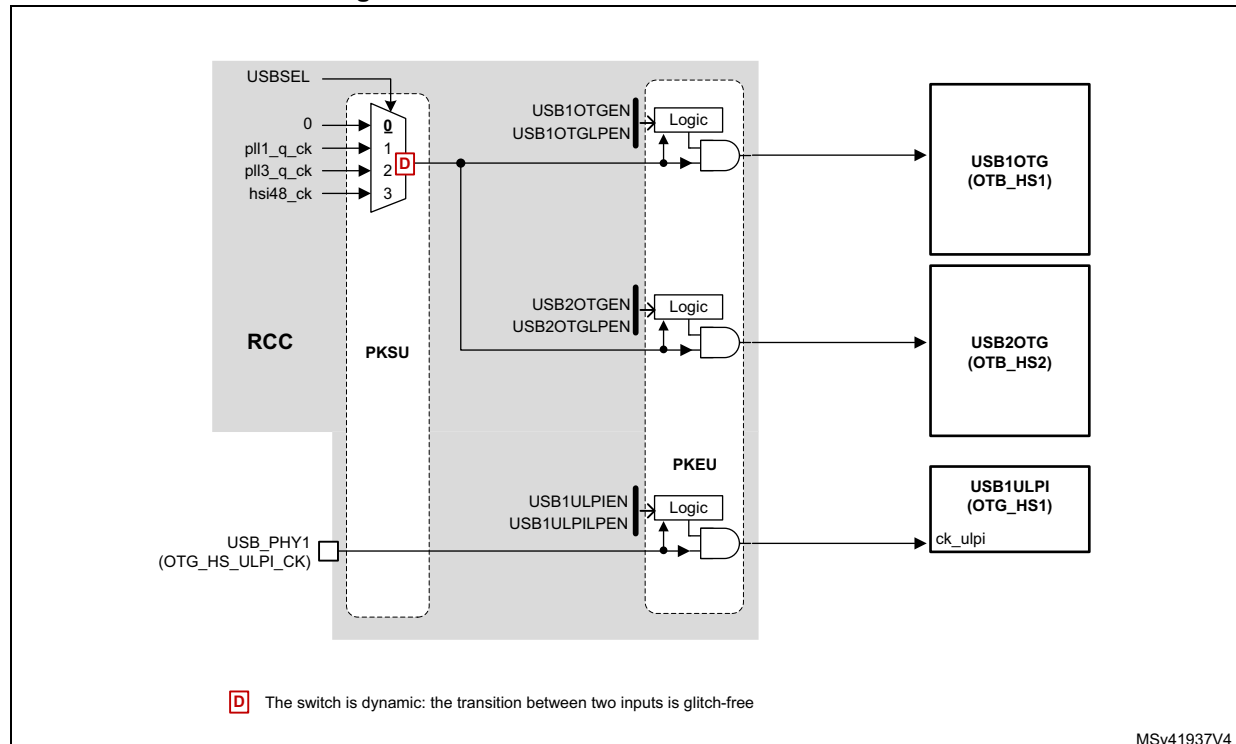


1. **X** represents the selected MUX input after a system reset.
2. This figure does not show the connection of the bus interface clock to the peripheral. For details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

Figure 56 shows the clock distribution for the USB blocks. The USB1ULPI block receives its clock from the external PHY.

The USBxOTG blocks receive the clock for USB communications which can be selected among different sources thanks to the MUX controlled by USBSEL.

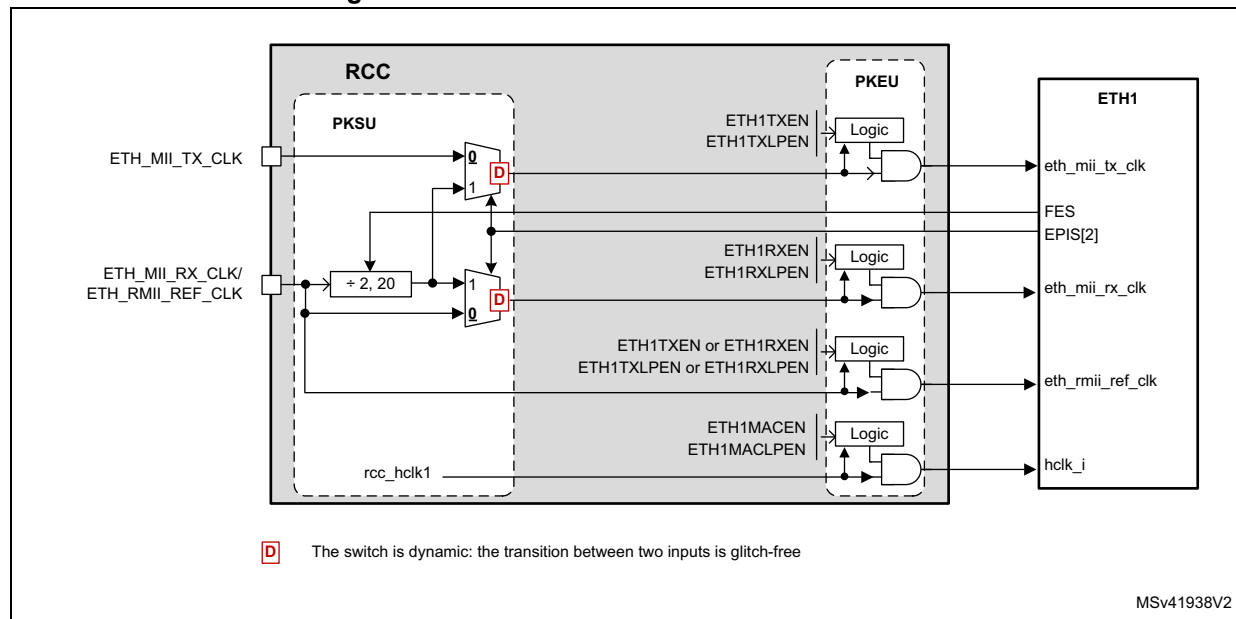
Figure 56. Kernel clock distribution for USB ⁽²⁾



1. **D** represents the selected MUX input after a system reset.
2. This figure does not show the connection of the bus interface clock to the peripheral. For details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

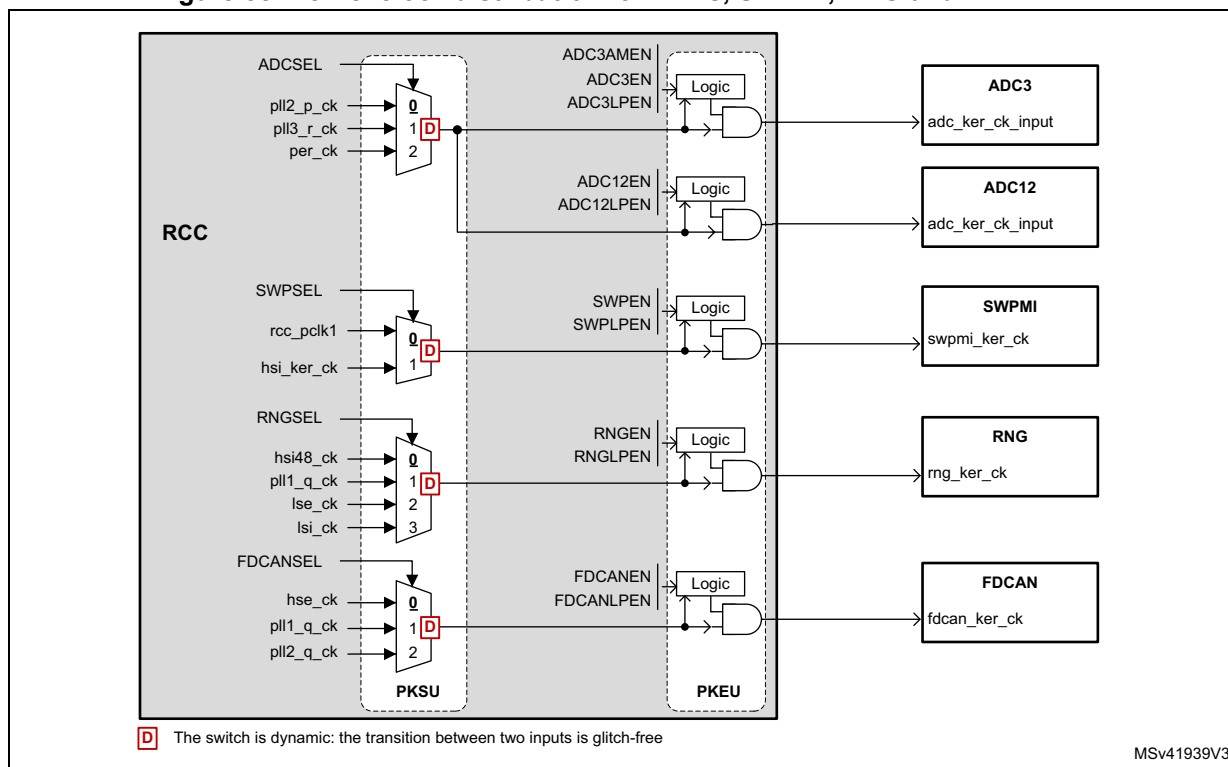
The Ethernet transmit and receive clocks shall be provided from an external Ethernet PHY. The clock selection for the RX and TX path is controlled via the SYSCFG block.

Figure 57. Kernel clock distribution for Ethernet



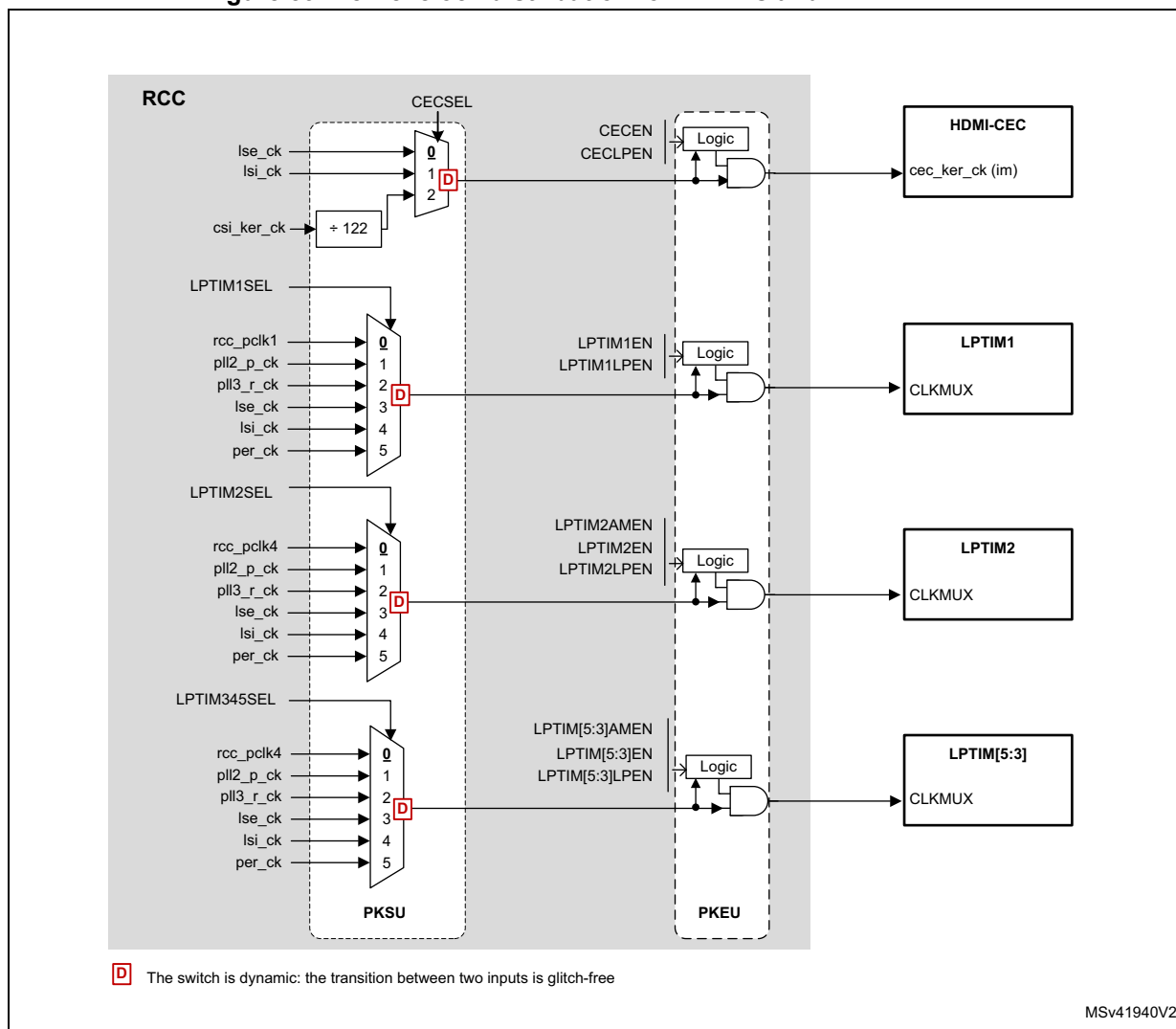
1. **X** represents the selected MUX input after a system reset.
2. This figure does not show the connection of the bus interface clock to the peripheral. For details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

Figure 58. Kernel clock distribution for ADCs, SWPMI, RNG and FDCAN (2)



1. **X** represents the selected MUX input after a system reset.
2. This figure does not show the connection of the bus interface clock to the peripheral. For details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

Figure 59. Kernel clock distribution for LPTIMs and HDMI-CEC (2)



1. **X** represents the selected MUX input after a system reset
2. This figure does not show the connection of the bus interface clock to the peripheral. For details on each enable cell, please refer to [Section 8.5.11: Peripheral clock gating control](#).

RTC/AWU clock

The **rtc_ck** clock source can be:

- the **hse_1M_ck** (**hse_ck** divided by a programmable prescaler)
- the **lse_ck**
- or the **lsi_ck** clock

The source clock is selected by programming the RTCSEL[1:0] bits in the [RCC Backup Domain Control Register \(RCC_BDCR\)](#) and the RTCPRE[5:0] bits in the [RCC Clock Configuration Register \(RCC_CFGR\)](#).

This selection cannot be modified without resetting the Backup domain.

If the LSE is selected as RTC clock, the RTC will work normally even if the backup or the V_{DD} supply disappears.

The LSE clock is in the Backup domain, whereas the other oscillators are not. As a consequence:

- If LSE is selected as RTC clock, the RTC continues working even if the V_{DD} supply is switched OFF, provided the V_{BAT} supply is maintained.
- If LSI is selected as the RTC clock, the AWU state is not guaranteed if the V_{DD} supply is powered off.
- If the HSE clock is used as RTC clock, the RTC state is not guaranteed if the V_{DD} supply is powered off or if the V_{CORE} supply is powered off.

The **rtc_ck** clock is enabled through RTCEN bit located in the [RCC Backup Domain Control Register \(RCC_BDCR\)](#).

The RTC bus interface clock (APB clock) is enabled through RTCAPBEN and RTCAPBLPEN bits located in RCC_APB4ENR/LPENR registers.

Note: *To read the RTC calendar register when the APB clock frequency is less than seven times the RTC clock frequency ($F_{APB} < 7 \times F_{RTCLK}$), the software must read the calendar time and date registers twice. The data are correct if the second read access to RTC_TR gives the same result than the first one. Otherwise a third read access must be performed.*

Watchdog clocks

The RCC provides the clock for the four watchdog blocks available on the circuit. The independent watchdog (IWDG1) is connected to the LSI. The window watchdog (WWDG1) are connected to the APB clock.

If an independent watchdog is started by either hardware option or software access, the LSI is forced ON and cannot be disabled. After the LSI oscillator setup delay, the clock is provided to the IWDGs.

Caution: Before enabling the WWDG1, the application must set the WW1RSC bit to '1'. If the WW1RSC remains to '0', when the WWDG1 is enabled, its the behavior is not guaranteed. The WW1RSC bit is located in [RCC Global Control Register \(RCC_GCR\)](#).

Clock frequency measurement using TIMx

Most of the clock source generator frequencies can be measured by means of the input capture of TIMx.

- Calibrating the HSI or CSI with the LSE

The primary purpose of having the LSE connected to a TIMx input capture is to be able to accurately measure the HSI or CSI. This requires to use the HSI or CSI as system clock source either directly or via PLL1. The number of system clock counts between consecutive edges of the LSE signal gives a measurement of the internal clock period. Taking advantage of the high precision of LSE crystals (typically a few tens of ppm) we can determine the internal clock frequency with the same resolution, and trim the source to compensate for manufacturing-process and/or temperature- and voltage-related frequency deviations.

The basic concept consists in providing a relative measurement (e.g. HSI/LSE ratio): the precision is therefore tightly linked to the ratio between the two clock sources. The greater the ratio is, the more accurate the measurement is.

The HSI and CSI oscillators have dedicated user-accessible calibration bits for this purpose (see [RCC Internal Clock Source Calibration Register \(RCC_ICSCR\)](#)) for revision Y devices and [RCC HSI configuration register \(RCC_HSI CFGR\)](#)/[RCC CSI configuration register \(RCC_CSI CFGR\)](#) for revision V devices). When the HSI or CSI

are used via the PLLx, the system *clock can also be fine-tuned* by using the fractional divider of the PLLs.

- Calibrating the LSI with the HSI

The LSI frequency can also be measured: this is useful for applications that do not have a crystal. The ultralow power LSI oscillator has a large manufacturing process deviation. By measuring it versus the HSI clock source, it is possible to determine its frequency with the precision of the HSI. The measured value can be used to have more accurate RTC time base timeouts (when LSI is used as the RTC clock source) and/or an IWDG timeout with an acceptable accuracy.

8.5.9 General clock concept overview

The RCC handles the distribution of the CPU, bus interface and peripheral clocks for the system (D1, D2 and D3 domains), according to the operating mode of each function (refer to [Section 8.5.1: Clock naming convention](#) for details on clock definitions).

For each peripheral, the application can control the activation/deactivation of its kernel and bus interface clock. Prior to use a peripheral, the CPU has to enable it (by setting PERxEN to '1'), and define if this peripheral remains active in CSleep mode (by setting PERxLPEN to '1'). This is called 'allocation' of a peripheral to the CPU (refer to [Section 8.5.10: Peripheral allocation](#) for more details).

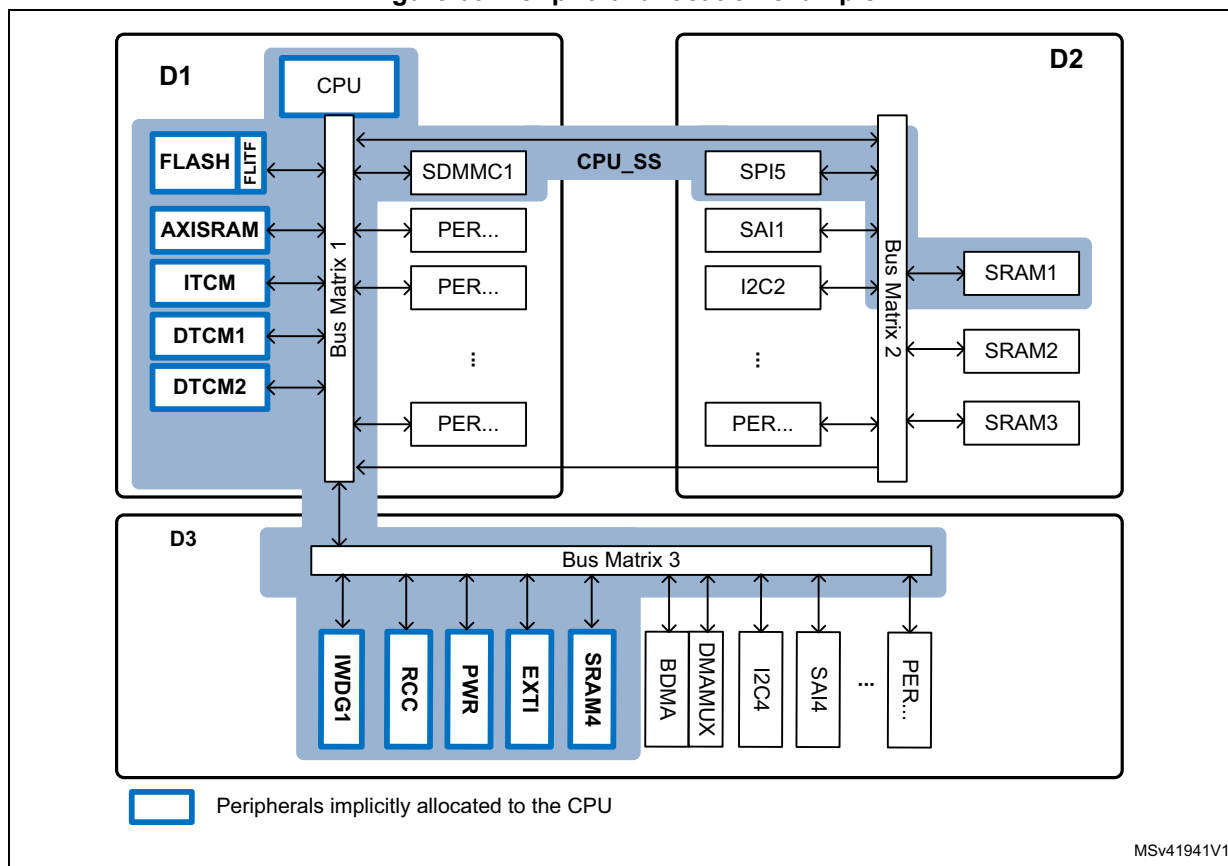
The peripheral allocation is used by the RCC to automatically control the clock gating according to the CPU and domain modes, and by the PWR to control the supply voltages of D1, D2 and D3 domains.

[Figure 60](#) gives an example of peripheral allocation:

- The CPU enabled SDMMC1, SPI5 and SRAM1, AXISRAM, ITCM, DTCM1, DTCM2 and SRAM4 are implicitly allocated to the CPU. The group composed of the CPU, bus matrix 1/2/3 and allocated peripherals makes up a sub-system (CPU_SS).

Note: The FLASH, AXISRAM, ITCM, DTCM1, DTCM2, SRAM4, IWGD1, IWGD2, PWR, EXTI and RCC are common resources and are implicitly allocated to the CPU.

Figure 60. Peripheral allocation example



When the CPU enters CStop mode, the RCC automatically disables the bus interface and kernel clocks of all the peripherals of the CPU_SS as well as the CPU clock. The PLLs, if enabled, are not disabled by the RCC since D3 is still running.

The D3 domain can be kept in DRun mode while the CPU is in CStop mode and D1 and D2 domains are in DStop or DStandby mode. This is done by setting RUN_D3 bit in PWR_CPUCR registers.

- If RUN_D3 is set to '1', then D3 is maintained in DRun mode, independently from the CPU modes (see [PWR CPU control register \(PWR_CPUCR\)](#)).
- If RUN_D3 is set to '0', then D3 domain enters DStop or DStandby mode when the CPU enters CStop mode (see [Table 60](#)).

Note that the CPU can control if D1, D2 or D3 domains are allowed to enter in DStandby when conditions are met, via bits PDDS_D1, PDDS_D2 and PDDS_D3 of [PWR CPU control register \(PWR_CPUCR\)](#).

A wakeup event will be able to exit D1, D2 and D3 domains from DStandby or DStop mode.

In addition, more autonomy can be given to some peripherals located into D3 domain (refer to [Section : D3 domain Autonomous mode](#) for details).

D3 domain Autonomous mode

The Autonomous mode allows to deliver the peripheral clocks to peripherals located in D3, even if the CPU is in CStop mode. When a peripheral has its autonomous bit enabled, it receives its peripheral clocks according to D3 domain state, if the CPU is in CStop mode:

- If the D3 domain is in DRun mode, peripherals with Autonomous mode activated receive their peripheral clocks,
- If the D3 domain is in DStop mode, no peripheral clock is provided.

The Autonomous mode does not prevent the D3 domain to enter DStop or DStandby mode.

The autonomous bits are located in [RCC D3 Autonomous mode Register \(RCC_D3AMR\)](#).

For example, the CPU can enter CStop mode, while the SAI4 is filling the SRAM4 with data received from an external device via BDMA. When the amount of received data is reached, the CPU can be activated by a wakeup event. This can be done by setting the SAI4, the BDMA, and SRAM4 in Autonomous mode, while keeping D3 in DRun mode (RUN_D3 set to '1'). In this example, the RCC does not switch off the PLLs as the D3 domain is always in DRun mode.

It is possible to go a step further with power consumption reduction by combining the Autonomous mode with the capability of some peripherals (UARTs, I2Cs) to request the kernel clock on their own, without waking-up the CPU. For example, if the system is expecting messages via I2C4, the whole system can be put in Stop mode. When the I2C4 peripheral detects a START bit, it will generate a “kernel clock request”. This request enables the HSI or CSI, and a kernel clock is provided only to the requester (in our example the I2C4). The I2C4 then decodes the incoming message. Several cases are then possible:

- If the device address of the message does not match, then I2C4 releases its “kernel clock request” until a new START condition is detected.
- If the device address of the incoming message matches, it has to be stored into D3 local memory. I2C4 is able to generate a wakeup event on address match to switch the D3 domain to DRun mode. The message is then transferred into memory via BDMA, and the D3 domain go back to DStop mode without any CPU activation. Note that if the amount of data transferred into memory reached the transfer count, the BDMA can also generate an interrupt to wake-up the CPU.
- If the device address of the incoming message matches and the peripheral is setup to wake up the CPU, then I2C4 generates a wakeup event to activate the CPU.

Please refer to the description of EXTI block in order to see which peripheral is able to perform a wake-up event to which domain.

Memory handling

The CPU can access all the memory areas available in the product:

- AXISRAM, ITCM, DTCM1, DTCM2 and FLASH,
- SRAM1, SRAM2 and SRAM3,
- SRAM4 and BKPRAM.

As shown in [Figure 60](#), FLASH, AXISRAM, SRAM4, ITCM, DTCM1 and DTCM2 are implicitly allocated to the CPU. As a result, there is no enable bit allowing the CPU to allocate these memories.

If the CPU wants to use memories located into D2 domain (SRAM1, SRAM2 and SRAM3), it has to enable them.

The BKPRAM has a dedicated enable in order to gate the bus interface clock. The CPU needs to enable the BKPRAM prior to use it.

Note: *The memory interface clocks (Flash and RAM interfaces) can be stopped by software during CSleep mode (via DxSRAMyLPEN bits).*

Refer to [Peripheral clock gating control](#) and [CPU and bus matrix clock gating control](#) sections for details on clock enabling.

System states overview

[Table 60](#) gives an overview of the system states with respect to the D1, D2 and D3 domain modes.

- The system remains in Run mode as long as D3 is in DRun mode. Several sub-states of system Run exist that are not detailed here (refer [Power control \(PWR\)](#) for more information).
- When the D1 domain is in DRun, the D2 domain can be in DRun, DStop or DStandby. When the D1 domain is in DStop or DStandby, the D2 domain can no longer remain in DRun it will switch to DStop or DStandby according to PDDS_D2 bit.
- D3 can run while D1 and D2 are in DStop/DStandby mode thanks to RUN_D3 bits of PWR_CPUUCR registers or when D3 is in Autonomous mode.
- The system remains in Stop mode as long as D3 is in DStop mode. This means implicitly that D1 and D2 are in DStop or DStandby. As soon as D1 or D2 exits DStop or DStandby, D3 switches to DRun mode.
- The system remains in Standby mode as long as D1, D2 and D3 are in DStandby.
- Domain states versus CPU states:
 - When the D1 domain is in DRun mode, it means that its bus matrix is clocked, and the CPU is in CRun mode.
 - When the D2 domain is in DRun mode, it means that its bus matrix is clocked, and the CPU is in CRun mode with at least a peripheral of D2 domain allocated.
 - When the D1 domain is in DStop mode it means that its bus matrix is no longer clocked, and the CPU is in CStop mode.
 - When the D2 domain is in DStop mode it means that its bus matrix is no longer clocked. This situation happens when:
 - the CPU did not allocate peripherals of D2 domain,
 - the CPU allocated peripherals of D2 domain, but the CPU is in CStop or CStandby,
 - When a domain is in DStandby mode, it means that the domain including its CPU are powered down.

Table 60. System states overview

System State	D1 State	D2 State	D3 State
Run	DRun	DRun/DStop/DStandby	DRun
	DStop/DStandby	DStop/DStandby	
Stop	DStop/DStandby	DStop/ DStandby	DStop
Standby	DStandby	DStandby	DStandby

8.5.10 Peripheral allocation

The CPU can allocate a peripheral and hence control its kernel and bus interface clock.

The CPU can allocate a peripheral by setting the dedicated PERxEN bit located into:

- RCC_xxxENR registers or
- RCC_C1_xxxENR registers.

The CPU can control the peripheral clocks gating when it is in CSleep mode via the PERxLPEN bits located into:

- RCC_xxxLPENR registers or
- RCC_C1_xxxLPENR registers.

Refer to [Section 8.7.1: Register mapping overview](#) for additional information.

The peripheral allocation bits (PERxEN bits) are used by the hardware to provide the kernel and bus interface clocks to the peripherals. However they are also used to link peripherals to the CPU (CPU sub-system). In this way, the hardware is able to safely gate the peripheral clocks and bus matrix clocks according to CPU states. The PWR block also uses this information to control properly the domain states.

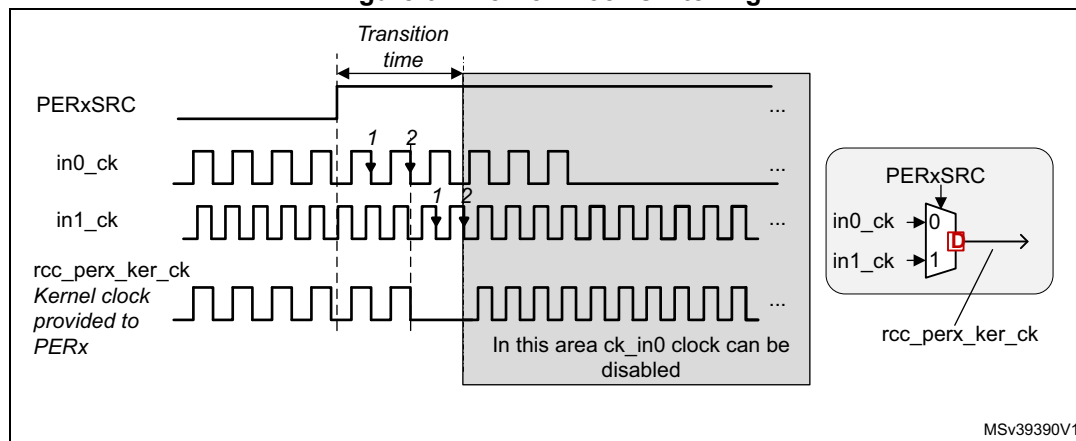
Clock switches and gating

- Clock switching delays

The input selected by the kernel clock switches can be changed dynamically without generating spurs or timing violation. As a consequence, switching from the original to the new input can only be performed if a clock is present on both inputs. If it not the case, no clock will be provided to the peripheral. To recover from this situation, the user has to provide a valid clock to both inputs.

During the transition from one input to another, the kernel clock provided to the peripheral will be gated, in the worst case, during 2 clock cycles of the previously selected clock and 2 clock cycles of the new selected clock. As shown in [Figure 61](#), both input clocks shall be present during transition time.

Figure 61. Kernel Clock switching



- Clock enabling delays

In the same way, the clock gating logic synchronizes the enable command (coming generally from a kernel clock request or PERxEN bits) with the selected clock, in order to avoid generation of spurs.

- A maximum delay of two periods of the enabled clock may occur between the enable command and the first rising edge of the clock. The enable command can be the rising edge of the PERxEN bits of RCC_xxxxENR registers, or a kernel clock request asserted by a peripheral.
- A maximum delay of 1.5 periods of the disabled clock may occur between the disable command and the last falling edge of the clock. The disable command can be the falling edge of the PERxEN bits of RCC_xxxxENR registers, or a kernel clock request released by a peripheral.

Note: *Both the kernel clock and the bus interface clock are affected by this re-synchronization delay.*

In addition, the clock enabling delay may strongly increase if the application is enabling for the first time a peripheral which is not located into the same domain. This is due to the fact that the domain where the peripheral is located could be in DStop or DStandby mode. The domain must be switched to DRun mode before the application can use this peripheral.

As an example, if the CPU enables a peripheral located in the D2 domain while the D2 domain is in DStop/DStandby mode, then the power controller (PWR) has first to provide a supply voltage to D2, then the RCC has to wait for an acknowledge from the PWR before enabling the clocks of the D2 domain. To handle properly this situation the RCC and the PWR blocks feature four flags:

- D1CKRDY/D2CKRDY located in [RCC Source Control Register \(RCC_CR\)](#)
- SBF_D1 and SBF_D2 located in [PWR CPU control register \(PWR_CPUCR\)](#).

The following sequence can be followed to avoid this issue:

- Enable the peripheral clocks (i.e. allocate the peripheral) by writing the corresponding PERxEN bit to '1' in the RCC_xxxxENR register,
- Read back the RCC_xxxxENR register to make sure that the previous write operation is not pending into a write buffer.
- If the peripheral is located in a different domain, perform the two next steps:
Read DxCKRDY until it is set to '1'.
Write SBF_Dx to zero and read-back the value, in order to check if the domain where the peripheral is located is still in DStandby. If the corresponding bit is read at '1', it means that the domain is still in DStandby. Repeat this operation until SBF_Dx is equal to '0', then continue the other steps.
- Perform a dummy read operation into a register of the enabled peripheral. This operation will take at least 2 clock cycles, which is equal to the max delay of the enable command.
- The peripheral can then be used.

Note: *When the bus interface clock is not active, read or write accesses to the peripheral registers are not supported. A read access will return invalid data. A write access will be ignored and will not create any bus errors.*

8.5.11 Peripheral clock gating control

As mentioned previously, each peripheral requires a bus interface clock, named **rcc_perx_bus_ck** (for peripheral 'x'). This clock can be an APB, AHB or AXI clock, according to which bus the peripheral is connected.

The clocks used as bus interface for peripherals located in D1 domain, could be **rcc_aclk**, **rcc_hclk3** or **rcc_pclk3**, depending on the bus connected to each peripheral. For simplicity sake, these clocks are named **rcc_bus_d1_ck**.

In the same way, the signal named **rcc_bus_d2_ck** represents **rcc_hclk1**, **rcc_hclk2**, **rcc_pclk1** or **rcc_pclk2**, depending on the bus connected to each peripheral of D2 domain.

Similarly, the signal **rcc_bus_d3_ck** represents **rcc_hclk4** or **rcc_pclk4** for peripherals located in D3.

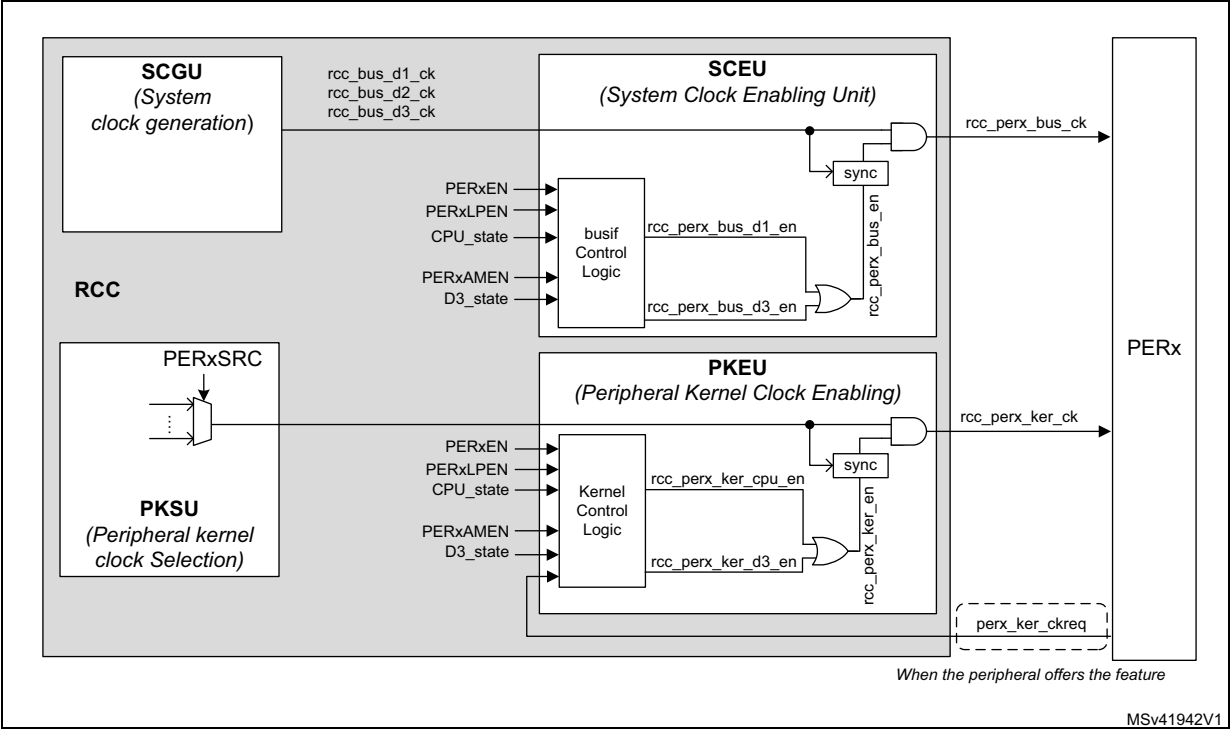
Some peripherals (SAI, UART...) also require a dedicated clock for their communication interface. This clock is generally asynchronous with respect to the bus interface clock. It is named kernel clock (**perx_ker_ckreq**). Both clocks can be gated according to several conditions detailed hereafter.

As shown in [Figure 62](#), enabling the kernel and bus interface clocks of each peripheral depends on several input signals:

- PERxEN and PERxLPEN bits
PERxEN represents the peripheral enable (allocation) bit for the CPU. The CPU can write these bits to '1' via RCC_C1_xxxxENR or RCC_xxxxENR registers.
- PERxAMEN bits
The PERxAMEN bits are belong to [RCC D3 Autonomous mode Register \(RCC_D3AMR\)](#).
- CPU state (**c_sleep** and **c_deepsleep** signals)
- D3 domain state (**d3_deepsleep** signal)
- The kernel clock request (**perx_ker_ckreq**) of the peripheral itself, when the feature is available.

Refer to [Section 8.5.10: Peripheral allocation](#) for more details.

Figure 62. Peripheral kernel clock enable logic details



[Table 61](#) gives a detailed description of the enabling logic of the peripheral clocks for peripherals located in D1 or D2 domain and allocated by the CPU.

Table 61. Peripheral clock enabling for D1 and D2 peripherals

PERxEN	PERxLPEN	PERxSRC	perx_ker_ckreq	CPU State	rcc_perx_ker_c_en	rcc_perx_bus_d1_en	Comments
0	X	X	X	X	0	0	No clock provided to the peripheral, because PERxEN='0'
1	X	X	X	CRun	1	1	Kernel and bus interface clocks are provided to the peripheral, because the CPU is in CRun, and PERxEN='1'
1	0	X	X	CSleep	0	0	No clock provided to the peripheral, because the CPU is in CSleep, and PERxLPEN='0'
1	1	X	X		1	1	Kernel and bus interface clocks are provided to the peripheral, because CPU is in CSleep, and PERxLPEN='1'
1	0	X	X	CStop	0	0	No clock provided to the peripheral because the PERxLPEN bit is set to '0'.
1	1	no lsi_ck and no lse_ck and no hsi_ker_ck and no csi_ker_ck	X		0	0	No clock provided to the peripheral because CPU is in CStop and lse_ck or lsi_ck or hsi_ker_ck or csi_ker_ck are not selected.
1	1	lsi_ck or lse_ck	X		1 (1)	0	Kernel clock is provided to the peripheral because PERxEN = PERxLPEN='1' and lsi_ck or lse_ck are selected. The bus interface clock is no provided as the CPU is in CStop
1	1	hsi_ker_ck or csi_ker_ck	1		1	0	Kernel clock is provided to the peripheral because req_ker_perx = '1', and PERxEN = PERxLPEN='1' and hsi_ker_ck or csi_ker_ck are selected. The bus interface clock is no provided as the CPU is in CStop
1	1	hsi_ker_ck or csi_ker_ck	0		0	0	No clock provided to the peripheral because CPU is in CStop, and no kernel clock request pending

1. For RNG block, the kernel clock is not delivered if the CPU to which it is allocated is in CStop mode, even if the clock selected is lsi_ck or lse_ck.

As a summary, we can state that the kernel clock is provided to the peripherals located on domains D1 and D2 when the following conditions are met:

1. The CPU is in CRun mode, and the peripheral is allocated.
2. The CPU is in CSleep mode, and the peripheral is allocated with PERxLPEN = '1'.
3. The CPU is in CStop mode, and the peripheral is allocated with PERxLPEN = '1', and the peripheral generates a kernel clock request, and the selected clock is **hsi_ker_ck** or **csi_ker_ck**.
4. The CPU is in CStop mode, and the peripheral is allocated with PERxLPEN = '1', and the kernel source clock of the peripheral is **lse_ck** or **lsi_ck**.

The bus interface clock will be provided to the peripherals only when conditions 1 or 2 are met.

[Table 62](#) gives a detailed description of the enabling logic of the kernel clock for all peripherals located in D3.

Table 62. Peripheral clock enabling for D3 peripherals

PERxEN	PERxLPEN	PERxAMEN	PERxSRC	perx_ker_ckreq	CPU State	D3 State	rcc_perx_ker_d3_en	rcc_perx_bus_d3_en	Comments
0	X	X	X	X	Any	Any	0	0	No clock provided to the peripheral, as PERxEN='0'
1	X	X	X	X	CRun	DRun	1	1	Kernel and bus interface clocks are provided to the peripheral, because the CPU is in CRun, and PERxEN='1'
1	0	X	X	X	CSleep		0	0	No clock provided to the peripheral, because the CPU is in CSleep, and PERxLPEN='0'
1	1	X	X	X			1	1	Kernel and bus interface clocks are provided to the peripheral, because the CPU is in CSleep, and PERxLPEN='1'
1	X	0	X	X	CStop		0	0	As the CPU is in CStop, and PERxEN='1', then the kernel clock gating depends on D3 state and PERxAMEN bits. No clock provided to the peripheral because PERxAMEN = '0'.
1	X	1	X	X			1	1	The kernel and bus interface clocks are provided because even if the CPU is in CStop mode, D3 is in DRun mode, with PERxEN and PERxAMEN bits set to '1'.
1	X	1	not lse_ck and not lsi_ck	0			DStop	0	0

Table 62. Peripheral clock enabling for D3 peripherals (continued)

PERxEN	PERxLPEN	PERxAMEN	PERxSRC	perx_ker_ckreq	CPU State	D3 State	rcc_perx_ker_d3_en	rcc_perx_bus_d3_en	Comments
1	X	1	not hsi_ker_ck and not csi_ker_ck and not lse_ck and not lsi_ck	1	CStop	DStop	0	0	No clock provided to the peripheral, because even if req_ker_perx = '0', lse_ck or lsi_ck or hsi_ker_ck or csi_ker_ck are not selected.
1	X	1	hsi_ker_ck or csi_ker_ck	1			1	0	Kernel clock is provided to the peripheral because req_ker_perx = '1', and PERxEN = PERxAMEN ='1', and the selected clock is hsi_ker_ck or csi_ker_ck . The bus interface clock is not provided as D3 is in DStop.
1	X	1	lse_ck or lsi_ck	X			1	0	Kernel clock is provided to the peripheral because PERxEN = PERxAMEN ='1' and lse_ck or lsi_ck are selected, while D3 is in STOP. The bus interface clock is not provided as D3 is in DSTOP.

As a summary, we can state that the kernel clock is provided to the peripherals of D3 if the following conditions are met:

1. The CPU is in CRun mode, and the peripheral is allocated.
2. The CPU is in CSleep mode, and the peripheral is allocated with **PERxLPEN** = '1'.
3. The CPU is in CStop mode, and the peripheral is allocated and D3 domain is in DRun mode with **PERxAMEN** = '1'.
4. The CPU is in CStop mode, and the peripheral is allocated, and D3 domain is in DStop mode with **PERxAMEN** = '1', and the peripheral is generating a kernel clock request and the kernel clock source is **hsi_ker_ck** or **csi_ker_ck**.
5. The CPU is in CStop mode, and the peripheral is allocated, and D3 domain is in DStop mode with **PERxAMEN** = '1', and the kernel clock source of the peripheral is **lse_ck** or **lsi_ck**.

The bus interface clock will be provided to the peripherals only when condition 1, 2 or 3 is met.

Note: When they are set to '1', the autonomous bits indicate that the associated peripheral will receive a kernel clock according to D3 state, and not according to the mode of the CPU. Only I2C, U(S)ART and LPUART peripherals are able to request the kernel clock. This feature gives to the peripheral the capability to transfer data with an optimal power consumption.

The autonomous bits dedicated to some peripherals located in D3 domain allow the data transfer with external devices without activating the CPU.

In order for the LPTIMER to operate with **lse_ck** or **lsi_ck** when the circuit is in Stop mode, the user application has to select the **lsi_ck** or **lse_ck** input via LPTIMxSEL fields, and set bit LPTIMxAMEN and LPTIMxLPEN to '1'.

8.5.12 CPU and bus matrix clock gating control

For each domain it is possible to control the activation/deactivation of the CPU clock and bus matrix clock.

For information about convention naming, refer to [Section 8.5.11: Peripheral clock gating control](#).

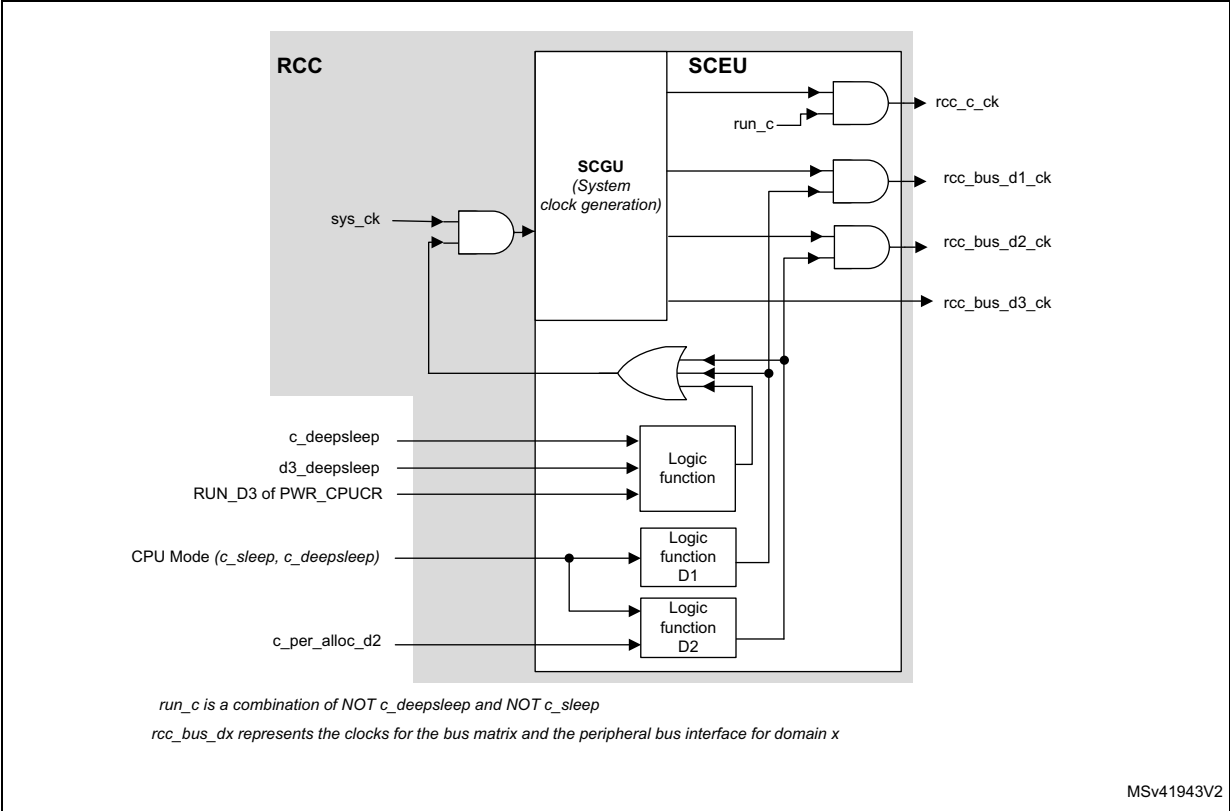
The clocks of the CPU, AHB and AXI bridges and APB busses are enabled according to the rules hereafter:

- The CPU clock **rcc_c_ck** is enabled when the CPU is in CRun mode.
- The AXI bridge clock is enabled when the CPU is in CRun mode.
- The D2 domain AHB bridges clocks are enabled when:
 - The CPU is in CRun or,
 - If the CPU is in CSleep with at least a peripheral (master) connected to this bus having both its PERxEN and PERxLPEN set to '1' or,
 - If the CPU is in CSleep with at least an APB bus having its clock enabled.
- The D3 domain AHB bridge clock is enabled when:
 - The CPU is in CRun or CSleep mode or,
 - When the RUN_D3 bit is set to '1', independently of CPU modes or,
 - When the **d3_deepsleep** signal is inactive ('0'), independently of CPU modes.
- The APB1,2,3 busses are enabled when:
 - The CPU is in CRun or,
 - If the CPU is in CSleep with at least a peripheral connected to this bus having both its PERxEN and PERxLPEN set to '1'.
- The APB4 bus is enabled when: the D3 domain is in DRun.

As shown in the [Figure 63](#), the enabling of the core and bus clock of each domain depends on several input signals:

- **c_sleep** and **c_deepsleep** signals from the CPU,
- **d3_sleepdeep** signal,
- RCC_xxxxENR.PERxEN bits of peripherals located on D2 domain

Figure 63. Bus clock enable logic



8.6 RCC Interrupts

The RCC provides 3 interrupt lines:

- **rcc_it**: a general interrupt line, providing events when the PLLs are ready, or when the oscillators are ready.
- **rcc_hsecss_it**: an interrupt line dedicated to the failure detection of the HSE Clock Security System.
- **rcc_lsecss_it**: an interrupt line dedicated to the failure detection of the LSE Clock Security System.

The interrupt enable is controlled via [RCC Clock Source Interrupt Enable Register \(RCC_CIER\)](#), except for the HSE CSS failure. When the HSE CSS feature is enabled, it not possible to mask the interrupt generation.

The interrupt flags can be checked via [RCC Clock Source Interrupt Flag Register \(RCC_CIFR\)](#), and those flags can be cleared via [RCC Clock Source Interrupt Clear Register \(RCC_CICR\)](#).

Note: The interrupt flags are not relevant if the corresponding interrupt enable bit is not set.

[Table 63](#) gives a summary of the interrupt sources, and the way to control them.

Table 63. Interrupt sources and control

Interrupt Source	Description	Interrupt enable	Action to clear interrupt	Interrupt Line
LSIRDYF	LSI ready	LSIRDYIE	Set LSIRDYC to '1'	rcc_it
LSERDYF	LSE ready	LSERDYIE	Set LSERDYC to '1'	
HSIDRYF	HSI ready	HSIDRYIE	Set HSIRDYC to '1'	
HSERDYF	HSE ready	HSERDYIE	Set HSERDYC to '1'	
CSIRDYF	CSI ready	CSIRDYIE	Set CSIRDYC to '1'	
HSI48RDYF	HSI48 ready	HSI48RDYIE	Set HSI48RDYC to '1'	
PLL1RDYF	PLL1 ready	PLL1RDYIE	Set PLL1RDYC to '1'	
PLL2RDYF	PLL2 ready	PLL2RDYIE	Set PLL2RDYC to '1'	
PLL3RDYF	PLL3 ready	PLL3RDYIE	Set PLL3RDYC to '1'	
LSECSSF	LSE Clock security system failure	LSECSSFIE ⁽¹⁾	Set LSECSSC to '1'	rcc_lsecss_it
HSECSSF	HSE Clock security system failure	_(²)	Set HSECSSC to '1'	rcc_hsecss_it

1. The security system feature must also be enabled (LSECSSON = '1'), in order to generate interrupts.

2. It is not possible to mask this interrupt when the security system feature is enabled (HSECSSON = '1').

8.7 RCC register description

8.7.1 Register mapping overview

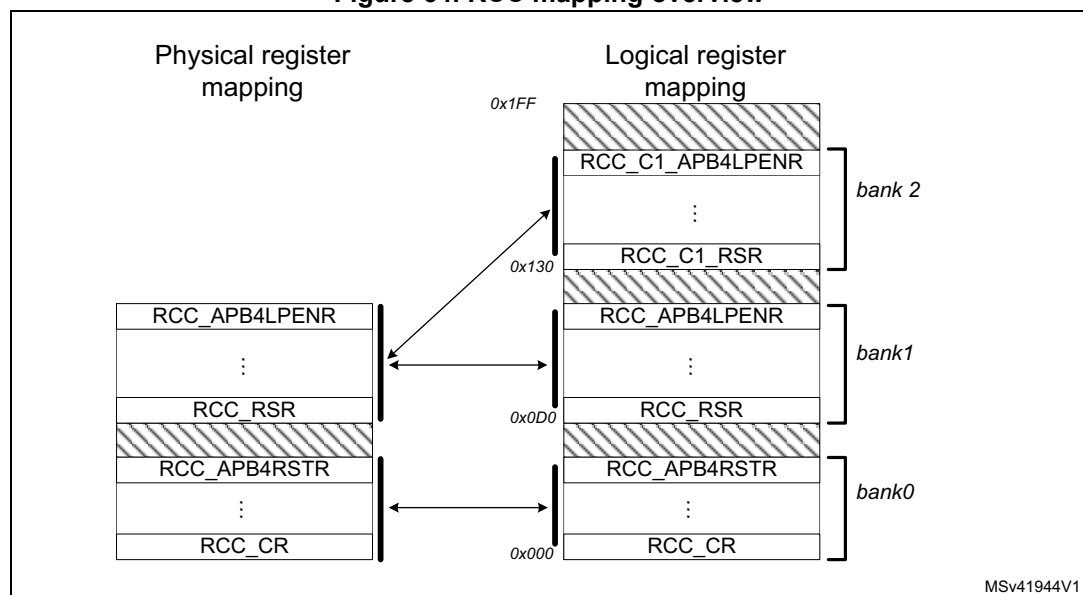
Note that the control of PERxEN and PERxLPEN bits can be performed at two different address offset: 0x0D0 and 0x130. So the application can use the registers named:

- RCC_xxxENR or RCC_C1_xxxENR to control the PERxEN bits
- RCC_xxxLPENR or RCC_C1_xxxLPENR to control the PERxLPEN bits
- RCC_RSR or RCC_C1_RSR to control the reset flag status bits.

This feature is provided to insure the compatibility with other products of this family.

Figure 64 shows the RCC mapping overview.

Figure 64. RCC mapping overview



8.7.2 RCC Source Control Register (RCC_CR)

Address offset: 0x000

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	PLL3RDY	PLL3ON	PLL2RDY	PLL2ON	PLL1RDY	PLL1ON	Res.	Res.	Res.	Res.	HSECSSON	HSEBYP	HSERDY	HSEON
		r	rw	r	rw	r	rw					rs	rw	r	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
D2CKRDY	D1CKRDY	HSI48RDY	HSI48ON	Res.	Res.	CSIKERON	CSIRDY	CSION	Res.	HSIDIVF	HSIDIV[1:0]	HSIRDY	HSIKERON	HSION	
r	r	r	rw			rw	r	rw		r	rw	rw	r	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **PLL3RDY**: PLL3 clock ready flag

Set by hardware to indicate that the PLL3 is locked.

0: PLL3 unlocked (default after reset)

1: PLL3 locked

Bit 28 **PLL3ON**: PLL3 enable

Set and cleared by software to enable PLL3.

Cleared by hardware when entering Stop or Standby mode.

0: PLL3 OFF (default after reset)

1: PLL3 ON

Bit 27 **PLL2RDY**: PLL2 clock ready flag

Set by hardware to indicate that the PLL2 is locked.

0: PLL2 unlocked (default after reset)

1: PLL2 locked

Bit 26 **PLL2ON**: PLL2 enable

Set and cleared by software to enable PLL2.

Cleared by hardware when entering Stop or Standby mode.

0: PLL2 OFF (default after reset)

1: PLL2 ON

Bit 25 **PLL1RDY**: PLL1 clock ready flag

Set by hardware to indicate that the PLL1 is locked.

0: PLL1 unlocked (default after reset)

1: PLL1 locked

Bit 24 **PLL1ON**: PLL1 enable

Set and cleared by software to enable PLL1.

Cleared by hardware when entering Stop or Standby mode. Note that the hardware prevents writing this bit to '0', if the PLL1 output is used as the system clock.

0: PLL1 OFF (default after reset)

1: PLL1 ON

Bits 23:20 Reserved, must be kept at reset value.

Bit 19 HSECSSON: HSE Clock Security System enable

Set by software to enable Clock Security System on HSE.

This bit is “set only” (disabled by a system reset or when the system enters in Standby mode).

When HSECSSON is set, the clock detector is enabled by hardware when the HSE is ready and disabled by hardware if an oscillator failure is detected.

0: Clock Security System on HSE OFF (Clock detector OFF) (default after reset)

1: Clock Security System on HSE ON (Clock detector ON if the HSE oscillator is stable, OFF if not).

Bit 18 HSEBYP: HSE clock bypass

Set and cleared by software to bypass the oscillator with an external clock. The external clock must be enabled with the HSEON bit, to be used by the device.

The HSEBYP bit can be written only if the HSE oscillator is disabled.

0: HSE oscillator not bypassed (default after reset)

1: HSE oscillator bypassed with an external clock

Bit 17 HSERDY: HSE clock ready flag

Set by hardware to indicate that the HSE oscillator is stable.

0: HSE clock is not ready (default after reset)

1: HSE clock is ready

Bit 16 HSEON: HSE clock enable

Set and cleared by software.

Cleared by hardware to stop the HSE when entering Stop or Standby mode.

This bit cannot be cleared if the HSE is used directly (via SW mux) as system clock or if the HSE is selected as reference clock for PLL1 with PLL1 enabled (PLL1ON bit set to ‘1’).

0: HSE is OFF (default after reset)

1: HSE is ON

Bit 15 D2CKRDY: D2 domain clocks ready flag

Set by hardware to indicate that the D2 domain clocks are available.

0: D2 domain clocks are not available (default after reset)

1: D2 domain clocks are available

Bit 14 D1CKRDY: D1 domain clocks ready flag

Set by hardware to indicate that the D1 domain clocks (CPU, bus and peripheral) are available.

0: D1 domain clocks are not available (default after reset)

1: D1 domain clocks are available

Bit 13 HSI48RDY: HSI48 clock ready flag

Set by hardware to indicate that the HSI48 oscillator is stable.

0: HSI48 clock is not ready (default after reset)

1: HSI48 clock is ready

Bit 12 HSI48ON: HSI48 clock enable

Set by software and cleared by software or by the hardware when the system enters to Stop or Standby mode.

0: HSI48 is OFF (default after reset)

1: HSI48 is ON

Bits 11:10 Reserved, must be kept at reset value.

Bit 9 **CSIKERON**: CSI clock enable in Stop mode

Set and reset by software to force the CSI to ON, even in Stop mode, in order to be quickly available as kernel clock for some peripherals. This bit has no effect on the value of CSION.

0: no effect on CSI (default after reset)

1: CSI is forced to ON even in Stop mode

Bit 8 **CSIRDY**: CSI clock ready flag

Set by hardware to indicate that the CSI oscillator is stable. This bit is activated only if the RC is enabled by CSION (it is not activated if the CSI is enabled by CSIKERON or by a peripheral request).

0: CSI clock is not ready (default after reset)

1: CSI clock is ready

Bit 7 **CSION**: CSI clock enable

Set and reset by software to enable/disable CSI clock for system and/or peripheral.

Set by hardware to force the CSI to ON when the system leaves Stop mode, if STOPWUCK = '1' or STOPKERWUCK = '1'.

This bit cannot be cleared if the CSI is used directly (via SW mux) as system clock or if the CSI is selected as reference clock for PLL1 with PLL1 enabled (PLL1ON bit set to '1').

0: CSI is OFF (default after reset)

1: CSI is ON

Bit 6 Reserved, must be kept at reset value.

Bit 5 **HSIDIVF**: HSI divider flag

Set and reset by hardware.

As a write operation to HSIDIV has not an immediate effect on the frequency, this flag indicates the current status of the HSI divider. HSIDIVF will go immediately to '0' when HSIDIV value is changed, and will be set back to '1' when the output frequency matches the value programmed into HSIDIV.

0: new division ratio not yet propagated to **hsi(_ker)_ck** (default after reset)

1: **hsi(_ker)_ck** clock frequency reflects the new HSIDIV value

Bits 4:3 **HSIDIV[1:0]**: HSI clock divider

Set and reset by software.

These bits allow selecting a division ratio in order to configure the wanted HSI clock frequency. The HSIDIV cannot be changed if the HSI is selected as reference clock for at least one enabled PLL (PLLxON bit set to '1'). In that case, the new HSIDIV value is ignored.

00: Division by 1, **hsi(_ker)_ck** = 64 MHz (default after reset)

01: Division by 2, **hsi(_ker)_ck** = 32 MHz

10: Division by 4, **hsi(_ker)_ck** = 16 MHz

11: Division by 8, **hsi(_ker)_ck** = 8 MHz

Bit 2 HSIRDY: HSI clock ready flag

Set by hardware to indicate that the HSI oscillator is stable.

0: HSI clock is not ready (default after reset)

1: HSI clock is ready

Bit 1 HSIKERON: High Speed Internal clock enable in Stop mode

Set and reset by software to force the HSI to ON, even in Stop mode, in order to be quickly available as kernel clock for peripherals. This bit has no effect on the value of HSION.

0: no effect on HSI (default after reset)

1: HSI is forced to ON even in Stop mode

Bit 0 HSION: High Speed Internal clock enable

Set and cleared by software.

Set by hardware to force the HSI to ON when the product leaves Stop mode, if STOPWUCK = '0' or STOPKERWUCK = '0'.

Set by hardware to force the HSI to ON when the product leaves Standby mode or in case of a failure of the HSE which is used as the system clock source.

This bit cannot be cleared if the HSI is used directly (via SW mux) as system clock or if the HSI is selected as reference clock for PLL1 with PLL1 enabled (PLL1ON bit set to '1').

0: HSI is OFF

1: HSI is ON (default after reset)

8.7.3 RCC Internal Clock Source Calibration Register (RCC_ICSCR)

Address offset: 0x004

Reset value: 0x4000 0xxx

This register is available only on revision Y devices.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	CSITRIM[4:0]					CSICAL[7:0]									
	rw	rw	rw	rw	rw	r	r	r	r	r	r	r	r	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSITRIM[3:0]					HSICAL[11:0]										
rw	rw	rw	rw		r	r	r	r	r	r	r	r	r	r	r

Bit 31 Reserved, must be kept at reset value.

Bits 30:26 **CSITRIM[4:0]**: CSI clock trimming

Set by software to adjust calibration.

CSITRIM field is added to the engineering option bytes loaded during reset phase (flash_csi_opt) in order to form the calibration trimming value.

CSICAL = CSITRIM + flash_csi_opt.

Note: The reset value of the field is 0x10.

Bits 25:18 **CSICAL[7:0]**: CSI clock calibration

Set by hardware by option byte loading during system reset **nreset**.

Adjusted by software through trimming bits CSITRIM.

This field represents the sum of engineering option byte calibration value and CSITRIM bits value.

Bits 17:12 **HSITRIM[5:0]**: HSI clock trimming

Set by software to adjust calibration.

HSITRIM field is added to the engineering option bytes loaded during reset phase (flash_hsi_opt) in order to form the calibration trimming value.

HSICAL = HSITRIM + flash_hsi_opt.

Note: The reset value of the field is 0x20.

Bits 11:0 **HSICAL[11:0]**: HSI clock calibration

Set by hardware by option byte loading during system reset **nreset**.

Adjusted by software through trimming bits HSITRIM.

This field represents the sum of engineering option byte calibration value and HSITRIM bits value.

8.7.4 RCC HSI configuration register (RCC_HSI CFGR)

Address offset: 0x004

Reset value: 0x4000 0xxx

This register is available only on revision V devices.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	HSITRIM[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	rw	rw	rw	rw	rw	rw	rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	HSICAL[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bit 31 Reserved, must be kept at reset value.

Bits 30:24 **HSITRIM[6:0]**: HSI clock trimming

Set by software to adjust calibration.

HSITRIM field is added to the engineering Option Bytes loaded during reset phase (FLASH_HSI_opt) in order to form the calibration trimming value.

HSICAL = HSITRIM + FLASH_HSI_opt.

Note: The reset value of the field is 0x40.

Bits 23:12 Reserved, must be kept at reset value.

Bits 11:0 **HSICAL[11:0]**: HSI clock calibration

Set by hardware by option byte loading during system reset **nreset**.

Adjusted by software through trimming bits HSITRIM.

This field represents the sum of engineering Option Byte calibration value and HSITRIM bits value.

8.7.5 **RCC Clock Recovery RC Register (RCC_CRRCR)**

Address offset: 0x008

Reset value: 0x0000 0xxx

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	HSI48CAL[9:0]									
						r									

- Bits 31:10 Reserved, must be kept at reset value.
- Bits 9:0 **HSI48CAL[9:0]**: Internal RC 48 MHz clock calibration
- Set by hardware by option byte loading during system reset **nreset**.
- Read-only.

8.7.6 RCC CSI configuration register (RCC_CSICFGR)

Address offset: 0x00C

Reset value: 0x2000 00xx

This register is available only on revision V devices.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	CSITRIM[5:0]						Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
		rw	rw	rw	rw	rw	rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	CSICAL[9:0]									
						r	r	r	r	r	r	r	r	r	r

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:24 **CSITRIM[5:0]**: CSI clock trimming

Set by software to adjust calibration.

CSITRIM bitfield is added to the engineering option bytes loaded during the reset phase (FLASH_CSI_opt) in order to build the calibration trimming value.

CSICAL = CSITRIM + FLASH_CSI_opt.

Note: The reset value of this bitfield is 0x10.

Bits 23:8 Reserved, must be kept at reset value.

Bits 9:0 **CSICAL[9:0]**: CSI clock calibration

Set by hardware by option byte loading during system reset **nreset**.

Adjusted by software through trimming bits CSITRIM.

This bitfield represents the sum of the engineering option byte calibration value and CSITRIM value.

8.7.7 RCC Clock Configuration Register (RCC_CFGR)

Address offset: 0x010

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MCO2[2:0]			MCO2PRE[3:0]			MCO1[2:0]			MCO11PRE[3:0]			Res.		Res.	
rw			rw			rw			rw						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMPRE	HRTIMSEL	RTCPRE[5:0]					STOPKERWUCK	STOPWUCK	SWS[2:0]			SW[2:0]			
rw	rw	rw					rw	rw	r			rw			

Bits 31:29 **MCO2[2:0]**: Micro-controller clock output 2

Set and cleared by software. Clock source selection may generate glitches on MCO2.

It is highly recommended to configure these bits only after reset, before enabling the external oscillators and the PLLs.

000: System clock selected (**sys_ck**) (default after reset)

001: PLL2 oscillator clock selected (**pll2_p_ck**)

010: HSE clock selected (**hse_ck**)

011: PLL1 clock selected (**pll1_p_ck**)

100: CSI clock selected (**csi_ck**)

101: LSI clock selected (**lsi_ck**)

others: reserved

Bits 28:25 **MCO2PRE[3:0]**: MCO2 prescaler

Set and cleared by software to configure the prescaler of the MCO2. Modification of this prescaler may generate glitches on MCO2. It is highly recommended to change this prescaler only after reset, before enabling the external oscillators and the PLLs.

0000: prescaler disabled (default after reset)

0001: division by 1 (bypass)

0010: division by 2

0011: division by 3

0100: division by 4

...

1111: division by 15

Bits 24:22 **MCO1[2:0]**: Micro-controller clock output 1

Set and cleared by software. Clock source selection may generate glitches on MCO1.

It is highly recommended to configure these bits only after reset, before enabling the external oscillators and the PLLs.

000: HSI clock selected (**hsi_ck**) (default after reset)

001: LSE oscillator clock selected (**lse_ck**)

010: HSE clock selected (**hse_ck**)

011: PLL1 clock selected (**pll1_q_ck**)

100: HSI48 clock selected (**hsi48_ck**)

others: reserved

Bits 21:18 **MCO1PRE[3:0]**: MCO1 prescaler

Set and cleared by software to configure the prescaler of the MCO1. Modification of this prescaler may generate glitches on MCO1. It is highly recommended to change this prescaler only after reset, before enabling the external oscillators and the PLLs.

0000: prescaler disabled (default after reset)

0001: division by 1 (bypass)

0010: division by 2

0011: division by 3

0100: division by 4

...

1111: division by 15

Bits 17:16 Reserved, must be kept at reset value.

Bit 15 **TIMPRE**: Timers clocks prescaler selection

This bit is set and reset by software to control the clock frequency of all the timers connected to APB1 and APB2 domains.

0: The Timers kernel clock is equal to **rcc_hclk1** if D2PPREx is corresponding to division by 1 or 2, else it is equal to $2 \times F_{\text{rcc_pclkx_d2}}$ (default after reset)

1: The Timers kernel clock is equal to **rcc_hclk1** if D2PPREx is corresponding to division by 1, 2 or 4, else it is equal to $4 \times F_{\text{rcc_pclkx_d2}}$

Please refer to [Table 56: Ratio between clock timer and pclk](#)

Bit 14 **HRTIMSEL**: High Resolution Timer clock prescaler selection

This bit is set and reset by software to control the clock frequency of high resolution the timer (HRTIM).

0: The HRTIM prescaler clock source is the same as other timers. (default after reset)

1: The HRTIM prescaler clock source is the CPU clock (**rcc_c_ck**).

Bits 13:8 **RTCPRE[5:0]**: HSE division factor for RTC clock

Set and cleared by software to divide the HSE to generate a clock for RTC.

Caution: The software has to set these bits correctly to ensure that the clock supplied to the RTC is lower than 1 MHz. These bits must be configured if needed before selecting the RTC clock source.

000000: no clock (default after reset)

000001: no clock

000010: HSE/2

000011: HSE/3

000100: HSE/4

...

111110: HSE/62

111111: HSE/63

Bit 7 **STOPKERWUCK**: Kernel clock selection after a wake up from system Stop

Set and reset by software to select the Kernel wakeup clock from system Stop.

0: The HSI is selected as wake up clock from system Stop (default after reset)

1: The CSI is selected as wake up clock from system Stop

See [Section 8.5.7: Handling clock generators in Stop and Standby mode](#) for details.

Bit 6 **STOPWUCK**: System clock selection after a wake up from system Stop

Set and reset by software to select the system wakeup clock from system Stop.

The selected clock is also used as emergency clock for the Clock Security System on HSE.

0: The HSI is selected as wake up clock from system Stop (default after reset)

1: The CSI is selected as wake up clock from system Stop

See [Section 8.5.7: Handling clock generators in Stop and Standby mode](#) for details.

Caution: STOPWUCK must not be modified when the Clock Security System is enabled (by HSECSSON bit) and the system clock is HSE (SWS="10") or a switch on HSE is requested (SW="10").

Bits 5:3 **SWS[2:0]**: System clock switch status

Set and reset by hardware to indicate which clock source is used as system clock.

000: HSI used as system clock (**hsi_ck**) (default after reset)

001: CSI used as system clock (**csi_ck**)

010: HSE used as system clock (**hse_ck**)

011: PLL1 used as system clock (**pll1_p_ck**)

others: Reserved

Bits 2:0 **SW[2:0]**: System clock switch

Set and reset by software to select system clock source (**sys_ck**).

Set by hardware in order to:

- force the selection of the HSI or CSI (depending on STOPWUCK selection) when leaving a system Stop mode
- force the selection of the HSI in case of failure of the HSE when used directly or indirectly as system clock.

000: HSI selected as system clock (**hsi_ck**) (default after reset)

001: CSI selected as system clock (**csi_ck**)

010: HSE selected as system clock (**hse_ck**)

011: PLL1 selected as system clock (**pll1_p_ck**)

others: Reserved

8.7.8 RCC Domain 1 Clock Configuration Register (RCC_D1CFGR)

Address offset: 0x018

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	D1CPRE[3:0]				Res.	D1PPRE[2:0]			HPRE[3:0]			
				rw					rw			rw			

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:8 **D1CPRE[3:0]**: D1 domain Core prescaler

Set and reset by software to control D1 domain CPU clock division factor.

Changing this division ratio has an impact on the frequency of the CPU clock, and all bus matrix clocks.

The clocks are divided by the new prescaler factor. This factor ranges from 1 to 16 periods of the slowest APB clock among **rcc_pclk[4:1]** after D1CPRE update. The application can check if the new division factor is taken into account by reading back this register.

0xxx: **sys_ck** not divided (default after reset)

1000: **sys_ck** divided by 2

1001: **sys_ck** divided by 4

1010: **sys_ck** divided by 8

1011: **sys_ck** divided by 16

1100: **sys_ck** divided by 64

1101: **sys_ck** divided by 128

1110: **sys_ck** divided by 256

1111: **sys_ck** divided by 512

Bit 7 Reserved, must be kept at reset value.

Bits 6:4 **D1PPRE[2:0]**: D1 domain APB3 prescaler

Set and reset by software to control the division factor of **rcc_pclk3**.

The clock is divided by the new prescaler factor from 1 to 16 cycles of **rcc_hclk3** after D1PPRE write.

0xx: **rcc_pclk3** = **rcc_hclk3** (default after reset)

100: **rcc_pclk3** = **rcc_hclk3** / 2

101: **rcc_pclk3** = **rcc_hclk3** / 4

110: **rcc_pclk3** = **rcc_hclk3** / 8

111: **rcc_pclk3** = **rcc_hclk3** / 16

Bits 3:0 **HPRE[3:0]**: D1 domain AHB prescaler

Set and reset by software to control the division factor of **rcc_hclk3** and **rcc_ahbclk**. Changing this division ratio has an impact on the frequency of all bus matrix clocks.

0xxx: **rcc_hclk3** = **sys_d1cpresck** (default after reset)

1000: **rcc_hclk3** = **sys_d1cpresck** / 2

1001: **rcc_hclk3** = **sys_d1cpresck** / 4

1010: **rcc_hclk3** = **sys_d1cpresck** / 8

1011: **rcc_hclk3** = **sys_d1cpresck** / 16

1100: **rcc_hclk3** = **sys_d1cpresck** / 64

1101: **rcc_hclk3** = **sys_d1cpresck** / 128

1110: **rcc_hclk3** = **sys_d1cpresck** / 256

1111: **rcc_hclk3** = **sys_d1cpresck** / 512

*Note: The clocks are divided by the new prescaler factor from 1 to 16 periods of the slowest APB clock among **rcc_pclk[4:1]** after HPRE update.*

*Note: Note also that **rcc_hclk3** = **rcc_ahbclk**.*

Caution: Care must be taken when using the voltage scaling. Due to the propagation delay of the new division factor, after a prescaler factor change and before lowering the V_{CORE} voltage, this register must be read in order to check that the new prescaler value has been taken into account.

Depending on the clock source frequency and the voltage range, the software application has to program a correct value in HPRE to make sure that the system frequency does not exceed the maximum frequency.

8.7.9 RCC Domain 2 Clock Configuration Register (RCC_D2CFGR)

Address offset: 0x01C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	D2PPRE2[2:0]			Res.	D2PPRE1[2:0]			Res.	Res.	Res.	Res.
					rw				rw						

Bits 31:11 Reserved, must be kept at reset value.

Bits 10:8 **D2PPRE2[2:0]**: D2 domain APB2 prescaler

Set and reset by software to control D2 domain APB2 clock division factor.

The clock is divided by the new prescaler factor from 1 to 16 cycles of **rcc_hclk1** after D2PPRE2 write.

0xx: **rcc_pclk2** = **rcc_hclk1** (default after reset)

100: **rcc_pclk2** = **rcc_hclk1** / 2

101: **rcc_pclk2** = **rcc_hclk1** / 4

110: **rcc_pclk2** = **rcc_hclk1** / 8

111: **rcc_pclk2** = **rcc_hclk1** / 16

Bit 7 Reserved, must be kept at reset value.

Bits 6:4 **D2PPRE1[2:0]**: D2 domain APB1 prescaler

Set and reset by software to control D2 domain APB1 clock division factor.

The clock is divided by the new prescaler factor from 1 to 16 cycles of **rcc_hclk1** after D2PPRE1 write.

0xx: **rcc_pclk1** = **rcc_hclk1** (default after reset)

100: **rcc_pclk1** = **rcc_hclk1** / 2

101: **rcc_pclk1** = **rcc_hclk1** / 4

110: **rcc_pclk1** = **rcc_hclk1** / 8

111: **rcc_pclk1** = **rcc_hclk1** / 16

Bits 3:0 Reserved, must be kept at reset value.

8.7.10 RCC Domain 3 Clock Configuration Register (RCC_D3CFGR)

Address offset: 0x020

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	D3PPRE[2:0]			Res.			
									rw						

Bits 31:7 Reserved, must be kept at reset value.

Bits 6:4 **D3PPRE[2:0]**: D3 domain APB4 prescaler

Set and reset by software to control D3 domain APB4 clock division factor.

The clock is divided by the new prescaler factor from 1 to 16 cycles of **rcc_hclk4** after D3PPRE write.

0xx: **rcc_pclk4** = **rcc_hclk4** (default after reset)

100: **rcc_pclk4** = **rcc_hclk4** / 2

101: **rcc_pclk4** = **rcc_hclk4** / 4

110: **rcc_pclk4** = **rcc_hclk4** / 8

111: **rcc_pclk4** = **rcc_hclk4** / 16

Bits 3:0 Reserved, must be kept at reset value.

8.7.11 RCC PLLs Clock Source Selection Register (RCC_PLLCKSELR)

Address offset: 0x028

Reset value: 0x0202 0200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	DIVM3[5:0]						Res.	Res.	DIVM2[5:4]	
						rw								rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIVM2[3:0]				Res.	Res.	DIVM1[5:0]						Res.	Res.	PLLSRC[1:0]	
rw						rw								rw	

Bits 31:26 Reserved, must be kept at reset value.

Bits 25:20 **DIVM3[5:0]**: Prescaler for PLL3

Set and cleared by software to configure the prescaler of the PLL3.

The hardware does not allow any modification of this prescaler when PLL3 is enabled (PLL3ON = '1').

In order to save power when PLL3 is not used, the value of DIVM3 must be set to '0'.

000000: prescaler disabled (default after reset)

000001: division by 1 (bypass)

000010: division by 2

000011: division by 3

...

100000: division by 32 (default after reset)

...

111111: division by 63

Bits 19:18 Reserved, must be kept at reset value.

Bits 17:12 **DIVM2[5:0]**: Prescaler for PLL2

Set and cleared by software to configure the prescaler of the PLL2.

The hardware does not allow any modification of this prescaler when PLL2 is enabled (PLL2ON = '1').

In order to save power when PLL2 is not used, the value of DIVM2 must be set to '0'.

000000: prescaler disabled

000001: division by 1 (bypass)

000010: division by 2

000011: division by 3

...

100000: division by 32 (default after reset)

...

111111: division by 63

Bits 11:10 Reserved, must be kept at reset value.

Bits 9:4 DIVM1[5:0]: Prescaler for PLL1

Set and cleared by software to configure the prescaler of the PLL1.

The hardware does not allow any modification of this prescaler when PLL1 is enabled (PLL1ON = '1').

In order to save power when PLL1 is not used, the value of DIVM1 must be set to '0'.

000000: prescaler disabled

000001: division by 1 (bypass)

000010: division by 2

000011: division by 3

...

100000: division by 32 (default after reset)

...

111111: division by 63

Bits 3:2 Reserved, must be kept at reset value.

Bits 1:0 PLLSRC[1:0]: DIVMx and PLLs clock source selection

Set and reset by software to select the PLL clock source.

These bits can be written only when all PLLs are disabled.

In order to save power, when no PLL is used, the value of PLLSRC must be set to '11'.

00: HSI selected as PLL clock (**hsi_ck**) (default after reset)

01: CSI selected as PLL clock (**csi_ck**)

10: HSE selected as PLL clock (**hse_ck**)

11: No clock send to DIVMx divider and PLLs

8.7.12 RCC PLLs Configuration Register (RCC_PLLCFGR)

Address offset: 0x02C

Reset value: 0x01FF 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	DIVR3EN	DIVQ3EN	DIVP3EN	DIVR2EN	DIVQ2EN	DIVP2EN	DIVR1EN	DIVQ1EN	DIVP1EN
							rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	PLL3RGE[[1:0]]		PLL3VCOSEL	PLL3FRACEN	PLL2RGE[[1:0]]		PLL2VCOSEL	PLL2FRACEN	PLL1RGE[[1:0]]		PLL1VCOSEL	PLL1FRACEN
				rw		rw	rw	rw		rw	rw	rw		rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bit 24 **DIVR3EN**: PLL3 DIVR divider output enableSet and reset by software to enable the **pll3_r_ck** output of the PLL3.To save power, DIVR3EN and DIVR3 bits must be set to '0' when the **pll3_r_ck** is not used.

This bit can be written only when the PLL3 is disabled (PLL3ON = '0' and PLL3RDY = '0').

0: **pll3_r_ck** output is disabled1: **pll3_r_ck** output is enabled (default after reset)Bit 23 **DIVQ3EN**: PLL3 DIVQ divider output enableSet and reset by software to enable the **pll3_q_ck** output of the PLL3.To save power, DIVR3EN and DIVR3 bits must be set to '0' when the **pll3_r_ck** is not used.

This bit can be written only when the PLL3 is disabled (PLL3ON = '0' and PLL3RDY = '0').

0: **pll3_q_ck** output is disabled1: **pll3_q_ck** output is enabled (default after reset)Bit 22 **DIVP3EN**: PLL3 DIVP divider output enableSet and reset by software to enable the **pll3_p_ck** output of the PLL3.

This bit can be written only when the PLL3 is disabled (PLL3ON = '0' and PLL3RDY = '0').

To save power, DIVR3EN and DIVR3 bits must be set to '0' when the **pll3_r_ck** is not used.0: **pll3_p_ck** output is disabled1: **pll3_p_ck** output is enabled (default after reset)Bit 21 **DIVR2EN**: PLL2 DIVR divider output enableSet and reset by software to enable the **pll2_r_ck** output of the PLL2.To save power, DIVR3EN and DIVR3 bits must be set to '0' when the **pll3_r_ck** is not used.

This bit can be written only when the PLL2 is disabled (PLL2ON = '0' and PLL2RDY = '0').

0: **pll2_r_ck** output is disabled1: **pll2_r_ck** output is enabled (default after reset)Bit 20 **DIVQ2EN**: PLL2 DIVQ divider output enableSet and reset by software to enable the **pll2_q_ck** output of the PLL2.To save power, DIVR3EN and DIVR3 bits must be set to '0' when the **pll3_r_ck** is not used.

This bit can be written only when the PLL2 is disabled (PLL2ON = '0' and PLL2RDY = '0').

0: **pll2_q_ck** output is disabled1: **pll2_q_ck** output is enabled (default after reset)

Bit 19 **DIVP2EN**: PLL2 DIVP divider output enable

Set and reset by software to enable the **pll2_p_ck** output of the PLL2.

This bit can be written only when the PLL2 is disabled (PLL2ON = '0' and PLL2RDY = '0').

To save power, DIVR3EN and DIVR3 bits must be set to '0' when the **pll3_r_ck** is not used.

0: **pll2_p_ck** output is disabled

1: **pll2_p_ck** output is enabled (default after reset)

Bit 18 **DIVR1EN**: PLL1 DIVR divider output enable

Set and reset by software to enable the **pll1_r_ck** output of the PLL1.

To save power, DIVR3EN and DIVR3 bits must be set to '0' when the **pll3_r_ck** is not used.

This bit can be written only when the PLL1 is disabled (PLL1ON = '0' and PLL1RDY = '0').

0: **pll1_r_ck** output is disabled

1: **pll1_r_ck** output is enabled (default after reset)

Bit 17 **DIVQ1EN**: PLL1 DIVQ divider output enable

Set and reset by software to enable the **pll1_q_ck** output of the PLL1.

In order to save power, when the **pll1_q_ck** output of the PLL1 is not used, the **pll1_q_ck** must be disabled.

This bit can be written only when the PLL1 is disabled (PLL1ON = '0' and PLL1RDY = '0').

0: **pll1_q_ck** output is disabled

1: **pll1_q_ck** output is enabled (default after reset)

Bit 16 **DIVP1EN**: PLL1 DIVP divider output enable

Set and reset by software to enable the **pll1_p_ck** output of the PLL1.

This bit can be written only when the PLL1 is disabled (PLL1ON = '0' and PLL1RDY = '0').

In order to save power, when the **pll1_p_ck** output of the PLL1 is not used, the **pll1_p_ck** must be disabled.

0: **pll1_p_ck** output is disabled

1: **pll1_p_ck** output is enabled (default after reset)

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:10 **PLL3RGE[1:0]**: PLL3 input frequency range

Set and reset by software to select the proper reference frequency range used for PLL3.

These bits must be written before enabling the PLL3.

00: The PLL3 input (**ref3_ck**) clock range frequency is between 1 and 2 MHz (default after reset)

01: The PLL3 input (**ref3_ck**) clock range frequency is between 2 and 4 MHz

10: The PLL3 input (**ref3_ck**) clock range frequency is between 4 and 8 MHz

11: The PLL3 input (**ref3_ck**) clock range frequency is between 8 and 16 MHz

Bit 9 **PLL3VCOSEL**: PLL3 VCO selection

Set and reset by software to select the proper VCO frequency range used for PLL3.

This bit must be written before enabling the PLL3.

0: Wide VCO range: 192 to 836 MHz (default after reset)

1: Medium VCO range: 150 to 420 MHz

Bit 8 **PLL3FRACEN**: PLL3 fractional latch enable

Set and reset by software to latch the content of FRACN3 into the Sigma-Delta modulator.

In order to latch the FRACN3 value into the Sigma-Delta modulator, PLL3FRACEN must be set to '0', then set to '1': the transition 0 to 1 transfers the content of FRACN3 into the modulator. Please refer to [Section : PLL initialization phase](#) for additional information.

Bits 7:6 PLL2RGE[1:0]: PLL2 input frequency range

Set and reset by software to select the proper reference frequency range used for PLL2.

These bits must be written before enabling the PLL2.

00: The PLL2 input (**ref2_ck**) clock range frequency is between 1 and 2 MHz (default after reset)

01: The PLL2 input (**ref2_ck**) clock range frequency is between 2 and 4 MHz

10: The PLL2 input (**ref2_ck**) clock range frequency is between 4 and 8 MHz

11: The PLL2 input (**ref2_ck**) clock range frequency is between 8 and 16 MHz

Bit 5 PLL2VCOSEL: PLL2 VCO selection

Set and reset by software to select the proper VCO frequency range used for PLL2.

This bit must be written before enabling the PLL2.

0: Wide VCO range: 192 to 836 MHz (default after reset)

1: Medium VCO range: 150 to 420 MHz

Bit 4 PLL2FRACEN: PLL2 fractional latch enable

Set and reset by software to latch the content of FRACN2 into the Sigma-Delta modulator.

In order to latch the FRACN2 value into the Sigma-Delta modulator, PLL2FRACEN must be set to '0', then set to '1': the transition 0 to 1 transfers the content of FRACN2 into the modulator. Please refer to [Section : PLL initialization phase](#) for additional information.

Bits 3:2 PLL1RGE[1:0]: PLL1 input frequency range

Set and reset by software to select the proper reference frequency range used for PLL1.

This bit must be written before enabling the PLL1.

00: The PLL1 input (**ref1_ck**) clock range frequency is between 1 and 2 MHz (default after reset)

01: The PLL1 input (**ref1_ck**) clock range frequency is between 2 and 4 MHz

10: The PLL1 input (**ref1_ck**) clock range frequency is between 4 and 8 MHz

11: The PLL1 input (**ref1_ck**) clock range frequency is between 8 and 16 MHz

Bit 1 PLL1VCOSEL: PLL1 VCO selection

Set and reset by software to select the proper VCO frequency range used for PLL1.

These bits must be written before enabling the PLL1.

0: Wide VCO range: 192 to 836 MHz (default after reset)

1: Medium VCO range: 150 to 420 MHz

Bit 0 PLL1FRACEN: PLL1 fractional latch enable

Set and reset by software to latch the content of FRACN1 into the Sigma-Delta modulator.

In order to latch the FRACN1 value into the Sigma-Delta modulator, PLL1FRACEN must be set to '0', then set to '1': the transition 0 to 1 transfers the content of FRACN1 into the modulator. Please refer to [Section : PLL initialization phase](#) for additional information.

8.7.13 RCC PLL1 Dividers Configuration Register (RCC_PLL1DIVR)

Address offset: 0x030

Reset value: 0x0101 0280

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	DIVR1[6:0]							Res.	DIVQ1[6:0]						
	rw								rw						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIVP1[6:0]								DIVN1[8:0]							
rw								rw							

Bit 31 Reserved, must be kept at reset value.

Bits 30:24 **DIVR1[6:0]**: PLL1 DIVR division factor

Set and reset by software to control the frequency of the **pll1_r_ck** clock.

These bits can be written only when the PLL1 is disabled (PLL1ON = '0' and PLL1RDY = '0').

0000000: **pll1_r_ck** = **vco1_ck**

0000001: **pll1_r_ck** = **vco1_ck** / 2 (default after reset)

0000010: **pll1_r_ck** = **vco1_ck** / 3

0000011: **pll1_r_ck** = **vco1_ck** / 4

...

1111111: **pll1_r_ck** = **vco1_ck** / 128

Bit 23 Reserved, must be kept at reset value.

Bits 22:16 **DIVQ1[6:0]**: PLL1 DIVQ division factor

Set and reset by software to control the frequency of the **pll1_q_ck** clock.

These bits can be written only when the PLL1 is disabled (PLL1ON = '0' and PLL1RDY = '0').

0000000: **pll1_q_ck** = **vco1_ck**

0000001: **pll1_q_ck** = **vco1_ck** / 2 (default after reset)

0000010: **pll1_q_ck** = **vco1_ck** / 3

0000011: **pll1_q_ck** = **vco1_ck** / 4

...

1111111: **pll1_q_ck** = **vco1_ck** / 128

Bits 15:9 **DIVP1[6:0]**: PLL1 DIVP division factor

Set and reset by software to control the frequency of the **pll1_p_ck** clock.

These bits can be written only when the PLL1 is disabled (PLL1ON = '0' and PLL1RDY = '0').

Note that odd division factors are not allowed.

0000000: **pll1_p_ck** = **vco1_ck**

0000001: **pll1_p_ck** = **vco1_ck** / 2 (default after reset)

0000010: Not allowed

0000011: **pll1_p_ck** = **vco1_ck** / 4

...

1111111: **pll1_p_ck** = **vco1_ck** / 128

Bits 8:0 **DIVN1[8:0]**: Multiplication factor for PLL1 VCO

Set and reset by software to control the multiplication factor of the VCO.

These bits can be written only when the PLL is disabled (PLL1ON = '0' and PLL1RDY = '0').

0x003: DIVN1 = 4

0x004: DIVN1 = 5

0x005: DIVN1 = 6

...

0x080: DIVN1 = 129 (default after reset)

...

0x1FF: DIVN1 = 512

Others: wrong configurations

Caution: The software has to set correctly these bits to insure that the VCO output frequency is between its valid frequency range, which is:

- 192 to 836 MHz if PLL1VCOSEL = '0'
- 150 to 420 MHz if PLL1VCOSEL = '1'
-

VCO output frequency = $F_{\text{ref1_ck}} \times \text{DIVN1}$, when fractional value 0 has been loaded into FRACN1, with:

- DIVN1 between 4 and 512
- The input frequency $F_{\text{ref1_ck}}$ between 1MHz and 16 MHz

8.7.14 RCC PLL1 Fractional Divider Register (RCC_PLL1FRACR)

Address offset: 0x034

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FRACN1[12:0]												Res.	Res.	Res.	
rw															

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:3 **FRACN1[12:0]**: Fractional part of the multiplication factor for PLL1 VCO

Set and reset by software to control the fractional part of the multiplication factor of the VCO.

These bits can be written at any time, allowing dynamic fine-tuning of the PLL1 VCO.

Caution: The software has to set correctly these bits to insure that the VCO output frequency is between its valid frequency range, which is:

- 192 to 836 MHz if PLL1VCOSEL = '0'
- 150 to 420 MHz if PLL1VCOSEL = '1'

VCO output frequency = $F_{ref1_ck} \times (DIVN1 + (FRACN1 / 2^{13}))$, with

- DIVN1 shall be between 4 and 512
- FRACN1 can be between 0 and $2^{13} - 1$
- The input frequency F_{ref1_ck} shall be between 1 and 16 MHz.

To change the FRACN value on-the-fly even if the PLL is enabled, the application has to proceed as follow:

- set the bit PLL1FRACEN to '0',
- write the new fractional value into FRACN1,
- set the bit PLL1FRACEN to '1'.

Bits 2:0 Reserved, must be kept at reset value.

8.7.15 RCC PLL2 Dividers Configuration Register (RCC_PLL2DIVR)

Address offset: 0x038

Reset value: 0x0101 0280

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	DIVR2[6:0]							Res.	DIVQ2[6:0]						
	rw								rw						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIVP2[6:0]								DIVN2[8:0]							
rw								rw							

Bit 31 Reserved, must be kept at reset value.

Bits 30:24 **DIVR2[6:0]**: PLL2 DIVR division factor

Set and reset by software to control the frequency of the **pll2_r_ck** clock.

These bits can be written only when the PLL2 is disabled (PLL2ON = '0' and PLL2RDY = '0').

0000000: **pll2_r_ck** = **vco2_ck**

0000001: **pll2_r_ck** = **vco2_ck** / 2 (default after reset)

0000010: **pll2_r_ck** = **vco2_ck** / 3

0000011: **pll2_r_ck** = **vco2_ck** / 4

...

1111111: **pll2_r_ck** = **vco2_ck** / 128

Bit 23 Reserved, must be kept at reset value.

Bits 22:16 **DIVQ2[6:0]**: PLL2 DIVQ division factor

Set and reset by software to control the frequency of the **pll2_q_ck** clock.

These bits can be written only when the PLL2 is disabled (PLL2ON = '0' and PLL2RDY = '0').

0000000: **pll2_q_ck** = **vco2_ck**

0000001: **pll2_q_ck** = **vco2_ck** / 2 (default after reset)

0000010: **pll2_q_ck** = **vco2_ck** / 3

0000011: **pll2_q_ck** = **vco2_ck** / 4

...

1111111: **pll2_q_ck** = **vco2_ck** / 128

Bits 15:9 **DIVP2[6:0]**: PLL2 DIVP division factor

Set and reset by software to control the frequency of the **pll2_p_ck** clock.

These bits can be written only when the PLL2 is disabled (PLL2ON = '0' and PLL2RDY = '0').

0000000: **pll2_p_ck** = **vco2_ck**

0000001: **pll2_p_ck** = **vco2_ck** / 2 (default after reset)

0000010: **pll2_p_ck** = **vco2_ck** / 3

0000011: **pll2_p_ck** = **vco2_ck** / 4

...

1111111: **pll2_p_ck** = **vco2_ck** / 128

Bits 8:0 **DIVN2[8:0]**: Multiplication factor for PLL2 VCO

Set and reset by software to control the multiplication factor of the VCO.

These bits can be written only when the PLL is disabled (PLL2ON = '0' and PLL2RDY = '0').

Caution: The software has to set correctly these bits to insure that the VCO output frequency is between its valid frequency range, which is:

- 192 to 836 MHz if PLL2VCOSEL = '0'
- 150 to 420 MHz if PLL2VCOSEL = '1'

VCO output frequency = $F_{ref2_ck} \times DIVN2$, when fractional value 0 has been loaded into FRACN2, with

- DIVN2 between 4 and 512
- The input frequency F_{ref2_ck} between 1MHz and 16MHz

0x003: DIVN2 = 4

0x004: DIVN2 = 5

0x005: DIVN2 = 6

...

0x080: DIVN2 = 129 (default after reset)

...

0x1FF: DIVN2 = 512

Others: wrong configurations

8.7.16 RCC PLL2 Fractional Divider Register (RCC_PLL2FRACR)

Address offset: 0x03C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FRACN2[12:0]												Res.	Res.	Res.	
rw															

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:3 **FRACN2[12:0]**: Fractional part of the multiplication factor for PLL2 VCO

Set and reset by software to control the fractional part of the multiplication factor of the VCO.

These bits can be written at any time, allowing dynamic fine-tuning of the PLL2 VCO.

Caution: The software has to set correctly these bits to insure that the VCO output frequency is between its valid frequency range, which is:

- 192 to 836 MHz if PLL2VCOSEL = '0'
- 150 to 420 MHz if PLL2VCOSEL = '1'

VCO output frequency = $F_{\text{ref2_ck}} \times (\text{DIVN2} + (\text{FRACN2} / 2^{13}))$, with

- DIVN2 shall be between 4 and 512
- FRACN2 can be between 0 and $2^{13} - 1$
- The input frequency $F_{\text{ref2_ck}}$ shall be between 1 and 16 MHz

In order to change the FRACN value on-the-fly even if the PLL is enabled, the application has to proceed as follow:

- set the bit PLL2FRACEN to '0',
- write the new fractional value into FRACN2,
- set the bit PLL2FRACEN to '1'.

Bits 2:0 Reserved, must be kept at reset value.

8.7.17 RCC PLL3 Dividers Configuration Register (RCC_PLL3DIVR)

Address offset: 0x040

Reset value: 0x0101 0280

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	DIVR3[6:0]							Res.	DIVQ3[6:0]						
	rw								rw						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIVP3[6:0]								DIVN3[8:0]							
rw								rw							

Bit 31 Reserved, must be kept at reset value.

Bits 30:24 **DIVR3[6:0]**: PLL3 DIVR division factor

Set and reset by software to control the frequency of the **pll3_r_ck** clock.

These bits can be written only when the PLL3 is disabled (PLL3ON = '0' and PLL3RDY = '0').

0000000: **pll3_r_ck** = **vco3_ck**

0000001: **pll3_r_ck** = **vco3_ck** / 2 (default after reset)

0000010: **pll3_r_ck** = **vco3_ck** / 3

0000011: **pll3_r_ck** = **vco3_ck** / 4

...

1111111: **pll3_r_ck** = **vco3_ck** / 128

Bit 23 Reserved, must be kept at reset value.

Bits 22:16 **DIVQ3[6:0]**: PLL3 DIVQ division factor

Set and reset by software to control the frequency of the **pll3_q_ck** clock.

These bits can be written only when the PLL3 is disabled (PLL3ON = '0' and PLL3RDY = '0').

0000000: **pll3_q_ck** = **vco3_ck**

0000001: **pll3_q_ck** = **vco3_ck** / 2 (default after reset)

0000010: **pll3_q_ck** = **vco3_ck** / 3

0000011: **pll3_q_ck** = **vco3_ck** / 4

...

1111111: **pll3_q_ck** = **vco3_ck** / 128

Bits 15:9 **DIVP3[6:0]**: PLL3 DIVP division factor

Set and reset by software to control the frequency of the **pll3_p_ck** clock.

These bits can be written only when the PLL3 is disabled (PLL3ON = '0' and PLL3RDY = '0').

0000000: **pll3_p_ck** = **vco3_ck**

0000001: **pll3_p_ck** = **vco3_ck** / 2 (default after reset)

0000010: **pll3_p_ck** = **vco3_ck** / 3

0000011: **pll3_p_ck** = **vco3_ck** / 4

...

1111111: **pll3_p_ck** = **vco3_ck** / 128

Bits 8:0 **DIVN3[7:0]**: Multiplication factor for PLL3 VCO

Set and reset by software to control the multiplication factor of the VCO.

These bits can be written only when the PLL is disabled (PLL3ON = '0' and PLL3RDY = '0').

Caution: The software has to set correctly these bits to insure that the VCO output frequency is between its valid frequency range, which is:

- 192 to 836 MHz if PLL3VCOSEL = '0'
- 150 to 420 MHz if PLL3VCOSEL = '1'

VCO output frequency = $F_{ref3_ck} \times DIVN3$, when fractional value 0 has been loaded into FRACN3, with

- DIVN3 between 4 and 512
- The input frequency F_{ref3_ck} between 1MHz and 16MHz

0x003: DIVN3 = 4

0x004: DIVN3 = 5

0x005: DIVN3 = 6

...

0x080: DIVN3 = 129 (default after reset)

...

0x1FF: DIVN3 = 512

Others: wrong configurations

8.7.18 RCC PLL3 Fractional Divider Register (RCC_PLL3FRACR)

Address offset: 0x044

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FRACN3[12:0]												Res.	Res.	Res.	
rw															

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:3 **FRACN3[12:0]**: Fractional part of the multiplication factor for PLL3 VCO

Set and reset by software to control the fractional part of the multiplication factor of the VCO.

These bits can be written at any time, allowing dynamic fine-tuning of the PLL3 VCO.

Caution: The software has to set correctly these bits to insure that the VCO output frequency is between its valid frequency range, which is:

- 192 to 836 MHz if PLL3VCOSEL = '0'
- 150 to 420 MHz if PLL3VCOSEL = '1'

VCO output frequency = $F_{\text{ref3_ck}} \times (\text{DIVN3} + (\text{FRACN3} / 2^{13}))$, with

- DIVN3 shall be between 4 and 512
- FRACN3 can be between 0 and $2^{13} - 1$
- The input frequency $F_{\text{ref3_ck}}$ shall be between 1 and 16 MHz

In order to change the FRACN value on-the-fly even if the PLL is enabled, the application has to proceed as follow:

- set the bit PLL1FRACEN to '0',
- write the new fractional value into FRACN1,
- set the bit PLL1FRACEN to '1'.

Bits 2:0 Reserved, must be kept at reset value.

8.7.19 RCC Domain 1 Kernel Clock Configuration Register (RCC_D1CCIPR)

Address offset: 0x04C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	CKPERSEL[1:0] ⁽¹⁾		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMCSEL ⁽¹⁾
		rw													rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	QSPISEL[1:0] ⁽¹⁾		Res.	Res.	FMCSEL[1:0] ⁽¹⁾	
										rw				rw	

1. Changing the clock source on-the-fly is allowed and will not generate any timing violation. However the user has to make use that both the previous and the new clock sources are present during the switching, and during the whole transition time. Please refer to [Section : Clock switches and gating](#).

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:28 **CKPERSEL[1:0]**: **per_ck** clock source selection

00: **hsi_ker_ck** clock selected as **per_ck** clock (default after reset)

01: **csi_ker_ck** clock selected as **per_ck** clock

10: **hse_ck** clock selected as **per_ck** clock

11: reserved, the **per_ck** clock is disabled

Bits 27:17 Reserved, must be kept at reset value.

Bit 16 **SDMMCSEL**: SDMMC kernel clock source selection

0: **pll1_q_ck** clock is selected as kernel peripheral clock (default after reset)

1: **pll2_r_ck** clock is selected as kernel peripheral clock

Bits 15:6 Reserved, must be kept at reset value.

Bits 5:4 **QSPISEL[1:0]**: QUADSPI kernel clock source selection

00: **rcc_hclk3** clock selected as kernel peripheral clock (default after reset)

01: **pll1_q_ck** clock selected as kernel peripheral clock

10: **pll2_r_ck** clock selected as kernel peripheral clock

11: **per_ck** clock selected as kernel peripheral clock

Bits 3:2 Reserved, must be kept at reset value.

Bits 1:0 **FMCSEL[1:0]**: FMC kernel clock source selection

00: **rcc_hclk3** clock selected as kernel peripheral clock (default after reset)

01: **pll1_q_ck** clock selected as kernel peripheral clock

10: **pll2_r_ck** clock selected as kernel peripheral clock

11: **per_ck** clock selected as kernel peripheral clock

8.7.20 RCC Domain 2 Kernel Clock Configuration Register (RCC_D2CCIP1R)

Address offset: 0x050

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWPSEL ⁽¹⁾	Res.	FDCANSEL[1:0] ⁽¹⁾	Res.	Res.	Res.	DFSDM1SEL ⁽¹⁾	Res.	Res.	SPDIFSEL[1:0] ⁽¹⁾	Res.	SPI45SEL[2:0] ⁽¹⁾				
rw		rw				rw			rw		rw				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	SPI123SEL[2:0] ⁽¹⁾	Res.	Res.	Res.	SAI23SEL[2:0] ⁽¹⁾	Res.	Res.	Res.	SAI1SEL[2:0] ⁽¹⁾						
	rw				rw				rw						

1. Changing the clock source on-the-fly is allowed and will not generate any timing violation. However the user has to make sure that both the previous and the new clock sources are present during the switching, and for the whole transition time. Please refer to [Section : Clock switches and gating](#).

Bit 31 **SWPSEL**: SWPMI kernel clock source selection

Set and reset by software.

0: **pclk** is selected as SWPMI kernel clock (default after reset)

1: **hsi_ker_ck** clock is selected as SWPMI kernel clock

Bit 30 Reserved, must be kept at reset value.

Bits 29:28 **FDCANSEL**: FDCAN kernel clock source selection

Set and reset by software.

00: **hse_ck** clock is selected as FDCAN kernel clock (default after reset)

01: **pll1_q_ck** clock is selected as FDCAN kernel clock

10: **pll2_q_ck** clock is selected as FDCAN kernel clock

11: reserved, the kernel clock is disabled

Bits 27:25 Reserved, must be kept at reset value.

Bit 24 **DFSDM1SEL**: DFSDM1 kernel **Clk** clock source selection

Set and reset by software.

Note: the DFSDM1 Aclk Clock Source Selection is done by SAI1SEL.

0: **rcc_pclk2** is selected as DFSDM1 Clk kernel clock (default after reset)

1: **sys_ck** clock is selected as DFSDM1 Clk kernel clock

Bits 23:22 Reserved, must be kept at reset value.

Bits 21:20 **SPDIFSEL[1:0]**: SPDIFRX kernel clock source selection

00: **pll1_q_ck** clock selected as SPDIFRX kernel clock (default after reset)

01: **pll2_r_ck** clock selected as SPDIFRX kernel clock

10: **pll3_r_ck** clock selected as SPDIFRX kernel clock

11: **hsi_ker_ck** clock selected as SPDIFRX kernel clock

Bit 19 Reserved, must be kept at reset value.

Bits 18:16 **SPI45SEL[2:0]**: SPI4 and 5 kernel clock source selection

Set and reset by software.

000: APB clock is selected as kernel clock (default after reset)

001: **pll2_q_ck** clock is selected as kernel clock

010: **pll3_q_ck** clock is selected as kernel clock

011: **hsi_ker_ck** clock is selected as kernel clock

100: **csi_ker_ck** clock is selected as kernel clock

101: **hse_ck** clock is selected as kernel clock

others: reserved, the kernel clock is disabled

Bit 15 Reserved, must be kept at reset value.

Bits 14:12 **SPI123SEL[2:0]**: SPI/I2S1,2 and 3 kernel clock source selection

Set and reset by software.

Caution: If the selected clock is the external clock and this clock is stopped, it will not be possible to switch to another clock. Refer to [Section : Clock switches and gating](#) for additional information.

000: **pll1_q_ck** clock selected as SPI/I2S1,2 and 3 kernel clock (default after reset)

001: **pll2_p_ck** clock selected as SPI/I2S1,2 and 3 kernel clock

010: **pll3_p_ck** clock selected as SPI/I2S1,2 and 3 kernel clock

011: I2S_CKIN clock selected as SPI/I2S1,2 and 3 kernel clock

100: **per_ck** clock selected as SPI/I2S1,2 and 3 kernel clock

others: reserved, the kernel clock is disabled

Note: I2S_CKIN is an external clock taken from a pin.

Bits 11:9 Reserved, must be kept at reset value.

Bits 8:6 **SAI23SEL[2:0]**: SAI2 and SAI3 kernel clock source selection
Set and reset by software.

Caution: If the selected clock is the external clock and this clock is stopped, it will not be possible to switch to another clock. Refer to [Section : Clock switches and gating](#) for additional information.

000: **pll1_q_ck** clock selected as SAI2 and SAI3 kernel clock (default after reset)
001: **pll2_p_ck** clock selected as SAI2 and SAI3 kernel clock
010: **pll3_p_ck** clock selected as SAI2 and SAI3 kernel clock
011: **I2S_CKIN** clock selected as SAI2 and SAI3 kernel clock
100: **per_ck** clock selected as SAI2 and SAI3 kernel clock
others: reserved, the kernel clock is disabled

Note: I2S_CKIN is an external clock taken from a pin.

Bits 5:3 Reserved, must be kept at reset value.

Bits 2:0 **SAI1SEL[2:0]**: SAI1 and DFSDM1 kernel **Aclk** clock source selection
Set and reset by software.

Caution: If the selected clock is the external clock and this clock is stopped, it will not be possible to switch to another clock. Refer to [Section : Clock switches and gating](#) for additional information.

Note: DFSDM1 Clock Source Selection is done by DFSDM1SEL.

000: **pll1_q_ck** clock selected as SAI1 and DFSDM1 **Aclk** kernel clock (default after reset)
001: **pll2_p_ck** clock selected as SAI1 and DFSDM1 **Aclk** kernel clock
010: **pll3_p_ck** clock selected as SAI1 and DFSDM1 **Aclk** kernel clock
011: **I2S_CKIN** clock selected as SAI1 and DFSDM1 **Aclk** kernel clock
100: **per_ck** clock selected as SAI1 and DFSDM1 **Aclk** kernel clock
others: reserved, the kernel clock is disabled

Note: I2S_CKIN is an external clock taken from a pin.

8.7.21 RCC Domain 2 Kernel Clock Configuration Register (RCC_D2CCIP2R)

Address offset: 0x054

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	LPTIM1SEL[2:0] ⁽¹⁾			Res.	Res.	Res.	Res.	CECSEL[1:0] ⁽¹⁾		USBSEL[1:0] ⁽¹⁾		Res.	Res.	Res.	Res.
	rw							rw		rw					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	I2C123SEL[1:0] ⁽¹⁾		Res.	Res.	RNGSEL[1:0] ⁽¹⁾		Res.	Res.	USART16SEL[2:0] ⁽¹⁾			USART234578SEL[2:0] ⁽¹⁾		
		rw				rw				rw			rw		

1. Changing the clock source on-the-fly is allowed and will not generate any timing violation. However the user has to make sure that both the previous and the new clock sources are present during the switching, and for the whole transition time. Please refer to [Section : Clock switches and gating](#).

Bit 31 Reserved, must be kept at reset value.

Bits 30:28 **LPTIM1SEL[2:0]**: LPTIM1 kernel clock source selection

Set and reset by software.

000: **rcc_pclk1** clock selected as kernel peripheral clock (default after reset)

001: **pll2_p_ck** clock selected as kernel peripheral clock

010: **pll3_r_ck** clock selected as kernel peripheral clock

011: **lse_ck** clock selected as kernel peripheral clock

100: **lsi_ck** clock selected as kernel peripheral clock

101: **per_ck** clock selected as kernel peripheral clock

others: reserved, the kernel clock is disabled

Bits 27:24 Reserved, must be kept at reset value.

Bits 23:22 **CECSEL[1:0]**: HDMI-CEC kernel clock source selection

Set and reset by software.

00: **lse_ck** clock is selected as kernel clock (default after reset)

01: **lsi_ck** clock is selected as kernel clock

10: **csi_ker_ck** divided by 122 is selected as kernel clock

11: reserved, the kernel clock is disabled

Bits 21:20 **USBSEL[1:0]**: USBOTG 1 and 2 kernel clock source selection

Set and reset by software.

00: Disable the kernel clock (default after reset)

01: **pll1_q_ck** clock is selected as kernel clock

10: **pll3_q_ck** clock is selected as kernel clock

11: **hsi48_ck** clock is selected as kernel clock

Bits 19:14 Reserved, must be kept at reset value.

Bits 13:12 **I2C123SEL[1:0]**: I2C1,2,3 kernel clock source selection

Set and reset by software.

00: **rcc_pclk1** clock is selected as kernel clock (default after reset)

01: **pll3_r_ck** clock is selected as kernel clock

10: **hsi_ker_ck** clock is selected as kernel clock

11: **csi_ker_ck** clock is selected as kernel clock

Bits 11:10 Reserved, must be kept at reset value.

Bits 9:8 **RNGSEL[1:0]**: RNG kernel clock source selection

Set and reset by software.

00: **hsi8_ck** clock is selected as kernel clock (default after reset)

01: **pll1_q_ck** clock is selected as kernel clock

10: **lse_ck** clock is selected as kernel clock

11: **lsi_ck** clock is selected as kernel clock

Bits 7:6 Reserved, must be kept at reset value.

Bits 5:3 **USART16SEL[2:0]**: USART1 and 6 kernel clock source selection

Set and reset by software.

000: **rcc_pclk2** clock is selected as kernel clock (default after reset)

001: **pll2_q_ck** clock is selected as kernel clock

010: **pll3_q_ck** clock is selected as kernel clock

011: **hsi_ker_ck** clock is selected as kernel clock

100: **csi_ker_ck** clock is selected as kernel clock

101: **lse_ck** clock is selected as kernel clock

others: reserved, the kernel clock is disabled

Bits 2:0 **USART234578SEL[2:0]**: USART2/3, UART4,5, 7/8 (APB1) kernel clock source selection

Set and reset by software.

000: **rcc_pclk1** clock is selected as kernel clock (default after reset)

001: **pll2_q_ck** clock is selected as kernel clock

010: **pll3_q_ck** clock is selected as kernel clock

011: **hsi_ker_ck** clock is selected as kernel clock

100: **csi_ker_ck** clock is selected as kernel clock

101: **lse_ck** clock is selected as kernel clock

others: reserved, the kernel clock is disabled

8.7.22 RCC Domain 3 Kernel Clock Configuration Register (RCC_D3CCIPR)

Address offset: 0x058

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	SPI6SEL[2:0] ⁽¹⁾			Res.	SAI4BSEL[2:0] ⁽¹⁾			SAI4ASEL[2:0] ⁽¹⁾			Res.	Res.	Res.	ADCSEL[1:0] ⁽¹⁾	
	rw				rw			rw						rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LPTIM345SEL[2:0] ⁽¹⁾				LPTIM2SEL[2:0] ⁽¹⁾			I2C4SEL[1:0] ⁽¹⁾		Res.	Res.	Res.	Res.	Res.	LPUART1SEL[2:0] ⁽¹⁾	
rw				rw			rw							rw	

1. Changing the clock source on-the-fly is allowed, and will not generate any timing violation. However the user has to make sure that both the previous and the new clock sources are present during the switching, and for the whole transition time. Please refer to [Section : Clock switches and gating](#).

Bit 31 Reserved, must be kept at reset value.

Bits 30:28 **SPI6SEL[2:0]**: SPI6 kernel clock source selection

Set and reset by software.

000: **rcc_pclk4** clock selected as kernel peripheral clock (default after reset)

001: **pll2_q_ck** clock selected as kernel peripheral clock

010: **pll3_q_ck** clock selected as kernel peripheral clock

011: **hsi_ker_ck** clock selected as kernel peripheral clock

100: **csi_ker_ck** clock selected as kernel peripheral clock

101: **hse_ck** clock selected as kernel peripheral clock

others: reserved, the kernel clock is disabled

Bit 27 Reserved, must be kept at reset value.

Bits 26:24 **SAI4BSEL[2:0]**: Sub-Block B of SAI4 kernel clock source selection

Set and reset by software.

Caution: If the selected clock is the external clock and this clock is stopped, it will not be possible to switch to another clock. Refer to [Section : Clock switches and gating](#) for additional information.

000: **pll1_q_ck** clock selected as kernel peripheral clock (default after reset)

001: **pll2_p_ck** clock selected as kernel peripheral clock

010: **pll3_p_ck** clock selected as kernel peripheral clock

011: **I2S_CKIN** clock selected as kernel peripheral clock

100: **per_ck** clock selected as kernel peripheral clock

others: reserved, the kernel clock is disabled

Note: I2S_CKIN is an external clock taken from a pin.

Bits 23:21 **SAI4ASEL[2:0]**: Sub-Block A of SAI4 kernel clock source selection

Set and reset by software.

Caution: If the selected clock is the external clock and this clock is stopped, it will not be possible to switch to another clock. Refer to [Section : Clock switches and gating](#) for additional information.

000: **pll1_q_ck** clock selected as kernel peripheral clock (default after reset)
 001: **pll2_p_ck** clock selected as kernel peripheral clock
 010: **pll3_p_ck** clock selected as kernel peripheral clock
 011: **I2S_CKIN** clock selected as kernel peripheral clock
 100: **per_ck** clock selected as kernel peripheral clock
 others: reserved, the kernel clock is disabled

Note: I2S_CKIN is an external clock taken from a pin.

Bits 20:18 Reserved, must be kept at reset value.

Bits 17:16 **ADCSEL[1:0]**: SAR ADC kernel clock source selection

Set and reset by software.

00: **pll2_p_ck** clock selected as kernel peripheral clock (default after reset)
 01: **pll3_r_ck** clock selected as kernel peripheral clock
 10: **per_ck** clock selected as kernel peripheral clock
 others: reserved, the kernel clock is disabled

Bits 15:13 **LPTIM345SEL[2:0]**: LPTIM3,4,5 kernel clock source selection

Set and reset by software.

000: **rcc_pclk4** clock selected as kernel peripheral clock (default after reset)
 001: **pll2_p_ck** clock selected as kernel peripheral clock
 010: **pll3_r_ck** clock selected as kernel peripheral clock
 011: **lse_ck** clock selected as kernel peripheral clock
 100: **lsi_ck** clock selected as kernel peripheral clock
 101: **per_ck** clock selected as kernel peripheral clock
 others: reserved, the kernel clock is disabled

Bits 12:10 **LPTIM2SEL[2:0]**: LPTIM2 kernel clock source selection

Set and reset by software.

000: **rcc_pclk4** clock selected as kernel peripheral clock (default after reset)
 001: **pll2_p_ck** clock selected as kernel peripheral clock
 010: **pll3_r_ck** clock selected as kernel peripheral clock
 011: **lse_ck** clock selected as kernel peripheral clock
 100: **lsi_ck** clock selected as kernel peripheral clock
 101: **per_ck** clock selected as kernel peripheral clock
 others: reserved, the kernel clock is disabled

Bits 9:8 **I2C4SEL[1:0]**: I2C4 kernel clock source selection

Set and reset by software.

00: **rcc_pclk4** clock selected as kernel peripheral clock (default after reset)

01: **pll3_r_ck** clock selected as kernel peripheral clock

10: **hsi_ker_ck** clock selected as kernel peripheral clock

11: **csi_ker_ck** clock selected as kernel peripheral clock

Bits 7:3 Reserved, must be kept at reset value.

Bits 2:0 **LPUART1SEL[2:0]**: LPUART1 kernel clock source selection

Set and reset by software.

000: **rcc_pclk_d3** clock is selected as kernel peripheral clock (default after reset)

001: **pll2_q_ck** clock is selected as kernel peripheral clock

010: **pll3_q_ck** clock is selected as kernel peripheral clock

011: **hsi_ker_ck** clock is selected as kernel peripheral clock

100: **csi_ker_ck** clock is selected as kernel peripheral clock

101: **lse_ck** clock is selected as kernel peripheral clock

others: reserved, the kernel clock is disabled

8.7.23 RCC Clock Source Interrupt Enable Register (RCC_CIER)

Address offset: 0x060

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	LSECSSIE	PLL3RDYIE	PLL2RDYIE	PLL1RDYIE	HSI48RDYIE	CSIRDYIE	HSE RDYIE	HSIRDYIE	LSERDYIE	LSIRDYIE
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:10 Reserved, must be kept at reset value.

Bit 9 LSECSSIE: LSE clock security system Interrupt Enable

Set and reset by software to enable/disable interrupt caused by the Clock Security System on external 32 kHz oscillator.

0: LSE CSS interrupt disabled (default after reset)

1: LSE CSS interrupt enabled

Bit 8 PLL3RDYIE: PLL3 ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by PLL3 lock.

0: PLL3 lock interrupt disabled (default after reset)

1: PLL3 lock interrupt enabled

Bit 7 PLL2RDYIE: PLL2 ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by PLL2 lock.

0: PLL2 lock interrupt disabled (default after reset)

1: PLL2 lock interrupt enabled

Bit 6 PLL1RDYIE: PLL1 ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by PLL1 lock.

0: PLL1 lock interrupt disabled (default after reset)

1: PLL1 lock interrupt enabled

Bit 5 HSI48RDYIE: HSI48 ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by the HSI48 oscillator stabilization.

0: HSI48 ready interrupt disabled (default after reset)

1: HSI48 ready interrupt enabled

Bit 4 CSIRDYIE: CSI ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by the CSI oscillator stabilization.

0: CSI ready interrupt disabled (default after reset)

1: CSI ready interrupt enabled

Bit 3 HSE RDYIE: HSE ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by the HSE oscillator stabilization.

0: HSE ready interrupt disabled (default after reset)

1: HSE ready interrupt enabled

Bit 2 **HSIRDYIE**: HSI ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by the HSI oscillator stabilization.

0: HSI ready interrupt disabled (default after reset)

1: HSI ready interrupt enabled

Bit 1 **LSERDYIE**: LSE ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by the LSE oscillator stabilization.

0: LSE ready interrupt disabled (default after reset)

1: LSE ready interrupt enabled

Bit 0 **LSIRDYIE**: LSI ready Interrupt Enable

Set and reset by software to enable/disable interrupt caused by the LSI oscillator stabilization.

0: LSI ready interrupt disabled (default after reset)

1: LSI ready interrupt enabled

8.7.24 RCC Clock Source Interrupt Flag Register (RCC_CIFR)

Address offset: 0x64

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	HSECSSF	LSECSSF	PLL3RDYF	PLL2RDYF	PLL1RDYF	HSI48RDYF	CSIRDYF	HSERDYF	HSIRDYF	LSERDYF	LSIRDYF
					r	r	r	r	r	r	r	r	r	r	r

Bits 31:11 Reserved, must be kept at reset value.

Bit 10 HSECSSF: HSE clock security system Interrupt Flag

Reset by software by writing HSECSSC bit.

Set by hardware in case of HSE clock failure.

0: No clock security interrupt caused by HSE clock failure (default after reset)

1: Clock security interrupt caused by HSE clock failure

Bit 9 LSECSSF: LSE clock security system Interrupt Flag

Reset by software by writing LSECSSC bit.

Set by hardware when a failure is detected on the external 32 kHz oscillator and LSECSSIE is set.

0: No failure detected on the external 32 kHz oscillator (default after reset)

1: A failure is detected on the external 32 kHz oscillator

Bit 8 PLL3RDYF: PLL3 ready Interrupt Flag

Reset by software by writing PLL3RDYC bit.

Set by hardware when the PLL3 locks and PLL3RDYIE is set.

0: No clock ready interrupt caused by PLL3 lock (default after reset)

1: Clock ready interrupt caused by PLL3 lock

Bit 7 PLL2RDYF: PLL2 ready Interrupt Flag

Reset by software by writing PLL2RDYC bit.

Set by hardware when the PLL2 locks and PLL2RDYIE is set.

0: No clock ready interrupt caused by PLL2 lock (default after reset)

1: Clock ready interrupt caused by PLL2 lock

Bit 6 PLL1RDYF: PLL1 ready Interrupt Flag

Reset by software by writing PLL1RDYC bit.

Set by hardware when the PLL1 locks and PLL1RDYIE is set.

0: No clock ready interrupt caused by PLL1 lock (default after reset)

1: Clock ready interrupt caused by PLL1 lock

Bit 5 HSI48RDYF: HSI48 ready Interrupt Flag

Reset by software by writing HSI48RDYC bit.

Set by hardware when the HSI48 clock becomes stable and HSI48RDYIE is set.

0: No clock ready interrupt caused by the HSI48 oscillator (default after reset)

1: Clock ready interrupt caused by the HSI48 oscillator

Bit 4 CSIRDYF: CSI ready Interrupt Flag

Reset by software by writing CSIRDYC bit.

Set by hardware when the CSI clock becomes stable and CSIRDYIE is set.

0: No clock ready interrupt caused by the CSI (default after reset)

1: Clock ready interrupt caused by the CSI

Bit 3 HSERDYF: HSE ready Interrupt Flag

Reset by software by writing HSERDYC bit.

Set by hardware when the HSE clock becomes stable and HSERDYIE is set.

0: No clock ready interrupt caused by the HSE (default after reset)

1: Clock ready interrupt caused by the HSE

Bit 2 HSIRDYF: HSI ready Interrupt Flag

Reset by software by writing HSIRDYC bit.

Set by hardware when the HSI clock becomes stable and HSIRDYIE is set.

0: No clock ready interrupt caused by the HSI (default after reset)

1: Clock ready interrupt caused by the HSI

Bit 1 LSERDYF: LSE ready Interrupt Flag

Reset by software by writing LSERDYC bit.

Set by hardware when the LSE clock becomes stable and LSERDYIE is set.

0: No clock ready interrupt caused by the LSE (default after reset)

1: Clock ready interrupt caused by the LSE

Bit 0 LSIRDYF: LSI ready Interrupt Flag

Reset by software by writing LSIRDYC bit.

Set by hardware when the LSI clock becomes stable and LSIRDYIE is set.

0: No clock ready interrupt caused by the LSI (default after reset)

1: Clock ready interrupt caused by the LSI

8.7.25 RCC Clock Source Interrupt Clear Register (RCC_CICR)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	HSECSSC	LSECSSC	PLL3RDYC	PLL2RDYC	PLL1RDYC	HSI48RDYC	CSIRDYC	HSERDYC	HSIRDYC	LSERDYC	LSIRDYC
					rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1

Bits 31:11 Reserved, must be kept at reset value.

Bit 10 HSECSSC: HSE clock security system Interrupt Clear

Set by software to clear HSECSSF.

Reset by hardware when clear done.

0: HSECSSF no effect (default after reset)

1: HSECSSF cleared

Bit 9 LSECSSC: LSE clock security system Interrupt Clear

Set by software to clear LSECSSF.

Reset by hardware when clear done.

0: LSECSSF no effect (default after reset)

1: LSECSSF cleared

Bit 8 PLL3RDYC: PLL3 ready Interrupt Clear

Set by software to clear PLL3RDYF.

Reset by hardware when clear done.

0: PLL3RDYF no effect (default after reset)

1: PLL3RDYF cleared

Bit 7 PLL2RDYC: PLL2 ready Interrupt Clear

Set by software to clear PLL2RDYF.

Reset by hardware when clear done.

0: PLL2RDYF no effect (default after reset)

1: PLL2RDYF cleared

Bit 6 PLL1RDYC: PLL1 ready Interrupt Clear

Set by software to clear PLL1RDYF.

Reset by hardware when clear done.

0: PLL1RDYF no effect (default after reset)

1: PLL1RDYF cleared

Bit 5 HSI48RDYC: HSI48 ready Interrupt Clear

Set by software to clear HSI48RDYF.

Reset by hardware when clear done.

0: HSI48RDYF no effect (default after reset)

1: HSI48RDYF cleared

Bit 4 **CSIRDYC**: CSI ready Interrupt Clear
Set by software to clear CSIRDYF.
Reset by hardware when clear done.
0: CSIRDYF no effect (default after reset)
1: CSIRDYF cleared

Bit 3 **HSERDYC**: HSE ready Interrupt Clear
Set by software to clear HSERDYF.
Reset by hardware when clear done.
0: HSERDYF no effect (default after reset)
1: HSERDYF cleared

Bit 2 **HSIRDYC**: HSI ready Interrupt Clear
Set by software to clear HSIRDYF.
Reset by hardware when clear done.
0: HSIRDYF no effect (default after reset)
1: HSIRDYF cleared

Bit 1 **LSERDYC**: LSE ready Interrupt Clear
Set by software to clear LSERDYF.
Reset by hardware when clear done.
0: LSERDYF no effect (default after reset)
1: LSERDYF cleared

Bit 0 **LSIRDYC**: LSI ready Interrupt Clear
Set by software to clear LSIRDYF.
Reset by hardware when clear done.
0: LSIRDYF no effect (default after reset)
1: LSIRDYF cleared

8.7.26 RCC Backup Domain Control Register (RCC_BDCR)

Address offset: 0x070

Reset value: 0x0000 0000, reset by Backup domain reset.

Access: $0 \leq \text{wait state} \leq 7$, word, half-word and byte access. Wait states are inserted in case of successive accesses to this register.

After a system reset, the RCC_BDCR register is write-protected. To modify this register, the DBP bit in the *PWR control register 1 (PWR_CR1)* has to be set to '1'. RCC_BDCR bits are only reset after a backup domain reset (see [Section 8.4.6: Backup domain reset](#)). Any other internal or external reset will not have any effect on these bits.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BDRST
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RTCEN	Res.	Res.	Res.	Res.	Res.	RTCSEL[1:0]		Res.	LSECSSD	LSECSSON	LSEDRV[1:0]		LSEBYP	LSEBDY	LSEON
rw						rwo			r	rs	rw		rw	r	rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **BDRST**: Backup domain software reset

Set and reset by software.

0: Reset not activated (default after backup domain reset)

1: Resets the entire VSW domain

Bit 15 **RTCEN**: RTC clock enable

Set and reset by software.

0: **rtc_ck** clock is disabled (default after backup domain reset)

1: **rtc_ck** clock enabled

Bits 14:10 Reserved, must be kept at reset value.

Bits 9:8 **RTCSEL[1:0]**: RTC clock source selection

Set by software to select the clock source for the RTC. These bits can be written only one time (except in case of failure detection on LSE). These bits must be written before LSECSSON is enabled. The BDRST bit can be used to reset them, then it can be written one time again.

If HSE is selected as RTC clock: this clock is lost when the system is in Stop mode or in case of a pin reset (NRST).

00: No clock (default after backup domain reset)

01: LSE clock used as RTC clock

10: LSI clock used as RTC clock

11: HSE clock divided by RTCPRE value is used as RTC clock

Bit 7 Reserved, must be kept at reset value.

Bit 6 LSECSSD: LSE clock security system failure detection

Set by hardware to indicate when a failure has been detected by the Clock Security System on the external 32 kHz oscillator.

0: No failure detected on 32 kHz oscillator (default after backup domain reset)

1: Failure detected on 32 kHz oscillator

Bit 5 LSECSSON: LSE clock security system enable

Set by software to enable the Clock Security System on 32 kHz oscillator.

LSECSSON must be enabled after LSE is enabled (LSEON enabled) and ready (LSERDY set by hardware), and after RTCSEL is selected.

Once enabled this bit cannot be disabled, except after a LSE failure detection (LSECSSD = '1'). In that case the software **must** disable LSECSSON.

0: Clock Security System on 32 kHz oscillator OFF (default after backup domain reset)

1: Clock Security System on 32 kHz oscillator ON

Bits 4:3 LSEDRV[1:0]: LSE oscillator driving capability

Set by software to select the driving capability of the LSE oscillator.

00: Lowest drive (default after backup domain reset)

01: Medium low drive

10: Medium high drive

11: Highest drive

Bit 2 LSEBYP: LSE oscillator bypass

Set and reset by software to bypass oscillator in debug mode. This bit must not be written when the LSE is enabled (by LSEON) or ready (LSERDY = '1')

0: LSE oscillator not bypassed (default after backup domain reset)

1: LSE oscillator bypassed

Bit 1 LSERDY: LSE oscillator ready

Set and reset by hardware to indicate when the LSE is stable. This bit needs 6 cycles of **lse_ck** clock to fall down after LSEON has been set to '0'.

0: LSE oscillator not ready (default after backup domain reset)

1: LSE oscillator ready

Bit 0 LSEON: LSE oscillator enabled

Set and reset by software.

0: LSE oscillator OFF (default after backup domain reset)

1: LSE oscillator ON

8.7.27 RCC Clock Control and Status Register (RCC_CSR)

Address offset: 0x074

Reset value: 0x0000 0000

Access: $0 \leq \text{wait state} \leq 7$, word, half-word and byte access

Wait states are inserted in case of successive accesses to this register.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LSIRDY	LSION
														r	rw

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **LSIRDY**: LSI oscillator ready

Set and reset by hardware to indicate when the Low Speed Internal RC oscillator is stable.

This bit needs 3 cycles of **lsi_ck** clock to fall down after LSION has been set to '0'.

This bit can be set even when LSION is not enabled if there is a request for LSI clock by the Clock Security System on LSE or by the Low Speed Watchdog or by the RTC.

0: LSI clock is not ready (default after reset)

1: LSI clock is ready

Bit 0 **LSION**: LSI oscillator enable

Set and reset by software.

0: LSI is OFF (default after reset)

1: LSI is ON

8.7.28 RCC AHB3 Reset Register (RCC_AHB3RSTR)

Address offset: 0x07C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC1RST
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	QSPIRST	Res.	FMCRST	Res.	Res.	Res.	Res.	Res.	Res.	JPGDECRST	DMA2DRST	Res.	Res.	Res.	MDMARST
	rw		rw							rw	rw				rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **SDMMC1RST**: SDMMC1 and SDMMC1 delay block reset

Set and reset by software.

0: does not reset SDMMC1 and SDMMC1 Delay block (default after reset)

1: resets SDMMC1 and SDMMC1 Delay block

Bit 15 Reserved, must be kept at reset value.

Bit 14 **QSPIRST**: QUADSPI and QUADSPI delay block reset

Set and reset by software.

0: does not reset QUADSPI and QUADSPI Delay block (default after reset)

1: resets QUADSPI and QUADSPI Delay block

Bit 13 Reserved, must be kept at reset value.

Bit 12 **FMCRST**: FMC block reset

Set and reset by software.

0: does not reset FMC block (default after reset)

1: resets FMC block

Bits 11:6 Reserved, must be kept at reset value.

Bit 5 **JPGDECRST**: JPGDEC block reset

Set and reset by software.

0: does not reset JPGDEC block (default after reset)

1: resets JPGDEC block

Bit 4 **DMA2DRST**: DMA2D block reset

Set and reset by software.

0: does not reset DMA2D block (default after reset)

1: resets DMA2D block

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **MDMARST**: MDMA block reset

Set and reset by software.

0: does not reset MDMA block (default after reset)

1: resets MDMA block

8.7.29 RCC AHB1 Peripheral Reset Register(RCC_AHB1RSTR)

Address offset: 0x080

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	USB2OTGRST	Res.	USB1OTGRST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
				rw		rw									
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETH1MACRST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ADC12RST	Res.	Res.	Res.	DMA2RST	DMA1RST
rw										rw				rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bit 27 **USB2OTGRST**: USB2OTG (OTG_HS2) block reset

Set and reset by software.

0: does not reset USB2OTG block (default after reset)

1: resets USB2OTG block

Bit 26 Reserved, must be kept at reset value.

Bit 25 **USB1OTGRST**: USB1OTG (OTG_HS1) block reset

Set and reset by software.

0: does not reset USB1OTG block (default after reset)

1: resets USB1OTG block

Bits 24:16 Reserved, must be kept at reset value.

Bit 15 **ETH1MACRST**: ETH1MAC block reset

Set and reset by software.

0: does not reset ETH1MAC block (default after reset)

1: resets ETH1MAC block

Bits 14:6 Reserved, must be kept at reset value.

Bit 5 **ADC12RST**: ADC1 and 2 block reset

Set and reset by software.

0: does not reset ADC1 and 2 block (default after reset)

1: resets ADC1 and 2 block

Bits 4:2 Reserved, must be kept at reset value.

Bit 1 **DMA2RST**: DMA2 block reset

Set and reset by software.

0: does not reset DMA2 block (default after reset)

1: resets DMA2 block

Bit 0 **DMA1RST**: DMA1 block reset

Set and reset by software.

0: does not reset DMA1 block (default after reset)

1: resets DMA1 block

8.7.30 RCC AHB2 Peripheral Reset Register (RCC_AHB2RSTR)

Address offset: 0x084

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	SDMMC2RST	Res.	Res.	RNGRST	HASHRST	CRYPTRST	Res.	Res.	Res.	CAMIFRST
						rw			rw	rw	rw				rw

Bits 31:10 Reserved, must be kept at reset value.

Bit 9 **SDMMC2RST**: SDMMC2 and SDMMC2 Delay block reset

Set and reset by software.

0: does not reset SDMMC2 and SDMMC2 Delay block (default after reset)

1: resets SDMMC2 and SDMMC2 Delay block

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **RNGRST**: Random Number Generator block reset

Set and reset by software.

0: does not reset RNG block (default after reset)

1: resets RNG block

Bit 5 **HASHRST**: Hash block reset

Set and reset by software.

0: does not reset hash block (default after reset)

1: resets hash block

Bit 4 **CRYPTRST**: Cryptography block reset

Set and reset by software.

0: does not reset cryptography block (default after reset)

1: resets cryptography block

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **CAMIFRST**: CAMITF block reset

Set and reset by software.

0: does not reset the CAMITF block (default after reset)

1: resets the CAMITF block

8.7.31 RCC AHB4 Peripheral Reset Register (RCC_AHB4RSTR)

Address offset: 0x088

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	HSEMRST	ADC3RST	Res.	Res.	BDMARST	Res.	CRCRST	Res.	Res.	Res.
						rw	rw			rw		rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	GPIOKRST	GPIOJRST	GPIORST	GPIOHRST	GPIOGRST	GPIOFRST	GPIOERST	GPIODRST	GPIOCRST	GPIOBRST	GPIOARST
					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:26 Reserved, must be kept at reset value.

Bit 25 **HSEMRST**: HSEM block reset

Set and reset by software.

0: does not reset the HSEM block (default after reset)

1: resets the HSEM block

Bit 24 **ADC3RST**: ADC3 block reset

Set and reset by software.

0: does not reset the ADC3 block (default after reset)

1: resets the ADC3 block

Bits 23:22 Reserved, must be kept at reset value.

Bit 21 **BDMARST**: BDMA block reset

Set and reset by software.

0: does not reset the BDMA block (default after reset)

1: resets the BDMA block

Bit 20 Reserved, must be kept at reset value.

Bit 19 **CRCRST**: CRC block reset

Set and reset by software.

0: does not reset the CRC block (default after reset)

1: resets the CRC block

Bits 18:11 Reserved, must be kept at reset value.

Bit 10 **GPIOKRST**: GPIOK block reset

Set and reset by software.

0: does not reset the GPIOK block (default after reset)

1: resets the GPIOK block

Bit 9 **GPIOJRST**: GPIOJ block reset

Set and reset by software.

0: does not reset the GPIOJ block (default after reset)

1: resets the GPIOJ block

- Bit 8 **GPIOIRST**: GPIOI block reset
Set and reset by software.
0: does not reset the GPIOI block (default after reset)
1: resets the GPIOI block
- Bit 7 **GPIOHRST**: GPIOH block reset
Set and reset by software.
0: does not reset the GPIOH block (default after reset)
1: resets the GPIOH block
- Bit 6 **GPIOGRST**: GPIOG block reset
Set and reset by software.
0: does not reset the GPIOG block (default after reset)
1: resets the GPIOG block
- Bit 5 **GPIOFRST**: GPIOF block reset
Set and reset by software.
0: does not reset the GPIOF block (default after reset)
1: resets the GPIOF block
- Bit 4 **GPIOERST**: GPIOE block reset
Set and reset by software.
0: does not reset the GPIOE block (default after reset)
1: resets the GPIOE block
- Bit 3 **GPIODRST**: GPIOD block reset
Set and reset by software.
0: does not reset the GPIOD block (default after reset)
1: resets the GPIOD block
- Bit 2 **GPIOCRST**: GPIOC block reset
Set and reset by software.
0: does not reset the GPIOC block (default after reset)
1: resets the GPIOC block
- Bit 1 **GPIOBRST**: GPIOB block reset
Set and reset by software.
0: does not reset the GPIOB block (default after reset)
1: resets the GPIOB block
- Bit 0 **GPIOARST**: GPIOA block reset
Set and reset by software.
0: does not reset the GPIOA block (default after reset)
1: resets the GPIOA block

8.7.32 **RCC APB3 Peripheral Reset Register (RCC_APB3RSTR)**

Address offset: 0x08C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LTD CRST	Res.	Res.	Res.
												rw			

Bits 31:4 Reserved, must be kept at reset value.

- Bit 3 **LTD****CRST**: LTDC block reset
Set and reset by software.
0: does not reset the LTDC block (default after reset)
1: resets the LTDC block

Bits 2:0 Reserved, must be kept at reset value.

8.7.33 RCC APB1 Peripheral Reset Register (RCC_APB1LRSTR)

Address offset: 0x090

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UART8RST	UART7RST	DAC12RST	Res.	CECRST	Res.	Res.	Res.	I2C3RST	I2C2RST	I2C1RST	UART5RST	UART4RST	USART3RST	USART2RST	SPDIFXRST
rw	rw	rw		rw				rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3RST	SPI2RST	Res.	Res.	Res.	Res.	LPTIM1RST	TIM14RST	TIM13RST	TIM12RST	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	TIM2RST
rw	rw					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **UART8RST**: UART8 block reset

Set and reset by software.

0: does not reset the UART8 block (default after reset)

1: resets the UART8 block

Bit 30 **UART7RST**: UART7 block reset

Set and reset by software.

0: does not reset the UART7 block (default after reset)

1: resets the UART7 block

Bit 29 **DAC12RST**: DAC1 and 2 Blocks Reset

Set and reset by software.

0: does not reset the DAC1 and 2 blocks (default after reset)

1: resets the DAC1 and 2 blocks

Bit 28 Reserved, must be kept at reset value.

Bit 27 **CECRST**: HDMI-CEC block reset

Set and reset by software.

0: does not reset the HDMI-CEC block (default after reset)

1: resets the HDMI-CEC block

Bits 26:24 Reserved, must be kept at reset value.

Bit 23 **I2C3RST**: I2C3 block reset

Set and reset by software.

0: does not reset the I2C3 block (default after reset)

1: resets the I2C3 block

Bit 22 **I2C2RST**: I2C2 block reset

Set and reset by software.

0: does not reset the I2C2 block (default after reset)

1: resets the I2C2 block

- Bit 21 **I2C1RST**: I2C1 block reset
Set and reset by software.
0: does not reset the I2C1 block (default after reset)
1: resets the I2C1 block
- Bit 20 **UART5RST**: UART5 block reset
Set and reset by software.
0: does not reset the UART5 block (default after reset)
1: resets the UART5 block
- Bit 19 **UART4RST**: UART4 block reset
Set and reset by software.
0: does not reset the UART4 block (default after reset)
1: resets the UART4 block
- Bit 18 **USART3RST**: USART3 block reset
Set and reset by software.
0: does not reset the USART3 block (default after reset)
1: resets the USART3 block
- Bit 17 **USART2RST**: USART2 block reset
Set and reset by software.
0: does not reset the USART2 block (default after reset)
1: resets the USART2 block
- Bit 16 **SPDIFRXRST**: SPDIFRX block reset
Set and reset by software.
0: does not reset the SPDIFRX block (default after reset)
1: resets the SPDIFRX block
- Bit 15 **SPI3RST**: SPI3 block reset
Set and reset by software.
0: does not reset the SPI3 block (default after reset)
1: resets the SPI3 block
- Bit 14 **SPI2RST**: SPI2 block reset
Set and reset by software.
0: does not reset the SPI2 block (default after reset)
1: resets the SPI2 block
- Bits 13:10 Reserved, must be kept at reset value.
- Bit 9 **LPTIM1RST**: LPTIM1 block reset
Set and reset by software.
0: does not reset the LPTIM1 block (default after reset)
1: resets the LPTIM1 block
- Bit 8 **TIM14RST**: TIM14 block reset
Set and reset by software.
0: does not reset the TIM14 block (default after reset)
1: resets the TIM14 block
- Bit 7 **TIM13RST**: TIM13 block reset
Set and reset by software.
0: does not reset the TIM13 block (default after reset)
1: resets the TIM13 block

- Bit 6 **TIM12RST**: TIM12 block reset
Set and reset by software.
0: does not reset the TIM12 block (default after reset)
1: resets the TIM12 block
- Bit 5 **TIM7RST**: TIM7 block reset
Set and reset by software.
0: does not reset the TIM7 block (default after reset)
1: resets the TIM7 block
- Bit 4 **TIM6RST**: TIM6 block reset
Set and reset by software.
0: does not reset the TIM6 block (default after reset)
1: resets the TIM6 block
- Bit 3 **TIM5RST**: TIM5 block reset
Set and reset by software.
0: does not reset the TIM5 block (default after reset)
1: resets the TIM5 block
- Bit 2 **TIM4RST**: TIM4 block reset
Set and reset by software.
0: does not reset the TIM4 block (default after reset)
1: resets the TIM4 block
- Bit 1 **TIM3RST**: TIM3 block reset
Set and reset by software.
0: does not reset the TIM3 block (default after reset)
1: resets the TIM3 block
- Bit 0 **TIM2RST**: TIM2 block reset
Set and reset by software.
0: does not reset the TIM2 block (default after reset)
1: resets the TIM2 block

8.7.34 RCC APB1 Peripheral Reset Register (RCC_APB1HRSTR)

Address offset: 0x094

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	FDCANRST	Res.	Res.	MDIOSRST	OPAMPRST	Res.	SWPRST	CRSRST	Res.
							rw			rw	rw		rw	rw	

Bits 31:9 Reserved, must be kept at reset value.

Bit 8 **FDCANRST**: FDCAN block reset

Set and reset by software.

0: does not reset the FDCAN block (default after reset)

1: resets the FDCAN block

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **MDIOSRST**: MDIOS block reset

Set and reset by software.

0: does not reset the MDIOS block (default after reset)

1: resets the MDIOS block

Bit 4 **OPAMPRST**: OPAMP block reset

Set and reset by software.

0: does not reset the OPAMP block (default after reset)

1: resets the OPAMP block

Bit 3 Reserved, must be kept at reset value.

Bit 2 **SWPRST**: SWPMI block reset

Set and reset by software.

0: does not reset the SWPMI block (default after reset)

1: resets the SWPMI block

Bit 1 **CRSRST**: Clock Recovery System reset

Set and reset by software.

0: does not reset CRS (default after reset)

1: resets CRS

Bit 0 Reserved, must be kept at reset value.

8.7.35 RCC APB2 Peripheral Reset Register (RCC_APB2RSTR)

Address offset: 0x098

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	HRTIMRST	DFSDM1RST	Res.	Res.	Res.	SAI3RST	SAI2RST	SAI1RST	Res.	SPI5RST	Res.	TIM17RST	TIM16RST	TIM15RST
		rw	rw				rw	rw	rw		rw		rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	SPI4RST	SPI1RST	Res.	Res.	Res.	Res.	Res.	Res.	USART6RST	USART1RST	Res.	Res.	TIM8RST	TIM1RST
		rw	rw							rw	rw			rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **HRTIMRST**: HRTIM block reset

Set and reset by software.

0: does not reset the HRTIM block (default after reset)

1: resets the HRTIM block

Bit 28 **DFSDM1RST**: DFSDM1 block reset

Set and reset by software.

0: does not reset DFSDM1 block (default after reset)

1: resets DFSDM1 block

Bits 27:25 Reserved, must be kept at reset value.

Bit 24 **SAI3RST**: SAI3 block reset

Set and reset by software.

0: does not reset the SAI3 block (default after reset)

1: resets the SAI3 block

Bit 23 **SAI2RST**: SAI2 block reset

Set and reset by software.

0: does not reset the SAI2 block (default after reset)

1: resets the SAI2 block

Bit 22 **SAI1RST**: SAI1 block reset

Set and reset by software.

0: does not reset the SAI1 (default after reset)

1: resets the SAI1

Bit 21 Reserved, must be kept at reset value.

Bit 20 **SPI5RST**: SPI5 block reset

Set and reset by software.

0: does not reset the SPI5 block (default after reset)

1: resets the SPI5 block

Bit 19 Reserved, must be kept at reset value.

Bit 18 **TIM17RST**: TIM17 block reset
Set and reset by software.
0: does not reset the TIM17 block (default after reset)
1: resets the TIM17 block

Bit 17 **TIM16RST**: TIM16 block reset
Set and reset by software.
0: does not reset the TIM16 block (default after reset)
1: resets the TIM16 block

Bit 16 **TIM15RST**: TIM15 block reset
Set and reset by software.
0: does not reset the TIM15 block (default after reset)
1: resets the TIM15 block

Bits 15:14 Reserved, must be kept at reset value.

Bit 13 **SPI4RST**: SPI4 block reset
Set and reset by software.
0: does not reset the SPI4 block (default after reset)
1: resets the SPI4 block

Bit 12 **SPI1RST**: SPI1 block reset
Set and reset by software.
0: does not reset the SPI1 block (default after reset)
1: resets the SPI1 block

Bits 11:6 Reserved, must be kept at reset value.

Bit 5 **USART6RST**: USART6 block reset
Set and reset by software.
0: does not reset the USART6 block (default after reset)
1: resets the USART6 block

Bit 4 **USART1RST**: USART1 block reset
Set and reset by software.
0: does not reset the USART1 block (default after reset)
1: resets the USART1 block

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8RST**: TIM8 block reset
Set and reset by software.
0: does not reset the TIM8 block (default after reset)
1: resets the TIM8 block

Bit 0 **TIM1RST**: TIM1 block reset
Set and reset by software.
0: does not reset the TIM1 block (default after reset)
1: resets the TIM1 block

8.7.36 RCC APB4 Peripheral Reset Register (RCC_APB4RSTR)

Address offset: 0x09C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SAI4RST	Res.	Res.	Res.	Res.	Res.
										rw					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VREFRST	COMP12RST	Res.	LPTIM5RST	LPTIM4RST	LPTIM3RST	LPTIM2RST	Res.	I2C4RST	Res.	SPI6RST	Res.	LPUART1RST	Res.	SYSCFGRST	Res.
rw	rw		rw	rw	rw	rw		rw		rw		rw		rw	

Bits 31:26 Reserved, must be kept at reset value.

Bits 25:22 Reserved, must be kept at reset value.

Bit 21 **SAI4RST**: SAI4 block reset

Set and reset by software.

0: does not reset the SAI4 block (default after reset)

1: resets the SAI4 block

Bits 20:16 Reserved, must be kept at reset value.

Bit 15 **VREFRST**: VREF block reset

Set and reset by software.

0: does not reset the VREF block (default after reset)

1: resets the VREF block

Bit 14 **COMP12RST**: COMP12 Blocks Reset

Set and reset by software.

0: does not reset the COMP1 and 2 blocks (default after reset)

1: resets the COMP1 and 2 blocks

Bit 13 Reserved, must be kept at reset value.

Bit 12 **LPTIM5RST**: LPTIM5 block reset

Set and reset by software.

0: does not reset the LPTIM5 block (default after reset)

1: resets the LPTIM5 block

Bit 11 **LPTIM4RST**: LPTIM4 block reset

Set and reset by software.

0: does not reset the LPTIM4 block (default after reset)

1: resets the LPTIM4 block

Bit 10 **LPTIM3RST**: LPTIM3 block reset

Set and reset by software.

0: does not reset the LPTIM3 block (default after reset)

1: resets the LPTIM3 block

- Bit 9 **LPTIM2RST**: LPTIM2 block reset
Set and reset by software.
0: does not reset the LPTIM2 block (default after reset)
1: resets the LPTIM2 block
- Bit 8 Reserved, must be kept at reset value.
- Bit 7 **I2C4RST**: I2C4 block reset
Set and reset by software.
0: does not reset the I2C4 block (default after reset)
1: resets the I2C4 block
- Bit 6 Reserved, must be kept at reset value.
- Bit 5 **SPI6RST**: SPI6 block reset
Set and reset by software.
0: does not reset the SPI6 block (default after reset)
1: resets the SPI6 block
- Bit 4 Reserved, must be kept at reset value.
- Bit 3 **LPUART1RST**: LPUART1 block reset
Set and reset by software.
0: does not reset the LPUART1 block (default after reset)
1: resets the LPUART1 block
- Bit 2 Reserved, must be kept at reset value.
- Bit 1 **SYSCFGRST**: SYSCFG block reset
Set and reset by software.
0: does not reset the SYSCFG block (default after reset)
1: resets the SYSCFG block
- Bit 0 Reserved, must be kept at reset value.

8.7.37 RCC Global Control Register (RCC_GCR)

Address offset: 0x0A0

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	WW1RSC
															rw1

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **WW1RSC**: WWDG1 reset scope control

This bit can be set by software but is cleared by hardware during a system reset
In order to work properly, before enabling the WWDG1, this bit must be set to '1'.

8.7.38 RCC D3 Autonomous mode Register (RCC_D3AMR)

The Autonomous mode allows providing the peripheral clocks to peripherals located in D3, even if the CPU is in CStop mode. When a peripheral is enabled, and has its autonomous bit enabled, it receives its peripheral clocks according to D3 domain state, if the CPU is in CStop mode.

Address offset: 0x0A8

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	SRAM4AMEN	BKPRAMAMEN	Res.	Res.	Res.	ADC3AMEN	Res.	Res.	SAI4AMEN	Res.	CRCAMEN	Res.	Res.	RTCAMEN
		rw	rw				rw			rw		rw			rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VREFAMEN	COMP12AMEN	Res.	LPTIM3AMEN	LPTIM4AMEN	LPTIM3AMEN	LPTIM2AMEN	Res.	I2C4AMEN	Res.	SPI6AMEN	Res.	LPUART1AMEN	Res.	Res.	BDMAAMEN
rw	rw		rw	rw	rw	rw		rw		rw		rw			rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **SRAM4AMEN**: SRAM4 Autonomous mode enable

Set and reset by software.

0: SRAM4 clock is disabled when the CPU is in CStop (default after reset)

1: SRAM4 peripheral bus clock enabled when D3 domain is in DRun.

Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information

Bit 28 **BKPRAMAMEN**: Backup RAM Autonomous mode enable

Set and reset by software.

0: Backup RAM clock is disabled when the CPU is in CStop (default after reset)

1: Backup RAM clock enabling is controlled by D3 domain state.

Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information

Bit 27 Reserved, must be kept at reset value.

Bits 26:25 Reserved, must be kept at reset value.

Bit 24 **ADC3AMEN**: ADC3 Autonomous mode enable

Set and reset by software.

0: ADC3 peripheral clocks are disabled when the CPU is in CStop (default after reset)

1: ADC3 peripheral clocks enabled when D3 domain is in DRun.

Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information

Bits 23:22 Reserved, must be kept at reset value.

Bit 21 **SAI4AMEN**: SAI4 Autonomous mode enable

Set and reset by software.

0: SAI4 peripheral clocks are disabled when the CPU is in CStop (default after reset)

1: SAI4 peripheral clocks enabled when D3 domain is in DRun.

Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information

Bit 20 Reserved, must be kept at reset value.

- Bit 19 **CRCAMEN**: CRC Autonomous mode enable
Set and reset by software.
0: CRC peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: CRC peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bits 18:17 Reserved, must be kept at reset value.
- Bit 16 **RTCAMEN**: RTC Autonomous mode enable
Set and reset by software.
0: RTC peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: RTC peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bit 15 **VREFAMEN**: VREF Autonomous mode enable
Set and reset by software.
0: VREF peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: VREF peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bit 14 **COMP12AMEN**: COMP12 Autonomous mode enable
Set and reset by software.
0: COMP12 peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: COMP12 peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bit 13 Reserved, must be kept at reset value.
- Bit 12 **LPTIM5AMEN**: LPTIM5 Autonomous mode enable
Set and reset by software.
0: LPTIM5 peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: LPTIM5 peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bit 11 **LPTIM4AMEN**: LPTIM4 Autonomous mode enable
Set and reset by software.
0: LPTIM4 peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: LPTIM4 peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bit 10 **LPTIM3AMEN**: LPTIM3 Autonomous mode enable
Set and reset by software.
0: LPTIM3 peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: LPTIM3 peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bit 9 **LPTIM2AMEN**: LPTIM2 Autonomous mode enable
Set and reset by software.
0: LPTIM2 peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: LPTIM2 peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bit 8 Reserved, must be kept at reset value.
- Bit 7 **I2C4AMEN**: I2C4 Autonomous mode enable
Set and reset by software.
0: I2C4 peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: I2C4 peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information

- Bit 6 Reserved, must be kept at reset value.
- Bit 5 **SPI6AMEN**: SPI6 Autonomous mode enable
Set and reset by software.
0: SPI6 peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: SPI6 peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bit 4 Reserved, must be kept at reset value.
- Bit 3 **LPUART1AMEN**: LPUART1 Autonomous mode enable
Set and reset by software.
0: LPUART1 peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: LPUART1 peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information
- Bits 2:1 Reserved, must be kept at reset value.
- Bit 0 **BDMAAMEN**: BDMA and DMAMUX Autonomous mode enable
Set and reset by software.
0: BDMA and DMAMUX peripheral clocks are disabled when the CPU is in CStop (default after reset)
1: BDMA and DMAMUX peripheral clocks enabled when D3 domain is in DRun.
Refer to [Section 8.5.11: Peripheral clock gating control](#) for additional information

8.7.39 RCC Reset Status Register (RCC_RSR)

This register can be accessed via two different offset address.

Table 64. RCC_RSR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_RSR	0x0D0	0x00FE 0000 ⁽¹⁾
RCC_C1_RSR	0x130	

1. Reset by power-on reset only

Access: $0 \leq \text{wait state} \leq 7$, word, half-word and byte access. Wait states are inserted in case of successive accesses to this register.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	LPWRRSTF	Res.	WWDG1RSTF	Res.	IWDG1RSTF	Res.	SFTRSTF	PORRSTF	PINRSTF	BORRSTF	D2RSTF	D1RSTF	Res.	CPURSTF	RMVF
	r		r		r		r	r	r	r	r	r		r	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bit 31 Reserved, must be kept at reset value.

Bit 30 **LPWRRSTF**: Reset due to illegal D1 DStandby or CPU CStop flag ⁽¹⁾

Reset by software by writing the RMVF bit.

Set by hardware when D1 domain goes erroneously in DStandby or when CPU goes erroneously in CStop.

0: No illegal reset occurred (default after power-on reset)

1: Illegal D1 DStandby or CPU CStop reset occurred

Bit 29 Reserved, must be kept at reset value.

Bit 28 **WWDG1RSTF**: Window Watchdog reset flag ⁽¹⁾

Reset by software by writing the RMVF bit.

Set by hardware when a window watchdog reset occurs.

0: No window watchdog reset occurred from WWDG1 (default after power-on reset)

1: window watchdog reset occurred from WWDG1

Bit 27 Reserved, must be kept at reset value.

Bit 26 **IWDG1RSTF**: Independent Watchdog reset flag ⁽¹⁾

Reset by software by writing the RMVF bit.

Set by hardware when an independent watchdog reset occurs.

0: No independent watchdog reset occurred (default after power-on reset)

1: Independent watchdog reset occurred

Bit 25 Reserved, must be kept at reset value.

- Bit 24 **SFTRSTF**: System reset from CPU reset flag ⁽¹⁾
 Reset by software by writing the RMVF bit.
 Set by hardware when the system reset is due to CPU. The CPU can generate a system reset by writing SYSRESETREQ bit of AIRCR register of the CM7.
 0: No CPU software reset occurred (default after power-on reset)
 1: A system reset has been generated by the CPU
- Bit 23 **PORRSTF**: POR/PDR reset flag ⁽¹⁾
 Reset by software by writing the RMVF bit.
 Set by hardware when a POR/PDR reset occurs.
 0: No POR/PDR reset occurred
 1: POR/PDR reset occurred (default after power-on reset)
- Bit 22 **PINRSTF**: Pin reset flag (NRST) ⁽¹⁾
 Reset by software by writing the RMVF bit.
 Set by hardware when a reset from pin occurs.
 0: No reset from pin occurred
 1: Reset from pin occurred (default after power-on reset)
- Bit 21 **BORRSTF**: BOR reset flag ⁽¹⁾
 Reset by software by writing the RMVF bit.
 Set by hardware when a BOR reset occurs (**pwr_bor_rst**).
 0: No BOR reset occurred
 1: BOR reset occurred (default after power-on reset)
- Bit 20 **D2RSTF**: D2 domain power switch reset flag ⁽¹⁾
 Reset by software by writing the RMVF bit.
 Set by hardware when a D2 domain exits from DStandby or after of power-on reset. Refer to [Table 55](#) for details.
 0: No D2 domain power switch reset occurred
 1: A D2 domain power switch (ePOD2) reset occurred (default after power-on reset)
- Bit 19 **D1RSTF**: D1 domain power switch reset flag ⁽¹⁾
 Reset by software by writing the RMVF bit.
 Set by hardware when a D1 domain exits from DStandby or after of power-on reset. Refer to [Table 55](#) for details.
 0: No D1 domain power switch reset occurred
 1: A D1 domain power switch (ePOD1) reset occurred (default after power-on reset)
- Bit 18 Reserved, must be kept at reset value.
- Bit 17 **CPURSTF**: CPU reset flag ⁽¹⁾
 Reset by software by writing the RMVF bit.
 Set by hardware every time a CPU reset occurs.
 0: No CPU reset occurred
 1: A CPU reset occurred (default after power-on reset)
- Bit 16 **RMVF**: Remove reset flag
 Set and reset by software to reset the value of the reset flags.
 0: Reset of the reset flags not activated (default after power-on reset)
 1: Reset the value of the reset flags
- Bits 15:0 Reserved, must be kept at reset value.

1. Refer to [Table 55: Reset source identification \(RCC_RSR\)](#) for details on flag behavior.

8.7.40 RCC AHB3 Clock Register (RCC_AHB3ENR)

This register can be accessed via two different offset address.

Table 65. RCC_AHB3ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_AHB3ENR	0x0D4	0x0000 0000
RCC_C1_AHB3ENR	0x134	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC1EN
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	QSPIEN	Res.	FMCEN	Res.	Res.	Res.	Res.	Res.	Res.	JPGDECEN	DMA2DEN	Res.	Res.	Res.	MDMAEN
	rw		rw							rw	rw				rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **SDMMC1EN**: SDMMC1 and SDMMC1 Delay Clock Enable

Set and reset by software.

0: SDMMC1 and SDMMC1 Delay clock disabled (default after reset)

1: SDMMC1 and SDMMC1 Delay clock enabled

Bit 15 Reserved, must be kept at reset value.

Bit 14 **QSPIEN**: QUADSPI and QUADSPI Delay Clock Enable

Set and reset by software.

0: QUADSPI and QUADSPI Delay clock disabled (default after reset)

1: QUADSPI and QUADSPI Delay clock enabled

Bit 13 Reserved, must be kept at reset value.

Bit 12 **FMCEN**: FMC Peripheral Clocks Enable

Set and reset by software.

0: FMC peripheral clocks disabled (default after reset)

1: FMC peripheral clocks enabled

The peripheral clocks of the FMC are: the kernel clock selected by FMCSEL and provided to `fmc_ker_ck` input, and the `rcc_hclk3` bus interface clock.

Bits 11:6 Reserved, must be kept at reset value.

Bit 5 **JPGDECEN**: JPGDEC Peripheral Clock Enable

Set and reset by software.

0: JPGDEC peripheral clock disabled (default after reset)

1: JPGDEC peripheral clock enabled

Bit 4 **DMA2DEN**: DMA2D Peripheral Clock Enable

Set and reset by software.

0: DMA2D peripheral clock disabled (default after reset)

1: DMA2D peripheral clock enabled

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **MDMAEN**: MDMA Peripheral Clock Enable

Set and reset by software.

0: MDMA peripheral clock disabled (default after reset)

1: MDMA peripheral clock enabled

8.7.41 RCC AHB1 Clock Register (RCC_AHB1ENR)

This register can be accessed via two different offset address.

Table 66. RCC_AHB1ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_AHB1ENR	0x0D8	0x0000 0000
RCC_C1_AHB1ENR	0x138	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	USB2OTGHSEN	USB1OTGHSULPIEN	USB1OTGHSEN	Res.	Res.	Res.	Res.	Res.	Res.	USB2OTGHSULPIEN	ETH1RXEN	ETH1TXEN
				rw	rw	rw							rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETH1MACEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ADC12EN	Res.	Res.	Res.	DMA2EN	DMA1EN
rw										rw				rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bit 27 **USB2OTGHSEN**: USB2OTG (OTG_HS2) Peripheral Clocks Enable

Set and reset by software.

0: USB2OTG peripheral clocks disabled (default after reset)

1: USB2OTG peripheral clocks enabled

The peripheral clocks of the USB2OTG are: the kernel clock selected by USBSEL and the **rcc_hclk1** bus interface clock.

Bit 26 **USB1OTGHSULPIEN**: USB_PHY1 Clocks Enable

Set and reset by software.

0: USB1ULPI PHY clocks disabled (default after reset)

1: USB1ULPI PHY clocks enabled

Bit 25 **USB1OTGHSEN**: USB1OTG (OTG_HS1) Peripheral Clocks Enable

Set and reset by software.

0: USB1OTG peripheral clocks disabled (default after reset)

1: USB1OTG peripheral clocks enabled

The peripheral clocks of the USB1OTG are: the kernel clock selected by USBSEL and the **rcc_hclk1** bus interface clock.

Bits 24:19 Reserved, must be kept at reset value.

Bit 18 **USB2OTGHSULPIEN**: Enable USB_PHY2 clocks

Set and reset by software.

0: USB2ULPI PHY clocks disabled (default after reset)

1: USB2ULPI PHY clocks enabled

Bit 17 **ETH1RXEN**: Ethernet Reception Clock Enable

Set and reset by software.

0: Ethernet Reception clock disabled (default after reset)

1: Ethernet Reception clock enabled

Bit 16 **ETH1TXEN**: Ethernet Transmission Clock Enable

Set and reset by software.

0: Ethernet Transmission clock disabled (default after reset)

1: Ethernet Transmission clock enabled

Bit 15 **ETH1MACEN**: Ethernet MAC bus interface Clock Enable

Set and reset by software.

0: Ethernet MAC bus interface clock disabled (default after reset)

1: Ethernet MAC bus interface clock enabled

Bits 14:6 Reserved, must be kept at reset value.

Bit 5 **ADC12EN**: ADC1/2 Peripheral Clocks Enable

Set and reset by software.

0: ADC1 and 2 peripheral clocks disabled (default after reset)

1: ADC1 and 2 peripheral clocks enabled

The peripheral clocks of the ADC1 and 2 are: the kernel clock selected by ADCSEL and provided to `adc_ker_ck_input`, and the **rcc_hclk1** bus interface clock.

Bits 4:2 Reserved, must be kept at reset value.

Bit 1 **DMA2EN**: DMA2 Clock Enable

Set and reset by software.

0: DMA2 clock disabled (default after reset)

1: DMA2 clock enabled

Bit 0 **DMA1EN**: DMA1 Clock Enable

Set and reset by software.

0: DMA1 clock disabled (default after reset)

1: DMA1 clock enabled

8.7.42 RCC AHB2 Clock Register (RCC_AHB2ENR)

This register can be accessed via two different offset address.

Table 67. RCC_AHB2ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_AHB2ENR	0x0DC	0x0000 0000
RCC_C1_AHB2ENR	0x13C	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SRAM3EN	SRAM2EN	SRAM1EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw	rw	rw													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	SDMMC2EN	Res.	Res.	RNGEN	HASHEN	CRYPTEN	Res.	Res.	Res.	DCMIEN
						rw			rw	rw	rw				rw

Bit 31 SRAM3EN: SRAM3 block enable

Set and reset by software.

When set, this bit indicates that the SRAM3 is allocated by the CPU. It causes the D2 domain to take into account also the CPU operation modes, i.e. keeping D2 domain in DRun when the CPU is in CRun.

0: SRAM3 interface clock is disabled. (default after reset)

1: SRAM3 interface clock is enabled.

Bit 30 SRAM2EN: SRAM2 block enable

Set and reset by software.

When set, this bit indicates that the SRAM2 is allocated by the CPU. It causes the D2 domain to take into account also the CPU operation modes, i.e. keeping D2 domain in DRun when the CPU is in CRun.

0: SRAM2 interface clock is disabled. (default after reset)

1: SRAM2 interface clock is enabled.

Bit 29 SRAM1EN: SRAM1 block enable

Set and reset by software.

When set, this bit indicates that the SRAM1 is allocated by the CPU. It causes the D2 domain to take into account also the CPU operation modes, i.e. keeping D2 domain in DRun when the CPU is in CRun.

0: SRAM1 interface clock is disabled. (default after reset)

1: SRAM1 interface clock is enabled.

Bits 28:10 Reserved, must be kept at reset value.

Bit 9 SDMMC2EN: SDMMC2 and SDMMC2 delay clock enable

Set and reset by software.

0: SDMMC2 and SDMMC2 Delay clock disabled (default after reset)

1: SDMMC2 and SDMMC2 Delay clock enabled

Bits 8:7 Reserved, must be kept at reset value.

- Bit 6 **RNGEN**: RNG peripheral clocks enable
Set and reset by software.
0: RNG peripheral clocks disabled (default after reset)
1: RNG peripheral clocks enabled:
The peripheral clocks of the RNG are: the kernel clock selected by RNGSEL and provided to **rng_ker_ck** input, and the **rcc_hclk2** bus interface clock.
- Bit 5 **HASHEN**: HASH peripheral clock enable
Set and reset by software.
0: HASH peripheral clock disabled (default after reset)
1: HASH peripheral clock enabled
- Bit 4 **CRYPTEN**: CRYPT peripheral clock enable
Set and reset by software.
0: CRYPT peripheral clock disabled (default after reset)
1: CRYPT peripheral clock enabled
- Bits 3:1 Reserved, must be kept at reset value.
- Bit 0 **DCMIEN**: DCMI peripheral clock enable
Set and reset by software.
0: DCMI peripheral clock disabled (default after reset)
1: DCMI peripheral clock enabled

8.7.43 RCC AHB4 Clock Register (RCC_AHB4ENR)

This register can be accessed via two different offset address.

Table 68. RCC_AHB4ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_AHB4ENR	0x0E0	0x0000 0000
RCC_C1_AHB4ENR	0x140	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	BKPRAMEN	Res.	Res.	HSEMEN	ADC3EN	Res.	Res.	BDMAEN	Res.	CRCEN	Res.	Res.	Res.
			rw			rw	rw			rw		rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	GPIOKEN	GPIOJEN	GPIOIEEN	GPIOHEN	GPIOGEN	GPIOFEN	GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN
					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:29 Reserved, must be kept at reset value.

Bit 28 **BKPRAMEN**: Backup RAM Clock Enable

Set and reset by software.

0: Backup RAM clock disabled (default after reset)

1: Backup RAM clock enabled

Bits 27:26 Reserved, must be kept at reset value.

Bit 25 **HSEMEN**: HSEM peripheral clock enable

Set and reset by software.

0: HSEM peripheral clock disabled (default after reset)

1: HSEM peripheral clock enabled

Bit 24 **ADC3EN**: ADC3 Peripheral Clocks Enable

Set and reset by software.

0: ADC3 peripheral clocks disabled (default after reset)

1: ADC3 peripheral clocks enabled

The peripheral clocks of the ADC3 are: the kernel clock selected by ADCSEL and provided to adc_ker_ck_input, and the **rcc_hclk4** bus interface clock.

Bits 23:22 Reserved, must be kept at reset value.

Bit 21 **BDMAEN**: BDMA and DMAMUX2 Clock Enable

Set and reset by software.

0: BDMA and DMAMUX2 clock disabled (default after reset)

1: BDMA and DMAMUX2 clock enabled

Bit 20 Reserved, must be kept at reset value.

Bit 19 **CRCCEN**: CRC peripheral clock enable

Set and reset by software.

0: CRC peripheral clock disabled (default after reset)

1: CRC peripheral clock enabled

Bits 18:11 Reserved, must be kept at reset value.

Bit 10 **GPIOKEN**: GPIOK peripheral clock enable

Set and reset by software.

0: GPIOK peripheral clock disabled (default after reset)

1: GPIOK peripheral clock enabled

Bit 9 **GPIOJEN**: GPIOJ peripheral clock enable

Set and reset by software.

0: GPIOJ peripheral clock disabled (default after reset)

1: GPIOJ peripheral clock enabled

Bit 8 **GPIOIEEN**: GPIOI peripheral clock enable

Set and reset by software.

0: GPIOI peripheral clock disabled (default after reset)

1: GPIOI peripheral clock enabled

Bit 7 **GPIOHEN**: GPIOH peripheral clock enable

Set and reset by software.

0: GPIOH peripheral clock disabled (default after reset)

1: GPIOH peripheral clock enabled

Bit 6 **GPIOGEN**: GPIOG peripheral clock enable

Set and reset by software.

0: GPIOG peripheral clock disabled (default after reset)

1: GPIOG peripheral clock enabled

Bit 5 **GPIOFEN**: GPIOF peripheral clock enable

Set and reset by software.

0: GPIOF peripheral clock disabled (default after reset)

1: GPIOF peripheral clock enabled

Bit 4 **GPIOEEN**: GPIOE peripheral clock enable

Set and reset by software.

0: GPIOE peripheral clock disabled (default after reset)

1: GPIOE peripheral clock enabled

Bit 3 **GPIODEN**: GPIOD peripheral clock enable

Set and reset by software.

0: GPIOD peripheral clock disabled (default after reset)

1: GPIOD peripheral clock enabled

- Bit 2 **GPIOCEN**: GPIOC peripheral clock enable
Set and reset by software.
0: GPIOC peripheral clock disabled (default after reset)
1: GPIOC peripheral clock enabled
- Bit 1 **GPIOBEN**: GPIOB peripheral clock enable
Set and reset by software.
0: GPIOB peripheral clock disabled (default after reset)
1: GPIOB peripheral clock enabled
- Bit 0 **GPIOAEN**: GPIOA peripheral clock enable
Set and reset by software.
0: GPIOA peripheral clock disabled (default after reset)
1: GPIOA peripheral clock enabled

8.7.44 RCC APB3 Clock Register (RCC_APB3ENR)

This register can be accessed via two different offset address.

Table 69. RCC_APB3ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB3ENR	0x0E4	0x0000 0000
RCC_C1_APB3ENR	0x144	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	WWDG1EN	Res.	Res.	LTDCCEN	Res.	Res.	Res.
									rs			rw			

Bits 31:7 Reserved, must be kept at reset value.

Bit 6 **WWDG1EN**: WWDG1 Clock Enable

Set by software, and reset by hardware when a system reset occurs.

Note that in order to work properly, before enabling the WWDG1, the bit WW1RSC must be set to '1'.

0: WWDG1 peripheral clock disable (default after reset)

1: WWDG1 peripheral clock enabled

Bits 5:4 Reserved, must be kept at reset value.

Bit 3 **LTDCCEN**: LTDC peripheral clock enable

Provides the pixel clock (**ltdc_ker_ck**) to the LTDC block.

Set and reset by software.

0: LTDC peripheral clock disabled (default after reset)

1: LTDC peripheral clock provided to the LTDC block

Bits 2:0 Reserved, must be kept at reset value.

8.7.45 RCC APB1 Clock Register (RCC_APB1LENR)

This register can be accessed via two different offset address.

Table 70. RCC_APB1ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB1LENR	0x0E8	0x0000 0000
RCC_C1_APB1LENR	0x148	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UART8EN	UART7EN	DAC12EN	Res.	CECEN	Res.	Res.	Res.	I2C3EN	I2C2EN	I2C1EN	UART5EN	UART4EN	USART3EN	USART2EN	SPDIFRXEN
rw	rw	rw		rw				rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3EN	SPI2EN	Res.	Res.	Res.	Res.	LPTIM1EN	TIM14EN	TIM13EN	TIM12EN	TIM7EN	TIM6EN	TIM5EN	TIM4EN	TIM3EN	TIM2EN
rw	rw					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 UART8EN: UART8 Peripheral Clocks Enable

Set and reset by software.

0: UART8 peripheral clocks disable (default after reset)

1: UART8 peripheral clocks enabled

The peripheral clocks of the UART8 are: the kernel clock selected by USART234578SEL and provided to usart_ker_ck input, and the **rcc_pclk1** bus interface clock.

Bit 30 UART7EN: UART7 Peripheral Clocks Enable

Set and reset by software.

0: UART7 peripheral clocks disable (default after reset)

1: UART7 peripheral clocks enabled

The peripheral clocks of the UART7 are: the kernel clock selected by USART234578SEL and provided to usart_ker_ck input, and the **rcc_pclk1** bus interface clock.

Bit 29 DAC12EN: DAC1 and 2 peripheral clock enable

Set and reset by software.

0: DAC1 and 2 peripheral clock disable (default after reset)

1: DAC1 and 2 peripheral clock enabled

Bit 28 Reserved, must be kept at reset value.

Bit 27 CECEN: HDMI-CEC peripheral clock enable

Set and reset by software.

0: HDMI-CEC peripheral clock disable (default after reset)

1: HDMI-CEC peripheral clock enabled

The peripheral clocks of the HDMI-CEC are: the kernel clock selected by CECSEL and provided to **cec_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bits 26:24 Reserved, must be kept at reset value.

- Bit 23 **I2C3EN**: I2C3 Peripheral Clocks Enable
Set and reset by software.
0: I2C3 peripheral clocks disable (default after reset)
1: I2C3 peripheral clocks enabled
The peripheral clocks of the I2C3 are: the kernel clock selected by I2C123SEL and provided to **i2c_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 22 **I2C2EN**: I2C2 Peripheral Clocks Enable
Set and reset by software.
0: I2C2 peripheral clocks disable (default after reset)
1: I2C2 peripheral clocks enabled
The peripheral clocks of the I2C2 are: the kernel clock selected by I2C123SEL and provided to **i2c_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 21 **I2C1EN**: I2C1 Peripheral Clocks Enable
Set and reset by software.
0: I2C1 peripheral clocks disable (default after reset)
1: I2C1 peripheral clocks enabled
The peripheral clocks of the I2C1 are: the kernel clock selected by I2C123SEL and provided to **i2c_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 20 **UART5EN**: UART5 Peripheral Clocks Enable
Set and reset by software.
0: UART5 peripheral clocks disable (default after reset)
1: UART5 peripheral clocks enabled
The peripheral clocks of the UART5 are: the kernel clock selected by USART234578SEL and provided to **uart_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 19 **UART4EN**: UART4 Peripheral Clocks Enable
Set and reset by software.
0: UART4 peripheral clocks disable (default after reset)
1: UART4 peripheral clocks enabled
The peripheral clocks of the UART4 are: the kernel clock selected by USART234578SEL and provided to **uart_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 18 **USART3EN**: USART3 Peripheral Clocks Enable
Set and reset by software.
0: USART3 peripheral clocks disable (default after reset)
1: USART3 peripheral clocks enabled
The peripheral clocks of the USART3 are: the kernel clock selected by USART234578SEL and provided to **usart_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 17 **USART2EN**: USART2 Peripheral Clocks Enable
Set and reset by software.
0: USART2 peripheral clocks disable (default after reset)
1: USART2 peripheral clocks enabled
The peripheral clocks of the USART2 are: the kernel clock selected by USART234578SEL and provided to **usart_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 16 **SPDIFRXEN**: SPDIFRX Peripheral Clocks Enable
Set and reset by software.
0: SPDIFRX peripheral clocks disable (default after reset)
1: SPDIFRX peripheral clocks enabled
The peripheral clocks of the SPDIFRX are: the kernel clock selected by SPDIFSEL and provided to **spdifrx_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 15 SPI3EN: SPI3 Peripheral Clocks Enable

Set and reset by software.

0: SPI3 peripheral clocks disable (default after reset)

1: SPI3 peripheral clocks enabled

The peripheral clocks of the SPI3 are: the kernel clock selected by I2S123SRC and provided to **spi_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 14 SPI2EN: SPI2 Peripheral Clocks Enable

Set and reset by software.

0: SPI2 peripheral clocks disable (default after reset)

1: SPI2 peripheral clocks enabled

The peripheral clocks of the SPI2 are: the kernel clock selected by I2S123SRC and provided to **spi_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bits 13:10 Reserved, must be kept at reset value.

Bit 9 LPTIM1EN: LPTIM1 Peripheral Clocks Enable

Set and reset by software.

0: LPTIM1 peripheral clocks disable (default after reset)

1: LPTIM1 peripheral clocks enabled

The peripheral clocks of the LPTIM1 are: the kernel clock selected by LPTIM1SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 8 TIM14EN: TIM14 peripheral clock enable

Set and reset by software.

0: TIM14 peripheral clock disable (default after reset)

1: TIM14 peripheral clock enabled

Bit 7 TIM13EN: TIM13 peripheral clock enable

Set and reset by software.

0: TIM13 peripheral clock disable (default after reset)

1: TIM13 peripheral clock enabled

Bit 6 TIM12EN: TIM12 peripheral clock enable

Set and reset by software.

0: TIM12 peripheral clock disable (default after reset)

1: TIM12 peripheral clock enabled

Bit 5 TIM7EN: TIM7 peripheral clock enable

Set and reset by software.

0: TIM7 peripheral clock disable (default after reset)

1: TIM7 peripheral clock enabled

Bit 4 TIM6EN: TIM6 peripheral clock enable

Set and reset by software.

0: TIM6 peripheral clock disable (default after reset)

1: TIM6 peripheral clock enabled

Bit 3 TIM5EN: TIM5 peripheral clock enable

Set and reset by software.

0: TIM5 peripheral clock disable (default after reset)

1: TIM5 peripheral clock enabled

- Bit 2 **TIM4EN**: TIM4 peripheral clock enable
Set and reset by software.
0: TIM4 peripheral clock disable (default after reset)
1: TIM4 peripheral clock enabled
- Bit 1 **TIM3EN**: TIM3 peripheral clock enable
Set and reset by software.
0: TIM3 peripheral clock disable (default after reset)
1: TIM3 peripheral clock enabled
- Bit 0 **TIM2EN**: TIM2 peripheral clock enable
Set and reset by software.
0: TIM2 peripheral clock disable (default after reset)
1: TIM2 peripheral clock enabled

8.7.46 RCC APB1 Clock Register (RCC_APB1HENR)

This register can be accessed via two different offset address.

Table 71. RCC_APB1ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB1HENR	0x0EC	0x0000 0000
RCC_C1_APB1HENR	0x14C	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	FDCANEN	Res.	Res.	MDIOSEN	OPAMPEN	Res.	SWPEN	CRSEN	Res.
							rw			rw	rw		rw	rw	

Bits 31:9 Reserved, must be kept at reset value.

Bit 8 FDCANEN: FDCAN Peripheral Clocks Enable

Set and reset by software.

0: FDCAN peripheral clocks disable (default after reset)

1: FDCAN peripheral clocks enabled:

The peripheral clocks of the FDCAN are: the kernel clock selected by FDCANSEL and provided to **fdcan_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 MDIOSEN: MDIOS peripheral clock enable

Set and reset by software.

0: MDIOS peripheral clock disable (default after reset)

1: MDIOS peripheral clock enabled

Bit 4 OPAMPEN: OPAMP peripheral clock enable

Set and reset by software.

0: OPAMP peripheral clock disable (default after reset)

1: OPAMP peripheral clock enabled

Bit 3 Reserved, must be kept at reset value.

Bit 2 **SWPEN**: SWPMI Peripheral Clocks Enable

Set and reset by software.

0: SWPMI peripheral clocks disable (default after reset)

1: SWPMI peripheral clocks enabled:

The peripheral clocks of the SWPMI are: the kernel clock selected by SWPSEL and provided to

swpmi_ker_ck input, and the **rcc_pclk1** bus interface clock.

Bit 1 **CRSEN**: Clock Recovery System peripheral clock enable

Set and reset by software.

0: CRS peripheral clock disable (default after reset)

1: CRS peripheral clock enabled

Bit 0 Reserved, must be kept at reset value.

8.7.47 RCC APB2 Clock Register (RCC_APB2ENR)

This register can be accessed via two different offset address.

Table 72. RCC_APB2ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB2ENR	0x0F0	0x0000 0000
RCC_C1_APB2ENR	0x150	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	HRTIMEN	DFSDM1EN	Res.	Res.	Res.	SAI3EN	SAI2EN	SAI1EN	Res.	SPI5EN	Res.	TIM17EN	TIM16EN	TIM15EN
		rw	rw				rw	rw	rw		rw		rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	SPI4EN	SPI1EN	Res.	Res.	Res.	Res.	Res.	Res.	USART6EN	USART1EN	Res.	Res.	TIM8EN	TIM1EN
		rw	rw							rw	rw			rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **HRTIMEN**: HRTIM peripheral clock enable

Set and reset by software.

0: HRTIM peripheral clock disabled (default after reset)

1: HRTIM peripheral clock enabled

Bit 28 **DFSDM1EN**: DFSDM1 Peripheral Clocks Enable

Set and reset by software.

0: DFSDM1 peripheral clocks disabled (default after reset)

1: DFSDM1 peripheral clocks enabled

DFSDM1 peripheral clocks are: the kernel clocks selected by SAI1SEL and DFSDM1SEL and provided to **Aclk** and **clk** inputs respectively, and the **rcc_pclk2** bus interface clock.

Bits 27:25 Reserved, must be kept at reset value.

Bit 24 **SAI3EN**: SAI3 Peripheral Clocks Enable

Set and reset by software.

0: SAI3 peripheral clocks disabled (default after reset)

1: SAI3 peripheral clocks enabled

The peripheral clocks of the SAI3 are: the kernel clock selected by SAI23SEL and provided to **sai_a_ker_ck** and **sai_b_ker_ck** inputs, and the **rcc_pclk2** bus interface clock.

Bit 23 **SAI2EN**: SAI2 Peripheral Clocks Enable

Set and reset by software.

0: SAI2 peripheral clocks disabled (default after reset)

1: SAI2 peripheral clocks enabled

The peripheral clocks of the SAI2 are: the kernel clock selected by SAI23SEL and provided to **sai_a_ker_ck** and **sai_b_ker_ck** inputs, and the **rcc_pclk2** bus interface clock.

- Bit 22 **SAI1EN**: SAI1 Peripheral Clocks Enable
 Set and reset by software.
 0: SAI1 peripheral clocks disabled (default after reset)
 1: SAI1 peripheral clocks enabled:
 The peripheral clocks of the SAI1 are: the kernel clock selected by SAI1SEL and provided to **sai_a_ker_ck** and **sai_b_ker_ck** inputs, and the **rcc_pclk2** bus interface clock.
- Bit 21 Reserved, must be kept at reset value.
- Bit 20 **SPI5EN**: SPI5 Peripheral Clocks Enable
 Set and reset by software.
 0: SPI5 peripheral clocks disabled (default after reset)
 1: SPI5 peripheral clocks enabled:
 The peripheral clocks of the SPI5 are: the kernel clock selected by SPI45SEL and provided to **spi_ker_ck** input, and the **rcc_pclk2** bus interface clock.
- Bit 19 Reserved, must be kept at reset value.
- Bit 18 **TIM17EN**: TIM17 peripheral clock enable
 Set and reset by software.
 0: TIM17 peripheral clock disabled (default after reset)
 1: TIM17 peripheral clock enabled
- Bit 17 **TIM16EN**: TIM16 peripheral clock enable
 Set and reset by software.
 0: TIM16 peripheral clock disabled (default after reset)
 1: TIM16 peripheral clock enabled
- Bit 16 **TIM15EN**: TIM15 peripheral clock enable
 Set and reset by software.
 0: TIM15 peripheral clock disabled (default after reset)
 1: TIM15 peripheral clock enabled
- Bits 15:14 Reserved, must be kept at reset value.
- Bit 13 **SPI4EN**: SPI4 Peripheral Clocks Enable
 Set and reset by software.
 0: SPI4 peripheral clocks disabled (default after reset)
 1: SPI4 peripheral clocks enabled:
 The peripheral clocks of the SPI4 are: the kernel clock selected by SPI45SEL and provided to **spi_ker_ck** input, and the **rcc_pclk2** bus interface clock.
- Bit 12 **SPI1EN**: SPI1 Peripheral Clocks Enable
 Set and reset by software.
 0: SPI1 peripheral clocks disabled (default after reset)
 1: SPI1 peripheral clocks enabled:
 The peripheral clocks of the SPI1 are: the kernel clock selected by I2S123SRC and provided to **spi_ker_ck** input, and the **rcc_pclk2** bus interface clock.
- Bits 11:6 Reserved, must be kept at reset value.
- Bit 5 **USART6EN**: USART6 Peripheral Clocks Enable
 Set and reset by software.
 0: USART6 peripheral clocks disabled (default after reset)
 1: USART6 peripheral clocks enabled:
 The peripheral clocks of the USART6 are: the kernel clock selected by USART16SEL and provided to **usart_ker_ck** input, and the **rcc_pclk2** bus interface clock.

Bit 4 **USART1EN**: USART1 Peripheral Clocks Enable

Set and reset by software.

0: USART1 peripheral clocks disabled (default after reset)

1: USART1 peripheral clocks enabled:

The peripheral clocks of the USART1 are: the kernel clock selected by USART16SEL and provided to **usart_ker_ck** input, and the **rcc_pclk2** bus interface clock.

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8EN**: TIM8 peripheral clock enable

Set and reset by software.

0: TIM8 peripheral clock disabled (default after reset)

1: TIM8 peripheral clock enabled

Bit 0 **TIM1EN**: TIM1 peripheral clock enable

Set and reset by software.

0: TIM1 peripheral clock disabled (default after reset)

1: TIM1 peripheral clock enabled

8.7.48 RCC APB4 Clock Register (RCC_APB4ENR)

This register can be accessed via two different offset address.

Table 73. RCC_APB4ENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB4ENR	0x0F4	0x0001 0000
RCC_C1_APB4ENR	0x154	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SAI4EN	Res.	Res.	Res.	Res.	RTCAPBEN
										rw					rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VREFEN	COMP12EN	Res.	LPTIM5EN	LPTIM4EN	LPTIM3EN	LPTIM2EN	Res.	I2C4EN	Res.	SPI6EN	Res.	LPUART1EN	Res.	SYSCFGEN	Res.
rw	rw		rw	rw	rw	rw		rw		rw		rw		rw	

Bits 31:26 Reserved, must be kept at reset value.

Bits 25:22 Reserved, must be kept at reset value.

Bit 21 **SAI4EN**: SAI4 Peripheral Clocks Enable

Set and reset by software.

0: SAI4 peripheral clocks disabled (default after reset)

1: SAI4 peripheral clocks enabled

The peripheral clocks of the SAI4 are: the kernel clocks selected by SAI4ASEL and SAI4BSEL, and provided to **sai_a_ker_ck** and **sai_b_ker_ck** inputs respectively, and the **rcc_pclk4** bus interface clock.

Bits 20:17 Reserved, must be kept at reset value.

Bit 16 **RTCAPBEN**: RTC APB Clock Enable

Set and reset by software.

0: The register clock interface of the RTC (APB) is disabled

1: The register clock interface of the RTC (APB) is enabled (default after reset)

Bit 15 **VREFEN**: VREF peripheral clock enable

Set and reset by software.

0: VREF peripheral clock disabled (default after reset)

1: VREF peripheral clock enabled

Bit 14 **COMP12EN**: COMP1/2 peripheral clock enable

Set and reset by software.

0: COMP1/2 peripheral clock disabled (default after reset)

1: COMP1/2 peripheral clock enabled

Bit 13 Reserved, must be kept at reset value.

Bit 12 LPTIM5EN: LPTIM5 Peripheral Clocks Enable

Set and reset by software.

0: LPTIM5 peripheral clocks disabled (default after reset)

1: LPTIM5 peripheral clocks enabled

The peripheral clocks of the LPTIM5 are: the kernel clock selected by LPTIM345SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 11 LPTIM4EN: LPTIM4 Peripheral Clocks Enable

Set and reset by software.

0: LPTIM4 peripheral clocks disabled (default after reset)

1: LPTIM4 peripheral clocks enabled

The peripheral clocks of the LPTIM4 are: the kernel clock selected by LPTIM345SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 10 LPTIM3EN: LPTIM3 Peripheral Clocks Enable

Set and reset by software.

0: LPTIM3 peripheral clocks disabled (default after reset)

1: LPTIM3 peripheral clocks enabled

The peripheral clocks of the LPTIM3 are: the kernel clock selected by LPTIM345SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 9 LPTIM2EN: LPTIM2 Peripheral Clocks Enable

Set and reset by software.

0: LPTIM2 peripheral clocks disabled (default after reset)

1: LPTIM2 peripheral clocks enabled

The peripheral clocks of the LPTIM2 are: the kernel clock selected by LPTIM2SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 8 Reserved, must be kept at reset value.

Bit 7 I2C4EN: I2C4 Peripheral Clocks Enable

Set and reset by software.

0: I2C4 peripheral clocks disabled (default after reset)

1: I2C4 peripheral clocks enabled

The peripheral clocks of the I2C4 are: the kernel clock selected by I2C4SEL and provided to **i2c_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 6 Reserved, must be kept at reset value.

Bit 5 SPI6EN: SPI6 Peripheral Clocks Enable

Set and reset by software.

0: SPI6 peripheral clocks disabled (default after reset)

1: SPI6 peripheral clocks enabled

The peripheral clocks of the SPI6 are: the kernel clock selected by SPI6SEL and provided to **spi_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 4 Reserved, must be kept at reset value.

Bit 3 LPUART1EN: LPUART1 Peripheral Clocks Enable

Set and reset by software.

0: LPUART1 peripheral clocks disabled (default after reset)

1: LPUART1 peripheral clocks enabled

The peripheral clocks of the LPUART1 are: the kernel clock selected by LPUART1SEL and provided to **lpuart_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 2 Reserved, must be kept at reset value.

Bit 1 **SYSCFGEN**: SYSCFG peripheral clock enable

Set and reset by software.

0: SYSCFG peripheral clock disabled (default after reset)

1: SYSCFG peripheral clock enabled

Bit 0 Reserved, must be kept at reset value.

8.7.49 RCC AHB3 Sleep Clock Register (RCC_AHB3LPENR)

This register can be accessed via two different offset address.

Table 74. RCC_AHB3LPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_AHB3LPENR	0x0FC	0xF001 5131
RCC_C1_AHB3LPENR	0x15C	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AXISRAMLPEN	ITCMLPEN	DTCM2LPEN	DTCM1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC1LPEN
rw	rw	rw	rw												rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	QSPILPEN	Res.	FMCLPEN	Res.	Res.	Res.	FLASHLPEN	Res.	Res.	JPGDECLPEN	DMA2DLPEN	Res.	Res.	Res.	MDMALPEN
	rw		rw				rw			rw	rw				rw

Bit 31 AXISRAMLPEN: AXISRAM Block Clock Enable During CSleep mode

Set and reset by software.

0: AXISRAM interface clock disabled during CSleep mode

1: AXISRAM interface clock enabled during CSleep mode (default after reset)

Bit 30 ITCMLPEN: D1ITCM Block Clock Enable During CSleep mode

Set and reset by software.

0: D1 ITCM interface clock disabled during CSleep mode

1: D1 ITCM interface clock enabled during CSleep mode (default after reset)

Bit 29 DTCM2LPEN: D1 DTCM2 Block Clock Enable During CSleep mode

Set and reset by software.

0: D1 DTCM2 interface clock disabled during CSleep mode

1: D1 DTCM2 interface clock enabled during CSleep mode (default after reset)

Bit 28 D1DTCM1LPEN: D1DTCM1 Block Clock Enable During CSleep mode

Set and reset by software.

0: D1DTCM1 interface clock disabled during CSleep mode

1: D1DTCM1 interface clock enabled during CSleep mode (default after reset)

Bits 27:17 Reserved, must be kept at reset value.

Bit 16 SDMMC1LPEN: SDMMC1 and SDMMC1 Delay Clock Enable During CSleep Mode

Set and reset by software.

0: SDMMC1 and SDMMC1 Delay clock disabled during CSleep mode

1: SDMMC1 and SDMMC1 Delay clock enabled during CSleep mode (default after reset)

Bit 15 Reserved, must be kept at reset value.

Bit 14 **QSPILPEN**: QUADSPI and QUADSPI Delay Clock Enable During CSleep Mode

Set and reset by software.

0: QUADSPI and QUADSPI Delay clock disabled during CSleep mode

1: QUADSPI and QUADSPI Delay clock enabled during CSleep mode (default after reset)

Bit 13 Reserved, must be kept at reset value.

Bit 12 **FMCLPEN**: FMC Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: FMC peripheral clocks disabled during CSleep mode

1: FMC peripheral clocks enabled during CSleep mode (default after reset):

The peripheral clocks of the FMC are: the kernel clock selected by FMCSEL and provided to **fmc_ker_ck** input, and the **rcc_hclk3** bus interface clock.

Bits 11:9 Reserved, must be kept at reset value.

Bit 8 **FLASHLPEN**: Flash interface Clock Enable During CSleep Mode

Set and reset by software.

0: Flash interface clock disabled during CSleep mode

1: Flash interface clock enabled during CSleep mode (default after reset)

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **JPGDECLPEN**: JPGDEC Clock Enable During CSleep Mode

Set and reset by software.

0: JPGDEC peripheral clock disabled during CSleep mode

1: JPGDEC peripheral clock enabled during CSleep mode (default after reset)

Bit 4 **DMA2DLPEN**: DMA2D Clock Enable During CSleep Mode

Set and reset by software.

0: DMA2D peripheral clock disabled during CSleep mode

1: DMA2D peripheral clock enabled during CSleep mode (default after reset)

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **MDMALPEN**: MDMA Clock Enable During CSleep Mode

Set and reset by software.

0: MDMA peripheral clock disabled during CSleep mode

1: MDMA peripheral clock enabled during CSleep mode (default after reset)

8.7.50 RCC AHB1 Sleep Clock Register (RCC_AHB1LPENR)

This register can be accessed via two different offset address.

Table 75. RCC_AHB1LPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_AHB1LPENR	0x100	0x1E03 C023
RCC_C1_AHB1LPENR	0x160	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	USB2OTGHSULPILPEN	USB2OTGHSLPEN	USB1OTGHSULPILPEN	USB1OTGHSLPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETH1RXLPEN	ETH1TXLPEN
			rw	rw	rw	rw								rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETH1MACLPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ADC12LPEN	Res.	Res.	Res.	DMA2LPEN	DMA1LPEN
rw										rw				rw	rw

Bits 31:29 Reserved, must be kept at reset value.

Bit 28 **USB2OTGHSULPILPEN**: USB_PHY1 clock enable during CSleep mode

Set and reset by software.

0: USB_PHY1 peripheral clock disabled during CSleep mode

1: USB_PHY1 peripheral clock enabled during CSleep mode (default after reset)

Bit 27 **USB2OTGHSLPEN**: USB2OTG (OTG_US2) peripheral clock enable during CSleep mode

Set and reset by software.

0: USB2OTG peripheral clocks disabled during CSleep mode

1: USB2OTG peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the USB2OTG are: the kernel clock selected by USBSEL and the **rcc_hclk1** bus interface clock.

Bit 26 **USB1OTGHSULPILPEN**: USB_PHY1 clock enable during CSleep mode

Set and reset by software.

0: USB_PHY1 peripheral clock disabled during CSleep mode

1: USB_PHY1 peripheral clock enabled during CSleep mode (default after reset)

Bit 25 **USB1OTGHSLPEN**: USB1OTG (OTG_HS1) peripheral clock enable during CSleep mode

Set and reset by software.

0: USB1OTG peripheral clock disabled during CSleep mode

1: USB1OTG peripheral clock enabled during CSleep mode (default after reset)

The peripheral clocks of the USB1OTG are: the kernel clock selected by USBSEL and the **rcc_hclk1** bus interface clock.

Bits 24:18 Reserved, must be kept at reset value.

Bit 17 **ETH1RXLPEN**: Ethernet Reception Clock Enable During CSleep Mode

Set and reset by software.

0: Ethernet Reception clock disabled during CSleep mode

1: Ethernet Reception clock enabled during CSleep mode (default after reset)

Bit 16 **ETH1TXLPEN**: Ethernet Transmission Clock Enable During CSleep Mode

Set and reset by software.

0: Ethernet Transmission clock disabled during CSleep mode

1: Ethernet Transmission clock enabled during CSleep mode (default after reset)

Bit 15 **ETH1MACLPEN**: Ethernet MAC bus interface Clock Enable During CSleep Mode

Set and reset by software.

0: Ethernet MAC bus interface clock disabled during CSleep mode

1: Ethernet MAC bus interface clock enabled during CSleep mode (default after reset)

Bits 14:6 Reserved, must be kept at reset value.

Bit 5 **ADC12LPEN**: ADC1/2 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: ADC1/2 peripheral clocks disabled during CSleep mode

1: ADC1/2 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the ADC1 and 2 are: the kernel clock selected by ADCSEL and provided to `adc_ker_ck_input`, and the **rcc_hclk1** bus interface clock.

Bits 4:2 Reserved, must be kept at reset value.

Bit 1 **DMA2LPEN**: DMA2 Clock Enable During CSleep Mode

Set and reset by software.

0: DMA2 clock disabled during CSleep mode

1: DMA2 clock enabled during CSleep mode (default after reset)

Bit 0 **DMA1LPEN**: DMA1 Clock Enable During CSleep Mode

Set and reset by software.

0: DMA1 clock disabled during CSleep mode

1: DMA1 clock enabled during CSleep mode (default after reset)

8.7.51 RCC AHB2 Sleep Clock Register (RCC_AHB2LPENR)

This register can be accessed via two different offset address.

Table 76. RCC_AHB2LPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_AHB2LPENR	0x104	0xE000 0271
RCC_C1_AHB2LPENR	0x164	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SRAM3LPEN	SRAM2LPEN	SRAM1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw	rw	rw													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	SDMMC2LPEN	Res.	Res.	RNGLPEN	HASHLPEN	CRYPTLPEN	Res.	Res.	Res.	CAMITLPEN
						rw			rw	rw	rw				rw

Bit 31 SRAM3LPEN: SRAM3 Clock Enable During CSleep Mode

Set and reset by software.

0: SRAM3 clock disabled during CSleep mode

1: SRAM3 clock enabled during CSleep mode (default after reset)

Bit 30 SRAM2LPEN: SRAM2 Clock Enable During CSleep Mode

Set and reset by software.

0: SRAM2 clock disabled during CSleep mode

1: SRAM2 clock enabled during CSleep mode (default after reset)

Bit 29 SRAM1LPEN: SRAM1 Clock Enable During CSleep Mode

Set and reset by software.

0: SRAM1 clock disabled during CSleep mode

1: SRAM1 clock enabled during CSleep mode (default after reset)

Bits 28:10 Reserved, must be kept at reset value.

Bit 9 SDMMC2LPEN: SDMMC2 and SDMMC2 Delay Clock Enable During CSleep Mode

Set and reset by software.

0: SDMMC2 and SDMMC2 Delay clock disabled during CSleep mode

1: SDMMC2 and SDMMC2 Delay clock enabled during CSleep mode (default after reset)

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 RNGLPEN: RNG peripheral clock enable during CSleep mode

Set and reset by software.

0: RNG peripheral clocks disabled during CSleep mode

1: RNG peripheral clock enabled during CSleep mode (default after reset)

The peripheral clocks of the RNG are: the kernel clock selected by RNGSEL and provided to **rng_ker_ck** input, and the **rcc_hclk2** bus interface clock.

Bit 5 **HASHLPEN**: HASH peripheral clock enable during CSleep mode

Set and reset by software.

0: HASH peripheral clock disabled during CSleep mode

1: HASH peripheral clock enabled during CSleep mode (default after reset)

Bit 4 **CRYPTLPEN**: CRYPT peripheral clock enable during CSleep mode

Set and reset by software.

0: CRYPT peripheral clock disabled during CSleep mode

1: CRYPT peripheral clock enabled during CSleep mode (default after reset)

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **DCMILPEN**: DCMI peripheral clock enable during CSleep mode

Set and reset by software.

0: DCMI peripheral clock disabled during CSleep mode

1: DCMI peripheral clock enabled during CSleep mode (default after reset)

8.7.52 RCC AHB4 Sleep Clock Register (RCC_AHB4LPENR)

This register can be accessed via two different offset address.

Table 77. RCC_AHB4LPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_AHB4LPENR	0x108	0x3128 07FF
RCC_C1_AHB4LPENR	0x168	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	SRAM4LPEN	BKPRAMLLEN	Res.	Res.	Res.	ADC3LPEN	Res.	Res.	BDMALLEN	Res.	CRCLLEN	Res.	Res.	Res.
		rw	rw				rw			rw		rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	GPIOILLEN	GPIOILLEN	GPIOILLEN	GPIOILLEN	GPIOILLEN	GPIOILLEN	GPIOILLEN	GPIOILLEN	GPIOILLEN	GPIOILLEN	GPIOILLEN
					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 SRAM4LPEN: SRAM4 Clock Enable During CSleep Mode

Set and reset by software.

0: SRAM4 clock disabled during CSleep mode

1: SRAM4 clock enabled during CSleep mode (default after reset)

Bit 28 BKPRAMLLEN: Backup RAM Clock Enable During CSleep Mode

Set and reset by software.

0: Backup RAM clock disabled during CSleep mode

1: Backup RAM clock enabled during CSleep mode (default after reset)

Bits 27:25 Reserved, must be kept at reset value.

Bit 24 ADC3LPEN: ADC3 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: ADC3 peripheral clocks disabled during CSleep mode

1: ADC3 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the ADC3 are: the kernel clock selected by ADCSEL and provided to adc_ker_ck_input, and the **rcc_hclk4** bus interface clock.

Bits 23:22 Reserved, must be kept at reset value.

Bit 21 BDMALLEN: BDMA Clock Enable During CSleep Mode

Set and reset by software.

0: BDMA clock disabled during CSleep mode

1: BDMA clock enabled during CSleep mode (default after reset)

Bit 20 Reserved, must be kept at reset value.

Bit 19 **CRCLPEN**: CRC peripheral clock enable during CSleep mode

Set and reset by software.

0: CRC peripheral clock disabled during CSleep mode

1: CRC peripheral clock enabled during CSleep mode (default after reset)

Bits 18:11 Reserved, must be kept at reset value.

Bit 10 **GPIOKLPEN**: GPIOK peripheral clock enable during CSleep mode

Set and reset by software.

0: GPIOK peripheral clock disabled during CSleep mode

1: GPIOK peripheral clock enabled during CSleep mode (default after reset)

Bit 9 **GPIOJLPEN**: GPIOJ peripheral clock enable during CSleep mode

Set and reset by software.

0: GPIOJ peripheral clock disabled during CSleep mode

1: GPIOJ peripheral clock enabled during CSleep mode (default after reset)

Bit 8 **GPIOILPEN**: GPIOI peripheral clock enable during CSleep mode

Set and reset by software.

0: GPIOI peripheral clock disabled during CSleep mode

1: GPIOI peripheral clock enabled during CSleep mode (default after reset)

Bit 7 **GPIOHLPEN**: GPIOH peripheral clock enable during CSleep mode

Set and reset by software.

0: GPIOH peripheral clock disabled during CSleep mode

1: GPIOH peripheral clock enabled during CSleep mode (default after reset)

Bit 6 **GPIOGLPEN**: GPIOG peripheral clock enable during CSleep mode

Set and reset by software.

0: GPIOG peripheral clock disabled during CSleep mode

1: GPIOG peripheral clock enabled during CSleep mode (default after reset)

Bit 5 **GPIOFLPEN**: GPIOF peripheral clock enable during CSleep mode

Set and reset by software.

0: GPIOF peripheral clock disabled during CSleep mode

1: GPIOF peripheral clock enabled during CSleep mode (default after reset)

Bit 4 **GPIOELPEN**: GPIOE peripheral clock enable during CSleep mode

Set and reset by software.

0: GPIOE peripheral clock disabled during CSleep mode

1: GPIOE peripheral clock enabled during CSleep mode (default after reset)

Bit 3 **GPIODLPEN**: GPIOD peripheral clock enable during CSleep mode

Set and reset by software.

0: GPIOD peripheral clock disabled during CSleep mode

1: GPIOD peripheral clock enabled during CSleep mode (default after reset)

- Bit 2 **GPIOCLPEN**: GPIOC peripheral clock enable during CSleep mode
Set and reset by software.
0: GPIOC peripheral clock disabled during CSleep mode
1: GPIOC peripheral clock enabled during CSleep mode (default after reset)
- Bit 1 **GPIOBLPEN**: GPIOB peripheral clock enable during CSleep mode
Set and reset by software.
0: GPIOB peripheral clock disabled during CSleep mode
1: GPIOB peripheral clock enabled during CSleep mode (default after reset)
- Bit 0 **GPIOALPEN**: GPIOA peripheral clock enable during CSleep mode
Set and reset by software.
0: GPIOA peripheral clock disabled during CSleep mode
1: GPIOA peripheral clock enabled during CSleep mode (default after reset)

8.7.53 RCC APB3 Sleep Clock Register (RCC_APB3LPENR)

This register can be accessed via two different offset address.

Table 78. RCC_APB3LPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB3LPENR	0x10C	0x0000 0058
RCC_C1_APB3LPENR	0x16C	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	WWDG1LPEN	Res.	Res.	LTDCLPEN	Res.	Res.	Res.
									rw			rw			

Bits 31:7 Reserved, must be kept at reset value.

Bit 6 **WWDG1LPEN**: WWDG1 Clock Enable During CSleep Mode

Set and reset by software.

0: WWDG1 clock disable during CSleep mode

1: WWDG1 clock enabled during CSleep mode (default after reset)

Bits 5:4 Reserved, must be kept at reset value.

Bit 3 **LTDCLPEN**: LTDC peripheral clock enable during CSleep mode

Provides the pixel clock (**ltdc_ker_ck**) to the LTDC block.

Set and reset by software.

0: LTDC clock disabled during CSleep mode

1: LTDC clock provided to the LTDC during CSleep mode (default after reset)

Bits 2:0 Reserved, must be kept at reset value.

8.7.54 RCC APB1 Low Sleep Clock Register (RCC_APB1LLPENR)

This register can be accessed via two different offset address.

Table 79. RCC_APB1LLPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB1LLPENR	0x110	0xE8FF CBFF
RCC_C1_APB1LLPENR	0x170	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UART8LPEN	UART7LPEN	DAC12LPEN	Res.	CECLPEN	Res.	Res.	Res.	I2C3LPEN	I2C2LPEN	I2C1LPEN	UART5LPEN	UART4LPEN	USART3LPEN	USART2LPEN	SPDIFRXLPEN
r/w	r/w	r/w		r/w				r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3LPEN	SPI2LPEN	Res.	Res.	Res.	Res.	LPTIM1LPEN	TIM14LPEN	TIM13LPEN	TIM12LPEN	TIM7LPEN	TIM6LPEN	TIM5LPEN	TIM4LPEN	TIM3LPEN	TIM2LPEN
r/w	r/w					r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bit 31 UART8LPEN: UART8 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: UART8 peripheral clocks disabled during CSleep mode

1: UART8 peripheral clocks enabled during CSleep mode (default after reset):

The peripheral clocks of the UART8 are: the kernel clock selected by USART234578SEL and provided to **usart_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 30 UART7LPEN: UART7 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: UART7 peripheral clocks disabled during CSleep mode

1: UART7 peripheral clocks enabled during CSleep mode (default after reset):

The peripheral clocks of the UART7 are: the kernel clock selected by USART234578SEL and provided to **usart_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 29 DAC12LPEN: DAC1/2 peripheral clock enable during CSleep mode

Set and reset by software.

0: DAC1/2 peripheral clock disabled during CSleep mode

1: DAC1/2 peripheral clock enabled during CSleep mode (default after reset)

Bit 28 Reserved, must be kept at reset value.

Bit 27 CECLPEN: HDMI-CEC Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: HDMI-CEC peripheral clocks disabled during CSleep mode

1: HDMI-CEC peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the HDMI-CEC are: the kernel clock selected by CECSEL and provided to **cec_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bits 26:24 Reserved, must be kept at reset value.

- Bit 23 **I2C3LPEN**: I2C3 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: I2C3 peripheral clocks disabled during CSleep mode
 1: I2C3 peripheral clocks enabled during CSleep mode (default after reset):
 The peripheral clocks of the I2C3 are: the kernel clock selected by I2C123SEL and provided to **i2c_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 22 **I2C2LPEN**: I2C2 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: I2C2 peripheral clocks disabled during CSleep mode
 1: I2C2 peripheral clocks enabled during CSleep mode (default after reset):
 The peripheral clocks of the I2C2 are: the kernel clock selected by I2C123SEL and provided to **i2c_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 21 **I2C1LPEN**: I2C1 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: I2C1 peripheral clocks disabled during CSleep mode
 1: I2C1 peripheral clocks enabled during CSleep mode (default after reset):
 The peripheral clocks of the I2C1 are: the kernel clock selected by I2C123SEL and provided to **i2c_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 20 **UART5LPEN**: UART5 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: UART5 peripheral clocks disabled during CSleep mode
 1: UART5 peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the UART5 are: the kernel clock selected by USART234578SEL and provided to **uart_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 19 **UART4LPEN**: UART4 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: UART4 peripheral clocks disabled during CSleep mode
 1: UART4 peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the UART4 are: the kernel clock selected by USART234578SEL and provided to **uart_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 18 **USART3LPEN**: USART3 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: USART3 peripheral clocks disabled during CSleep mode
 1: USART3 peripheral clocks enabled during CSleep mode (default after reset):
 The peripheral clocks of the USART3 are: the kernel clock selected by USART234578SEL and provided to **usart_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 17 **USART2LPEN**: USART2 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: USART2 peripheral clocks disabled during CSleep mode
 1: USART2 peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the USART2 are: the kernel clock selected by USART234578SEL and provided to **usart_ker_ck** input, and the **rcc_pclk1** bus interface clock.
- Bit 16 **SPDIFRXLPEN**: SPDIFRX Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: SPDIFRX peripheral clocks disabled during CSleep mode
 1: SPDIFRX peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the SPDIFRX are: the kernel clock selected by SPDIFSEL and provided to **spdifrx_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 15 SPI3LPEN: SPI3 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: SPI3 peripheral clocks disabled during CSleep mode

1: SPI3 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the SPI3 are: the kernel clock selected by I2S123SRC and provided to **spi_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 14 SPI2LPEN: SPI2 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: SPI2 peripheral clocks disabled during CSleep mode

1: SPI2 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the SPI2 are: the kernel clock selected by I2S123SRC and provided to **spi_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bits 13:10 Reserved, must be kept at reset value.

Bit 9 LPTIM1LPEN: LPTIM1 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: LPTIM1 peripheral clocks disabled during CSleep mode

1: LPTIM1 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the LPTIM1 are: the kernel clock selected by LPTIM1SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 8 TIM14LPEN: TIM14 peripheral clock enable during CSleep mode

Set and reset by software.

0: TIM14 peripheral clock disabled during CSleep mode

1: TIM14 peripheral clock enabled during CSleep mode (default after reset)

Bit 7 TIM13LPEN: TIM13 peripheral clock enable during CSleep mode

Set and reset by software.

0: TIM13 peripheral clock disabled during CSleep mode

1: TIM13 peripheral clock enabled during CSleep mode (default after reset)

Bit 6 TIM12LPEN: TIM12 peripheral clock enable during CSleep mode

Set and reset by software.

0: TIM12 peripheral clock disabled during CSleep mode

1: TIM12 peripheral clock enabled during CSleep mode (default after reset)

Bit 5 TIM7LPEN: TIM7 peripheral clock enable during CSleep mode

Set and reset by software.

0: TIM7 peripheral clock disabled during CSleep mode

1: TIM7 peripheral clock enabled during CSleep mode (default after reset)

Bit 4 TIM6LPEN: TIM6 peripheral clock enable during CSleep mode

Set and reset by software.

0: TIM6 peripheral clock disabled during CSleep mode

1: TIM6 peripheral clock enabled during CSleep mode (default after reset)

Bit 3 TIM5LPEN: TIM5 peripheral clock enable during CSleep mode

Set and reset by software.

0: TIM5 peripheral clock disabled during CSleep mode

1: TIM5 peripheral clock enabled during CSleep mode (default after reset)

- Bit 2 **TIM4LPEN**: TIM4 peripheral clock enable during CSleep mode
Set and reset by software.
0: TIM4 peripheral clock disabled during CSleep mode
1: TIM4 peripheral clock enabled during CSleep mode (default after reset)
- Bit 1 **TIM3LPEN**: TIM3 peripheral clock enable during CSleep mode
Set and reset by software.
0: TIM3 peripheral clock disabled during CSleep mode
1: TIM3 peripheral clock enabled during CSleep mode (default after reset)
- Bit 0 **TIM2LPEN**: TIM2 peripheral clock enable during CSleep mode
Set and reset by software.
0: TIM2 peripheral clock disabled during CSleep mode
1: TIM2 peripheral clock enabled during CSleep mode (default after reset)

8.7.55 RCC APB1 High Sleep Clock Register (RCC_APB1HLPENR)

This register can be accessed via two different offset address.

Table 80. RCC_APB1HLPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB1HLPENR	0x114	0x0000 0136
RCC_C1_APB1HLPENR	0x174	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	FDCANLPEN	Res.	Res.	MDIOSLPEN	OPAMPLPEN	Res.	SWPLPEN	CRSLPEN	Res.
							rw			rw	rw		rw	rw	

Bits 31:9 Reserved, must be kept at reset value.

Bit 8 FDCANLPEN: FDCAN Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: FDCAN peripheral clocks disabled during CSleep mode

1: FDCAN peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the FDCAN are: the kernel clock selected by FDCANSEL and provided to **fdcan_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 MDIOSLPEN: MDIOS peripheral clock enable during CSleep mode

Set and reset by software.

0: MDIOS peripheral clock disabled during CSleep mode

1: MDIOS peripheral clock enabled during CSleep mode (default after reset)

Bit 4 OPAMPLPEN: OPAMP peripheral clock enable during CSleep mode

Set and reset by software.

0: OPAMP peripheral clock disabled during CSleep mode

1: OPAMP peripheral clock enabled during CSleep mode (default after reset)

Bit 3 Reserved, must be kept at reset value.

Bit 2 **SWPLPEN**: SWPMI Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: SWPMI peripheral clocks disabled during CSleep mode

1: SWPMI peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the SWPMI are: the kernel clock selected by SWPSEL and provided to **swpmi_ker_ck** input, and the **rcc_pclk1** bus interface clock.

Bit 1 **CRSLPEN**: Clock Recovery System peripheral clock enable during CSleep mode

Set and reset by software.

0: CRS peripheral clock disabled during CSleep mode

1: CRS peripheral clock enabled during CSleep mode (default after reset)

Bit 0 Reserved, must be kept at reset value.

8.7.56 RCC APB2 Sleep Clock Register (RCC_APB2LPENR)

This register can be accessed via two different offset address.

Table 81. RCC_APB2LPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB2LPENR	0x118	0x31D7 3033
RCC_C1_APB2LPENR	0x178	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	HRTIMLPEN	DFSDM1LPEN	Res.	Res.	Res.	SAI3LPEN	SAI2LPEN	SAI1LPEN	Res.	SP15LPEN	Res.	TIM17LPEN	TIM16LPEN	TIM15LPEN
		rw	rw				rw	rw	rw		rw		rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	SPI4LPEN	SPI1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	USART6LPEN	USART1LPEN	Res.	Res.	TIM8LPEN	TIM1LPEN
		rw	rw							rw	rw			rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **HRTIMLPEN**: HRTIM peripheral clock enable during CSleep mode

Set and reset by software.

0: HRTIM peripheral clock disabled during CSleep mode

1: HRTIM peripheral clock enabled during CSleep mode (default after reset)

Bit 28 **DFSDM1LPEN**: DFSDM1 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: DFSDM1 peripheral clocks disabled during CSleep mode

1: DFSDM1 peripheral clocks enabled during CSleep mode (default after reset)

DFSDM1 peripheral clocks are: the kernel clocks selected by SAI1SEL and DFSDM1SEL and provided to **Aclk** and **clk** inputs respectively, and the **rcc_pclk2** bus interface clock.

Bits 27:25 Reserved, must be kept at reset value.

Bit 24 **SAI3LPEN**: SAI3 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: SAI3 peripheral clocks disabled during CSleep mode

1: SAI3 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the SAI3 are: the kernel clock selected by SAI23SEL and provided to **sai_a_ker_ck** and **sai_b_ker_ck** inputs, and the **rcc_pclk2** bus interface clock.

Bit 23 **SAI2LPEN**: SAI2 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: SAI2 peripheral clocks disabled during CSleep mode

1: SAI2 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the SAI2 are: the kernel clock selected by SAI23SEL and provided to **sai_a_ker_ck** and **sai_b_ker_ck** inputs, and the **rcc_pclk2** bus interface clock.

- Bit 22 **SAI1LPEN**: SAI1 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: SAI1 peripheral clocks disabled during CSleep mode
 1: SAI1 peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the SAI1 are: the kernel clock selected by SAI1SEL and provided to **sai_a_ker_ck** and **sai_b_ker_ck** inputs, and the **rcc_pclk2** bus interface clock.
- Bit 21 Reserved, must be kept at reset value.
- Bit 20 **SPI5LPEN**: SPI5 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: SPI5 peripheral clocks disabled during CSleep mode
 1: SPI5 peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the SPI5 are: the kernel clock selected by SPI45SEL and provided to **spi_ker_ck** input, and the **rcc_pclk2** bus interface clock.
- Bit 19 Reserved, must be kept at reset value.
- Bit 18 **TIM17LPEN**: TIM17 peripheral clock enable during CSleep mode
 Set and reset by software.
 0: TIM17 peripheral clock disabled during CSleep mode
 1: TIM17 peripheral clock enabled during CSleep mode (default after reset)
- Bit 17 **TIM16LPEN**: TIM16 peripheral clock enable during CSleep mode
 Set and reset by software.
 0: TIM16 peripheral clock disabled during CSleep mode
 1: TIM16 peripheral clock enabled during CSleep mode (default after reset)
- Bit 16 **TIM15LPEN**: TIM15 peripheral clock enable during CSleep mode
 Set and reset by software.
 0: TIM15 peripheral clock disabled during CSleep mode
 1: TIM15 peripheral clock enabled during CSleep mode (default after reset)
- Bits 15:14 Reserved, must be kept at reset value.
- Bit 13 **SPI4LPEN**: SPI4 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: SPI4 peripheral clocks disabled during CSleep mode
 1: SPI4 peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the SPI4 are: the kernel clock selected by SPI45SEL and provided to **spi_ker_ck** input, and the **rcc_pclk2** bus interface clock.
- Bit 12 **SPI1LPEN**: SPI1 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: SPI1 peripheral clocks disabled during CSleep mode
 1: SPI1 peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the SPI1 are: the kernel clock selected by I2S123SRC and provided to **spi_ker_ck** input, and the **rcc_pclk2** bus interface clock.
- Bits 11:6 Reserved, must be kept at reset value.
- Bit 5 **USART6LPEN**: USART6 Peripheral Clocks Enable During CSleep Mode
 Set and reset by software.
 0: USART6 peripheral clocks disabled during CSleep mode
 1: USART6 peripheral clocks enabled during CSleep mode (default after reset)
 The peripheral clocks of the USART6 are: the kernel clock selected by USART16SEL and provided to **usart_ker_ck** input, and the **rcc_pclk2** bus interface clock.

Bit 4 **USART1LPEN**: USART1 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: USART1 peripheral clocks disabled during CSleep mode

1: USART1 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the USART1 are: the kernel clock selected by USART16SEL and provided to **usart_ker_ck** inputs, and the **rcc_pclk2** bus interface clock.

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8LPEN**: TIM8 peripheral clock enable during CSleep mode

Set and reset by software.

0: TIM8 peripheral clock disabled during CSleep mode

1: TIM8 peripheral clock enabled during CSleep mode (default after reset)

Bit 0 **TIM1LPEN**: TIM1 peripheral clock enable during CSleep mode

Set and reset by software.

0: TIM1 peripheral clock disabled during CSleep mode

1: TIM1 peripheral clock enabled during CSleep mode (default after reset)

8.7.57 RCC APB4 Sleep Clock Register (RCC_APB4LPENR)

This register can be accessed via two different offset address.

Table 82. RCC_APB4LPENR address offset and reset value

Register Name	Address Offset	Reset Value
RCC_APB4LPENR	0x11C	0x0421 DEAA
RCC_C1_APB4LPENR	0x17C	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SAI4LPEN	Res.	Res.	Res.	Res.	RTCAPBLPEN
										rw					rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VREFLPEN	COMP12LPEN	Res.	LPTIM5LPEN	LPTIM4LPEN	LPTIM3LPEN	LPTIM2LPEN	Res.	I2C4LPEN	Res.	SPI6LPEN	Res.	LPUART1LPEN	Res.	SYSCFGLPEN	Res.
rw	rw		rw	rw	rw	rw		rw		rw		rw		rw	

Bits 31:26 Reserved, must be kept at reset value.

Bits 25:22 Reserved, must be kept at reset value.

Bit 21 SAI4LPEN: SAI4 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: SAI4 peripheral clocks disabled during CSleep mode

1: SAI4 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the SAI4 are: the kernel clocks selected by SAI4ASEL and SAI4BSEL, and provided to **sai_a_ker_ck** and **sai_b_ker_ck** inputs respectively, and the **rcc_pclk4** bus interface clock.

Bits 20:17 Reserved, must be kept at reset value.

Bit 16 RTCAPBLPEN: RTC APB Clock Enable During CSleep Mode

Set and reset by software.

0: The register clock interface of the RTC (APB) is disabled during CSleep mode

1: The register clock interface of the RTC (APB) is enabled during CSleep mode (default after reset)

Bit 15 VREFLPEN: VREF peripheral clock enable during CSleep mode

Set and reset by software.

0: VREF peripheral clock disabled during CSleep mode

1: VREF peripheral clock enabled during CSleep mode (default after reset)

Bit 14 COMP12LPEN: COMP1/2 peripheral clock enable during CSleep mode

Set and reset by software.

0: COMP1/2 peripheral clock disabled during CSleep mode

1: COMP1/2 peripheral clock enabled during CSleep mode (default after reset)

Bit 13 Reserved, must be kept at reset value.

Bit 12 LPTIM5LPEN: LPTIM5 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: LPTIM5 peripheral clocks disabled during CSleep mode

1: LPTIM5 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the LPTIM5 are: the kernel clock selected by LPTIM345SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 11 LPTIM4LPEN: LPTIM4 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: LPTIM4 peripheral clocks disabled during CSleep mode

1: LPTIM4 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the LPTIM4 are: the kernel clock selected by LPTIM345SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 10 LPTIM3LPEN: LPTIM3 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: LPTIM3 peripheral clocks disabled during CSleep mode

1: LPTIM3 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the LPTIM3 are: the kernel clock selected by LPTIM345SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 9 LPTIM2LPEN: LPTIM2 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: LPTIM2 peripheral clocks disabled during CSleep mode

1: LPTIM2 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the LPTIM5 are: the kernel clock selected by LPTIM2SEL and provided to **lptim_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 8 Reserved, must be kept at reset value.

Bit 7 I2C4LPEN: I2C4 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: I2C4 peripheral clocks disabled during CSleep mode

1: I2C4 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the I2C4 are: the kernel clock selected by I2C4SEL and provided to **i2c_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 6 Reserved, must be kept at reset value.

Bit 5 SPI6LPEN: SPI6 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: SPI6 peripheral clocks disabled during CSleep mode

1: SPI6 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the SPI6 are: the kernel clock selected by SPI6SEL and provided to **spi_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 4 Reserved, must be kept at reset value.

Bit 3 LPUART1LPEN: LPUART1 Peripheral Clocks Enable During CSleep Mode

Set and reset by software.

0: LPUART1 peripheral clocks disabled during CSleep mode

1: LPUART1 peripheral clocks enabled during CSleep mode (default after reset)

The peripheral clocks of the LPUART1 are: the kernel clock selected by LPUART1SEL and provided to **lpuart_ker_ck** input, and the **rcc_pclk4** bus interface clock.

Bit 2 Reserved, must be kept at reset value.

Bit 1 **SYSCFGLPEN**: SYSCFG peripheral clock enable during CSleep mode

Set and reset by software.

0: SYSCFG peripheral clock disabled during CSleep mode

1: SYSCFG peripheral clock enabled during CSleep mode (default after reset)

Bit 0 Reserved, must be kept at reset value.

8.8 RCC register map

Table 83. RCC register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x000	RCC_CR	Res.	Res.	PLL3RDY	PLL3ON	PLL2RDY	PLL2ON	PLL1RDY	PLL1ON	Res.	Res.	Res.	Res.	HSECSSON	HSEBYP	HSERDY	HSEON	D2CKRDY	D1CKRDY	HSI48RDY	HSI48ON	Res.	Res.	CSIKERON	CSIRDY	CSION	Res.	HSIDIVF	HSIDIV[1:0]	HSIRDY	HSIKERON	HSION			
	Reset value			0	0	0	0	0	0					0	0	0	0	0	0	0	0			0	0	0		0	0	0	0	0	1		
0x004	RCC_ICSCR ⁽¹⁾	Res.	CSITRIM[4:0]				CSICAL[7:0]				HSITRIM[5:0]				HSICAL[11:0]																				
	Reset value		1	0	0	0	0	-	-	-	-	-	-	-	-	1	0	0	0	0	0	-	-	-	-	-	-	-	-	-	-	-	-		
0x004	RCC_HSI4CFGR ⁽²⁾	Res.	HSITRIM[6:0]				Res.				Res.				Res.				HSICAL[11:0]																
	Reset value		1	0	0	0	0	0														X	X	X	X	X	X	X	X	X	X	X			
0x008	RCC_CRRCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HSI48CAL[9:0]													
	Reset value																							-	-	-	-	-	-	-	-	-			
0x00C	reserved	Reserved																																	
0x00C	RCC_CSICFGR ⁽²⁾	Res.	Res.	CSITRIM[5:0]				Res.				Res.				Res.				Res.				CSICAL[9:0]											
	Reset value			1	0	0	0	0	0																	X	X	X	X	X	X	X			
0x010	RCC_CFGR	MCO2[2:0]		MCO2PRE[3:0]				MCO1[2:0]		MCO1PRE[3:0]				Res.		TIMPRE HRTIMSEL		RTCPRE[5:0]				STOPKERWUCK		STOPWUCK		SWS[2:0]		SW[2:0]							
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x014	reserved	Reserved																																	
0x018	RCC_D1CFGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	D1CPRE[3:0]				Res.		D1PPRE[2:0]				HPRE[3:0]				
	Reset value																					0	0	0	0		0	0	0	0	0	0			
0x01C	RCC_D2CFGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	D2PPRE2[2:0]				Res.		D2PPRE1[2:0]							
	Reset value																					0	0	0	0		0	0							
0x020	RCC_D3CFGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	D3PPRE[2:0]				Res.									
	Reset value																					0	0	0			0								
0x024	reserved	Reserved																																	

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x028	RCC_PLLCKSELR	Res.	Res.	Res.	Res.	Res.	Res.			DIVM3[5:0]					Res.	Res.	DIVM2[5:0]					Res.	Res.	DIVM1[5:0]					Res.	Res.	PLLSRC[1:0]			
	Reset value							1	0	0	0	0	0		Res.	Res.		1	0	0	0	0	0		Res.	Res.	1	0	0	0	0	0	0	0
0x02C	RCC_PLLCFGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DIVR3EN	DIVQ3EN	DIVP3EN	DIVR2EN	DIVQ2EN	DIVP2EN	DIVR1EN	DIVQ1EN	DIVP1EN	Res.	Res.	Res.	Res.	PLL3RGE[1:0]	PLL3VCOSSEL	PLL3FRACEN	PLL2RGE[1:0]	PLL2VCOSSEL	PLL2FRACEN	PLL1RGE[1:0]	PLL1VCOSSEL	PLL1FRACEN				
	Reset value								1	1	1	1	1	1	1	1	1					0	0	0	0	0	0	0	0	0	0	0	0	0
0x030	RCC_PLL1DIVR	Res.	DIVR1[6:0]					Res.	DIVQ1[6:0]					DIVP1[6:0]					DIVN1[8:0]															
	Reset value		0	0	0	0	0	0	1		0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0
0x034	RCC_PLL1FRACR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FRACN1[12:0]										Res.	Res.	Res.				
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x038	RCC_PLL2DIVR	Res.	DIVR2[6:0]					Res.	DIVQ2[6:0]					DIVP2[6:0]					DIVN2[8:0]															
	Reset value		0	0	0	0	0	0	1		0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0
0x03C	RCC_PLL2FRACR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FRACN2[12:0]										Res.	Res.	Res.				
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x040	RCC_PLL3DIVR	Res.	DIVR3[6:0]					Res.	DIVQ3[6:0]					DIVP3[6:0]					DIVN3[8:0]															
	Reset value		0	0	0	0	0	0	1		0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0
0x044	RCC_PLL3FRACR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FRACN3[12:0]										Res.	Res.	Res.				
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x048	reserved	Reserved																																
0x04C	RCC_D1CCIPR	Res.	Res.	CKPERSEL[1:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMCSEL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	QSPISEL[1:0]	Res.	Res.	FMCSEL[1:0]			
	Reset value			0	0												0										0	0			0	0		
0x050	RCC_D2CCIP1R	SWPSEL	Res.	FDCANSEL[1:0]	Res.	Res.	Res.	DFSDM1SEL	Res.	Res.	SPDIFSEL[1:0]	Res.	SPI45SEL[2:0]	Res.	Res.	Res.	Res.	SPDIFSEL[2:0]	Res.	Res.	Res.	Res.	Res.	Res.	SAI23SEL[2:0]	Res.	Res.	Res.	SAI1SEL[2:0]	Res.	Res.	Res.		
	Reset value	0		0	0			0			0	0	0	0	0	0	0		0	0	0	0			0	0	0				0	0	0	0

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x054	RCC_D2CCIP2R	Res.	LPTIM1SEL[2:0]				Res.	Res.	Res.	Res.	CECSEL[1:0]		USBSEL[1:0]		Res.	Res.	Res.	Res.	Res.	I2C123SEL[1:0]		Res.	Res.	RNGSEL[1:0]		Res.	Res.	USART16SEL[2:0]				USART234578SEL[2:0]					
	Reset value	0	0	0	0					0	0	0	0								0	0			0	0			0	0	0	0	0	0			
0x058	RCC_D3CCIPR	Res.	SPI6SEL[2:0]				Res.	SAI4BSEL[2:0]		SAI4ASEL[2:0]		Res.	Res.	Res.	ADCSEL[1:0]		LPTIM345SEL[2:0]		LPTIM2SEL[2:0]		I2C4SEL[1:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LPUART1SEL[2:0]				0	0	0
	Reset value	0	0	0	0		0	0	0	0	0	0			0	0	0	0	0	0	0	0	0	0	0							0	0	0			
0x05C	reserved	Reserved																																			
0x060	RCC_CIER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LSECSSIE	PLL3RDYIE	PLL2RDYIE	PLL1RDYIE	HSI48RDYIE	CSIRDYIE	HSERDYIE	HSIRDYIE	LSERDYIE	LSIRDYIE			
	Reset value																								0	0	0	0	0	0	0	0	0	0			
0x064	RCC_CIFR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HSECSSF	LSECSSF	PLL3RDYF	PLL2RDYF	PLL1RDYF	HSI48RDYF	CSIRDYF	HSERDYF	HSIRDYF	LSERDYF	LSIRDYF			
	Reset value																							0	0	0	0	0	0	0	0	0	0	0			
0x068	RCC_CICR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HSECSSC	LSECSSC	PLL3RDYC	PLL2RDYC	PLL1RDYC	HSI48RDYC	CSIRDYC	HSERDYC	HSIRDYC	LSERDYC	LSIRDYC			
	Reset value																							0	0	0	0	0	0	0	0	0	0	0			
0x06C	reserved	Reserved																																			
0x070	RCC_BDCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BDRST	RTCN	Res.	Res.	Res.	Res.	Res.	Res.	RTCSEL[1:0]		Res.	LSECSSD	LSECSSON	LSEDRV[1:0]		LSEBYP	LSERDY	LSEON			
	Reset value																0	0							0	0	0	0	0	0	0	0	0	0			
0x074	RCC_CSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.			
	Reset value																																0	0	0		
0x078	reserved	Reserved																																			

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x07C	RCC_AHB3RSTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC1RST	Res.	QSPIRST	Res.	FMC RST	Res.	Res.	Res.	Res.	Res.	Res.	JPGDECRST	Res.	DMA2DRST	Res.	Res.	Res.	MDMARST
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x080	RCC_AHB1RSTR	Res.	Res.	Res.	Res.	USB2OTGRST	Res.	USB1OTGRST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETH1MACRST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ADC12RST	Res.	Res.	DMA2RST	DMA1RST		
	Reset value					0		0									0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x084	RCC_AHB2RSTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC2RST	Res.	Res.	Res.	RNGRST	HASHRST	CRYPTRST	Res.	Res.	Res.	CAMITFRST
	Reset value																							0			0	0	0	0	0	0	0	0
0x088	RCC_AHB4RSTR	Res.	Res.	Res.	Res.	Res.	Res.	HSEMRST	ADC3RST	Res.	Res.	BDMARST	Res.	CRCRST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	GPIOKRST	GPIORST	GPIORST	GPIORST	GPIORST	GPIORST	GPIORST	GPIORST	GPIORST	GPIORST	GPIORST	GPIORST
	Reset value							0	0			0		0									0	0	0	0	0	0	0	0	0	0	0	0
0x08C	RCC_APB3RSTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value																																	
0x090	RCC_APB1LRSTR	UART8RST	UART7RST	DAC12RST	Res.	CECRST	Res.	Res.	Res.	I2C3RST	I2C2RST	I2C1RST	UART5RST	UART4RST	USART3RST	USART2RST	SPDIFRXRST	SPI3RST	SPI2RST	Res.	Res.	Res.	Res.	LPTIM1RST	TIM14RST	TIM13RST	TIM12RST	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	TIM2RST	
	Reset value	0	0	0		0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x094	RCC_APB1HRSTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FDCANRST	Res.	Res.	Res.	MDIOSRST	OPAMP RST	Res.	RST	CRSRST	Res.	Res.
	Reset value																							0			0	0	0		0	0	0	0
0x098	RCC_APB2RSTR	Res.	Res.	HRTIMRST	DFSDM1RST	Res.	Res.	Res.	SAI3RST	SAI2RST	SAI1RST	Res.	SPI5RST	Res.	TIM17RST	TIM16RST	TIM15RST	Res.	Res.	SPI4RST	SPI1RST	Res.	Res.	Res.	Res.	Res.	Res.	USART6RST	USART1RST	Res.	Res.	TIM8RST	TIM1RST	
	Reset value			0	0				0	0	0		0		0	0	0			0	0							0	0			0	0	0

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x09C	RCC_APB4RSTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SAI4RST	Res.	Res.	Res.	Res.	Res.	VREFRST	COMP12RST	Res.	LPTIM5RST	LPTIM4RST	LPTIM3RST	LPTIM2RST	Res.	I2C4RST	Res.	SPI6RST	Res.	LPUART1RST	Res.	SYSCFG RST	Res.		
	Reset value											0						0	0		0	0	0	0		0		0		0		0	0		
0x0A0	RCC_GCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	0	
	Reset value																																	0	
0x0A4	reserved	Reserved																																	
0x0A8	RCC_D3AMR	Res.	Res.	SRAM4AMEN	BKPRAMAMEN	Res.	Res.	Res.	ADC3AMEN	Res.	Res.	SAI4AMEN	Res.	CRCAMEN	Res.	Res.	RTCAMEN	VREFAMEN	COMP12AMEN	Res.	LPTIM5AMEN	LPTIM4AMEN	LPTIM3AMEN	LPTIM2AMEN	Res.	I2C4AMEN	Res.	SPI6AMEN	Res.	LPUART1AMEN	Res.	Res.	BDMAAMEN		
	Reset value			0	0				0			0		0			0	0	0		0	0	0	0		0		0		0			0		
0x0AC to 0x0CC	reserved	Reserved																																	
0x0D0	RCC_RSR	Res.	LPWR RSTF	Res.	Res.	Res.	Res.	Res.	WDG1RSTF	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.		
	Reset value		0			0			0			0	1	1	1	1	1																		
0x0D4	RCC_AHB3ENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC1EN	Res.	QSPIEN	Res.	FMCEN	Res.	Res.	Res.	Res.	Res.	Res.	JPGDECEN	DMA2DEN	Res.	Res.	Res.	Res.	MDMAEN	
	Reset value																0		0		0							0	0					0	
0x0D8	RCC_AHB1ENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value					0	0	0								0	0	0										0					0	0	
0x0DC	RCC_AHB2ENR	SRAM3EN	SRAM2EN	SRAM1EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC2EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DCMIEN
	Reset value	0	0	0																			0					0	0	0	0	0	0	0	

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x0E0	RCC_AHB4ENR	Res.	Res.	Res.	BKPRAMEN	Res.	Res.	HSEMEN	ADC3EN	Res.	Res.	BDMAEN	Res.	CRCEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	GPIOKEN	GPIOJEN	GPIOJEN	GPIOJEN	GPIOJEN	GPIOGEN	GPIOGEN	GPIOGEN	GPIODEN	GPIODEN	GPIODEN	GPIODEN
	Reset value				0			0	0			0		0									0	0	0	0	0	0	0	0	0	0	0	0
0x0E4	RCC_APB3ENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	WWDG1EN	Res.	Res.	Res.	LTDCCEN	Res.	Res.	Res.
	Reset value																									0				0				
0x0E8	RCC_APB1LENR	UART8EN	UART7EN	DAC12EN	Res.	CECEN	Res.	Res.	Res.	I2C3EN	I2C2EN	I2C1EN	UART5EN	UART4EN	USART13EN	USART12EN	SPDIFRXEN	SP13EN	SP12EN	Res.	Res.	Res.	Res.	LPTIM1EN	TIM14EN	TIM13EN	TIM12EN	TIM7EN	TIM6EN	TIM5EN	TIM4EN	TIM3EN	TIM2EN	
	Reset value	0	0	0		0				0	0	0	0	0	0	0	0	0	0	0				0	0	0	0	0	0	0	0	0	0	0
0x0EC	RCC_APB1HENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FDCANEN	Res.	MDIOSEN	OPAMPEN	Res.	SWPEN	CRSEN	Res.	Res.	
	Reset value																								0		0	0	0	0	0	0	0	0
0x0F0	RCC_APB2ENR	Res.	Res.	HRTIMEN	DFSDM1EN	Res.	Res.	Res.	SAI3EN	SAI2EN	SAI1EN	Res.	SPI5EN	Res.	TIM17EN	TIM16EN	TIM15EN	Res.	Res.	Res.	SPI4EN	SPI1EN	Res.	Res.	Res.	Res.	Res.	USART6EN	USART1EN	Res.	Res.	TIM8EN	TIM1EN	
	Reset value			0	0				0	0	0		0		0	0	0				0	0					0	0	0			0	0	
0x0F4	RCC_APB4ENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SAI4EN	Res.	Res.	Res.	Res.	Res.	RTCAPBEN	VREFEN	COMP12EN	Res.	Res.	LPTIM5EN	LPTIM4EN	LPTIM3EN	LPTIM2EN	Res.	I2C4EN	Reserved	SPI6EN	Res.	LPUART1EN	Res.	Res.	SYSCFGEN
	Reset value										0						1	0	0			0	0	0	0	0	0	0	0	0	0	0	0	0
0x0F8	reserved	Reserved																																
0x0FC	RCC_AHB3LPENR	AXISRAMLPEN	ITCMLPEN	DTCM2LPEN	DTCM1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC1LPEN	Res.	QSPILPEN	Res.	Res.	FMCLPEN	Res.	Res.	Res.	Res.	FLASHLPEN	Res.	JPGDECLPEN	DMA2DLPEN	Res.	Res.	Res.	MDMALPEN
	Reset value	1	1	1	1												1		1			1				1		1	1				1	
0x100	RCC_AHB1LPENR	Res.	Res.	Res.	USB2OTGHSULPILPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETH1RXLPEN	ETH1TXLPEN	ETH1MACLPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ADC12LPEN	Res.	Res.	Res.	DMA2LPEN	DMA1LPEN	
	Reset value				1												1	1	1								1						1	1

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x104	RCC_AHB2LPENR	SRAM3LPEN	SRAM2LPEN	SRAM1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC2LPEN	Res.	Res.	Res.	RNGLPEN	HASHLPEN	CRYPTLPEN	Res.	Res.	Res.	CAMITFLPEN
	Reset value	1	1	1																				1	1		1	1	1				1	
0x108	RCC_AHB4LPENR	Res.	Res.	SRAM4LPEN	BKPRAMLLEN	Res.	Res.	Res.	ADC3LPEN	Res.	Res.	BDMALPEN	Res.	CRCLPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	GPIOKLPEN	GPIOLPEN	GPIOLPEN	GPIOLPEN	GPIOLPEN	GPIOLPEN	GPIODLPEN	GPIODLPEN	GPIODLPEN	GPIODLPEN	GPIODLPEN	GPIODLPEN
	Reset value			1	1				1			1		1									1	1	1	1	1	1	1	1	1	1	1	1
0x10C	RCC_APB3LPENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value																										1							
0x110	RCC_APB1LLPENR	UART8LPEN	UART7LPEN	DAC12LPEN	Res.	CECLPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LPTIM1LPEN	TIM14LPEN	TIM13LPEN	TIM12LPEN	TIM7LPEN	TIM6LPEN	TIM5LPEN	TIM4LPEN	TIM3LPEN	TIM2LPEN	
	Reset value	1	1	1		1					1	1	1	1	1	1	1	1	1	1	1	1		1	1	1	1	1	1	1	1	1	1	1
0x114	RCC_APB1HLPENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FDCANLPEN	Res.	Res.	MDIOSLPEN	OPAMPLPEN	Res.	SWPLPEN	CRSLPEN	Res.	
	Reset value																								1		1	1	1	1	1	1	1	
0x118	RCC_APB2LPENR	Res.	Res.	HRTIMLPEN	DFSDM1LPEN	Res.	Res.	Res.	SAI3LPEN	SAI2LPEN	SAI1LPEN	Res.	SPI5LPEN	Res.	TIM17LPEN	TIM16LPEN	TIM15LPEN	Res.	Res.	SPI4LPEN	SPI1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	USART6LPEN	USART1LPEN	Res.	Res.	TIM8LPEN	TIM1LPEN	Res.
	Reset value			1	1				1	1	1		1		1	1	1			1	1						1	1	1			1	1	1
0x11C	RCC_APB4LPENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SAI4LPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LPTIM5LPEN	LPTIM4LPEN	LPTIM3LPEN	LPTIM2LPEN	Res.	SPI6LPEN	Res.	LPUART1LPEN	Res.	SYSCFGLPEN	Res.
	Reset value																											1					1	
0x120 to 0x130	reserved	Reserved																																

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x134	RCC_C1_AHB3ENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC1EN	Res.	QSPIEN	Res.	FMCEN	Res.	Res.	Res.	Res.	Res.	Res.	JPGDECEN	DMA2DEN	Res.	Res.	Res.	MDMAEN	
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x138	RCC_C1_AHB1ENR	Res.	Res.	Res.	Res.	USB2OTGHSEN	USB1OTGHSULPIEN	USB1OTGHSEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETH1RXEN	ETH1TXEN	ETH1MACEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ADC12EN	Res.	Res.	DMA2EN	DMA1EN	
	Reset value					0	0	0							0	0	0											0			0	0	0	0
0x13C	RCC_C1_AHB2ENR	SRAM3EN	SRAM2EN	SRAM1EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC2EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DCMIEN
	Reset value	0	0	0																				0									0	0
0x140	RCC_C1_AHB4ENR	Res.	Res.	Res.	BKPRAMEN	Res.	Res.	HSEMEN	ADC3EN	Res.	Res.	DMA1EN	Res.	CRCEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	GPIOKEN	GPIOJEN	GPIOIEN	GPIOHEN	GPIOGEN	GPIOFEN	GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN
	Reset value				0			0	0			0		0									0	0	0	0	0	0	0	0	0	0	0	0
0x144	RCC_C1_APB3ENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
	Reset value																																	
0x148	RCC_C1_APB1LENR	UART8EN	UART7EN	DAC12EN	Res.	HDMICECEN	Res.	Res.	Res.	I2C3EN	I2C2EN	I2C1EN	UART5EN	UART4EN	USART3EN	USART2EN	SPDIFRXEN	SPI3EN	SPI2EN	Res.	Res.	Res.	Res.	LPTIM1EN	TIM14EN	TIM13EN	TIM12EN	TIM7EN	TIM6EN	TIM5EN	TIM4EN	TIM3EN	TIM2EN	
	Reset value	0	0	0		0				0	0	0	0	0	0	0	0	0	0	0				0	0	0	0	0	0	0	0	0	0	0
0x14C	RCC_C1_APB1HENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FDCANEN	Res.	Res.	MDIOSEN	OPAMPEN	Res.	SWPEN	CRSEN	Res.	
	Reset value																								0			0			0	0		
0x150	RCC_C1_APB2ENR	Res.	Res.	HRTIMEN	DFSDM1EN	Res.	Res.	SAI3EN	SAI2EN	SAI1EN	Res.	SPI5EN	TIM17EN	TIM16EN	TIM15EN	Res.	Res.	Res.	SPI4EN	SPI1EN	Res.	Res.	Res.	Res.	Res.	Res.	USART6EN	USART11EN	Res.	Res.	TIM8EN	TIM1EN		
	Reset value			0	0			0	0	0		0	0	0	0	0			0	0							0	0			0	0	0	0

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x154	RCC_C1_ APB4ENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SAI4EN	Res.	Res.	Res.	Res.	RTCAPEN	VREFEN	COMP12EN	Res.	LPTIM5EN	LPTIM4EN	LPTIM3EN	LPTIM2EN	Res.	I2C4EN	Res.	SPI6EN	Res.	LPUART1EN	Res.	SYSCFGEN	Res.		
	Reset value										0						1	0	0		0	0	0	0		0		0		0		0			
0x158	reserved	Reserved																																	
0x15C	RCC_C1_ AHB3LPENR	AXISRAMLPEN	ITCMLPEN	DTCM2LPEN	DTCM1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC1LPEN	Res.	QSPILPEN	Res.	FMCLPEN	Res.	Res.	Res.	FLASHLPEN	Res.	Res.	JPGDECLPEN	DMA2DLPEN	Res.	Res.	Res.	MDMALPEN		
	Reset value	1	1	1	1												1		1		1				1			1	1				1		
0x160	RCC_C1_ AHB1LPENR	Res.	Res.	Res.	Res.	USB2OTGHSLPEN	USB1OTGHSLPEN	USB1OTGHSLPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETH1RXLPEN	ETH1TXLPEN	ETH1MACLPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ADC12LPEN	Res.	Res.	Res.	DMA2LPEN	DMA1LPEN		
	Reset value					1	1	1								1	1	1										1					1	1	
0x164	RCC_C1_ AHB2LPENR	SRAM3LPEN	SRAM2LPEN	SRAM1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDMMC2LPEN	Res.	Res.	RNGLPEN	HASHLPEN	CRYPTLPEN	Res.	Res.	Res.	CAMITFLPEN		
	Reset value	1	1	1																				1			1	1	1				1	1	
0x168	RCC_C1_ AHB4LPENR	Res.	Res.	SRAM4LPEN	BKPRAML PEN	Res.	Res.	Res.	ADC3LPEN	Res.	Res.	DMA1LPEN	Res.	CRCLPEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	GPIOKLPEN	GPIOJLPEN	GPIOILPEN	GPIOHPEN	GPIODLPEN	GPIODLPEN	GPIODLPEN	GPIODLPEN	GPIODLPEN	GPIODLPEN		
	Reset value			1	1				1			1		1										1	1	1	1	1	1	1	1	1	1	1	
0x16C	RCC_C1_ APB3LPENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.		
	Reset value																																		
0x170	RCC_C1_ APB1LLPENR	UART8LPEN	UART7LPEN	DAC12LPEN	Res.	CECLPEN	Res.	Res.	Res.	Res.	Res.	I2C3LPEN	I2C2LPEN	I2C1LPEN	UART5LPEN	UART4LPEN	USART3LPEN	USART2LPEN	SPDIFRXLPEN	SP13LPEN	SP12LPEN	Res.	Res.	Res.	Res.	LPTIM1LPEN	TIM14LPEN	TIM13LPEN	TIM12LPEN	TIM7LPEN	TIM6LPEN	TIM5LPEN	TIM4LPEN	TIM3LPEN	TIM2LPEN
	Reset value	1	1	1		1						1	1	1	1	1	1	1	1	1	1					1	1	1	1	1	1	1	1	1	

Table 83. RCC register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x174	RCC_C1_ APB1HLPENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FDCANILPEN		Res.	Res.	MDIOSLPEN	OPAMPLPEN	Res.	SWPLPEN	CRSLPEN	Res.
	Reset value																								1			1	1		1	1		
0x178	RCC_C1_ APB2LPENR	Res.	Res.	HRTIMLPEN	DFSDM1LPEN	Res.	Res.	Res.	SA3LPEN	SAI2LPEN	SAI1LPEN	Res.	SPI5LPEN	Res.	TIM17LPEN	TIM16LPEN	TIM15LPEN	Res.	Res.	SPI4LPEN	SPI1LPEN	Res.	Res.	Res.	Res.	Res.	Res.	USART6LPEN	USART1LPEN	Res.	Res.	TIM8LPEN	TIM1LPEN	
	Reset value			1	1				1	1	1		1		1	1	1			1	1							1	1				1	
0x17C	RCC_C1_ APB4LPENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SAI4LPEN	Res.	Res.	Res.	Res.	Res.	RTCAPBLPEN	VREFLPEN	COMP12LPEN	Res.	LPTIM5LPEN	LPTIM4LPEN	LPTIM3LPEN	LPTIM2LPEN	Res.	I2C4LPEN	Res.	SPI6LPEN	Res.	LPUART1LPEN	Res.	SYSCFGLPEN	Res.	
	Reset value										1						1	1	1		1	1	1	1		1		1		1				
0x180 to 0x1FC	reserved	Reserved																																

1. Available only on revision Y devices.
2. Available only on revision V devices.

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.